Bankruptcy Data Analysis Report

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— Aniv Mazumder, Koushan Saha, Shashwat Asthana

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1. Introduction

This study uses the classic Bankruptcy Dataset, a well-known benchmark in multivariate statistics from Johnson and Wichern's *Applied Multivariate Statistical Analysis*, based on Edward Altman's work on bankruptcy prediction. The dataset classifies firms into two groups using a binary response variable Y: financially sound firms ($Y = 1$) and firms observed within the final two years prior to bankruptcy ($Y = 0$).

The primary objective is to evaluate multivariate classification methods. This analysis is based on the assumption that financial distress develops over time, leading to systematic differences in both the population mean vectors ($\boldsymbol{\mu}$) and covariance structures ($\boldsymbol{\Sigma}$) of the financial ratios between the two groups.

The predictor space consists of four continuous financial ratios, denoted as $\mathbf{X} = (X_1, X_2, X_3, X_4)^T$. These ratios serve as established proxies for a firm's liquidity, profitability, and overall structural stability. The expectation is that the multivariate distribution of these ratios contains significant predictive information regarding the probability of default.

The variables are formally defined as follows:

- X_1: Cash Flow to Total Debt. This is a primary solvency metric. It evaluates the firm's capacity to service its total debt obligations using its operating cash flows. Lower values indicate a higher risk of insolvency.
- X_2: Net Income to Total Assets (Return on Assets). This ratio measures overall profitability and asset utilization efficiency. It reflects the firm's ability to generate earnings independent of its capital structure.
- X_3: Current Assets to Total Liabilities. A measure of aggregate liquidity. This ratio assesses the firm's short-term financial resilience by comparing its most liquid assets to its total outstanding obligations.
- X_4: Current Assets to Net Sales. An activity and liquidity turnover metric. It indicates the proportion of short-term assets required to support the firm's sales volume, providing insight into working capital management.

Objective and Scope of the Analysis

The primary objective of this comprehensive analysis is to develop statistical models capable of effectively predicting corporate bankruptcy status based on a multidimensional feature space of financial ratios. Rather than relying on a single modeling paradigm, this study employs a rigorous sequence of data preprocessing, dimensional reduction, probabilistic classification to fully map the structural differences between bankrupt and financially sound cohorts.

The analytical pipeline proceeds through several phases:

- **Exploratory Analysis:** The analysis initiates with a rigorous assessment of marginal and multivariate normality. To mathematically support the application of variance-sensitive multivariate techniques—particularly Linear Discriminant Analysis (LDA) and Quadratic Discriminant Analysis (QDA)—we execute transformations (e.g., Box-Cox) and multivariate outlier excision to ensure homoscedasticity and adherence to theoretical Gaussian distribution. Logistic Regression is concurrently deployed as a discriminative benchmark that relaxes these strict distributional assumptions.
- **Dimensionality Reduction and Latent Structures:** To combat dimensional redundancy and isolate the core predictive signals, the transformed feature space is subjected to Principal Component Analysis (PCA). The efficacy of the classification models is subsequently re-evaluated within this lower-dimensional, orthogonal subspace. Parallel to this, Exploratory Factor Analysis (EFA) is applied to investigate whether the underlying covariance structure can be adequately explained by unobserved, shared latent financial constructs.
- **Probabilistic Classification:** Expanding beyond linear and quadratic hyperplanes, the study incorporates Naive Bayes classification to evaluate predictive performance under the assumption of conditional independence among the financial metrics.

- **Unsupervised Clustering and Geometric Discovery:** Finally, the analysis shifts to an unsupervised framework to investigate the natural geometric grouping of the data. Three distinct clustering strategies—standard stochastic k-means, k-means with supervised group-mean initialization, and a hybrid methodology utilizing LDA-derived discriminant scores—are executed to assess how closely the inherent topological clusters align with the true, unobserved bankruptcy labels.

Ultimately, by systematically evaluating the data through these diverse methodological lenses, this report aims to establish the most structurally sound and mathematically reliable framework for financial distress classification.

2. Data Exploration and Marginal Normality

2.1 Dataset Description and Initial Analysis

The dataset contains no missing values, and all observations are complete across the four financial ratio variables and the bankruptcy indicator. The final column is binary data (0 or 1).

```
> head(data)
     x1    x2   x3   x4 y
1 -0.45 -0.41 1.09 0.45 0
2 -0.56 -0.31 1.51 0.16 0
3  0.06  0.02 1.01 0.40 0
4 -0.07 -0.09 1.45 0.26 0
5 -0.10 -0.09 1.56 0.67 0
6 -0.14 -0.07 0.71 0.28 0
```

Figure 2.1: Dataset snapshot

```
> print(sum(is.na(data))) # no missing values
[1] 0
```

2.2 Histograms and Skewness

2.2.1 Columnwise Histograms

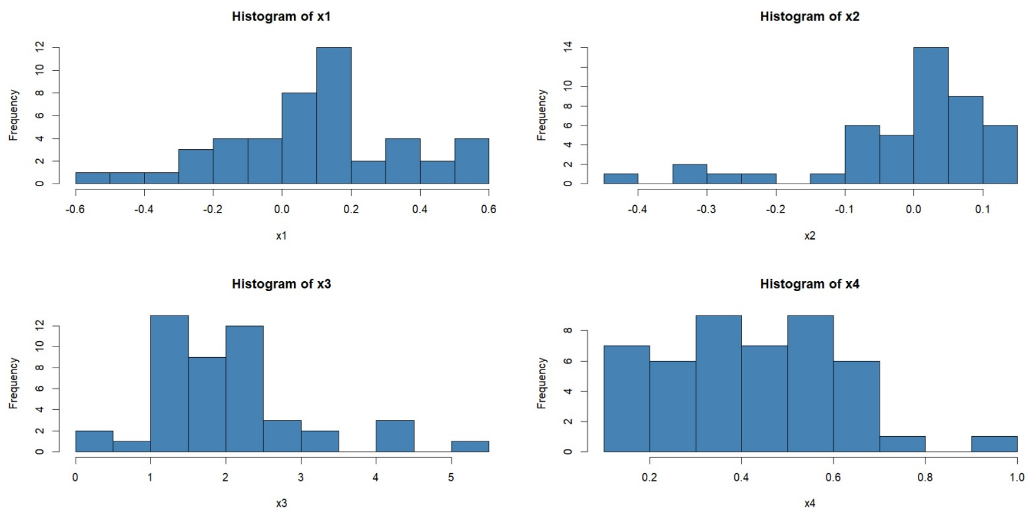


Figure 2.2: Overall distribution of variables.

```
> apply(df_vars, 2, skewness)
        x1         x2         x3         x4 
-0.2041031 -1.5364893  1.0759011  0.3469426 
```

- X_1 (Cash flow to total debt): The distribution exhibits a slight negative skewness (-0.204). Visually, it is the most symmetric of the four variables, though it possesses a minor left tail with the central mass concentrated between 0.0 and 0.2.
- X_2 (Net income to total assets): This variable displays a severe negative skewness (-1.536). The histogram confirms a heavy left tail; the bulk of the data is concentrated on the right (near 0.0 to 0.1), indicating that while most firms have moderate profitability, a subset exhibits extreme negative returns.
- X_3 (Current assets to total liabilities): The distribution is characterized by a pronounced positive skewness (1.076). The empirical mass is heavily concentrated at the lower end (between 1.0 and 2.5), with a long right tail driven by a few firms possessing exceptionally high aggregate liquidity.
- X_4 (Current assets to net sales): This ratio shows a mild positive skewness (0.347). The distribution appears somewhat platykurtic and roughly symmetric, with observations relatively well-dispersed between 0.1 and 0.7 before tapering off in the right tail.

2.2.2 Groupwise Histograms

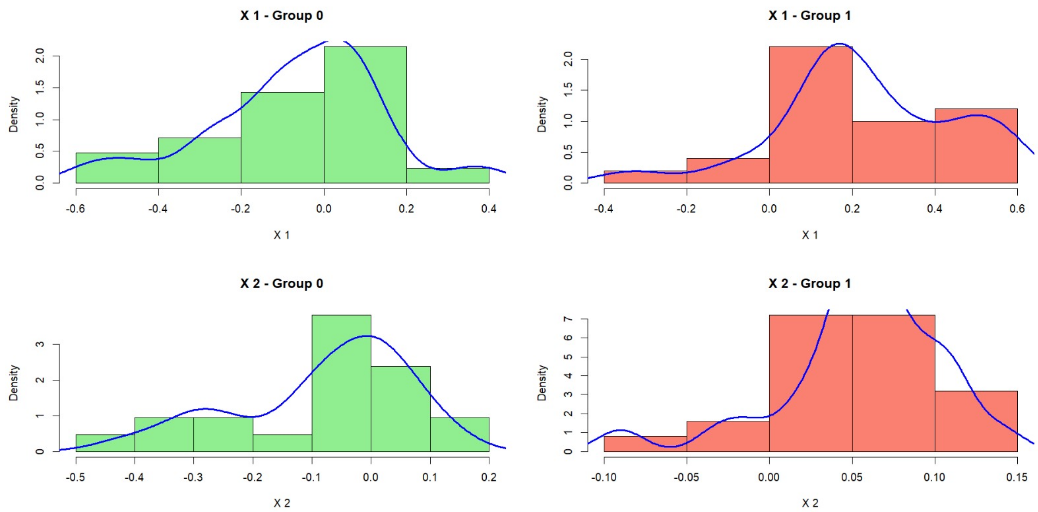


Figure 2.3: Group-wise histograms for Columns 1 and 2. Both variables show shifts in location between groups, with Group 1 generally taking higher values.

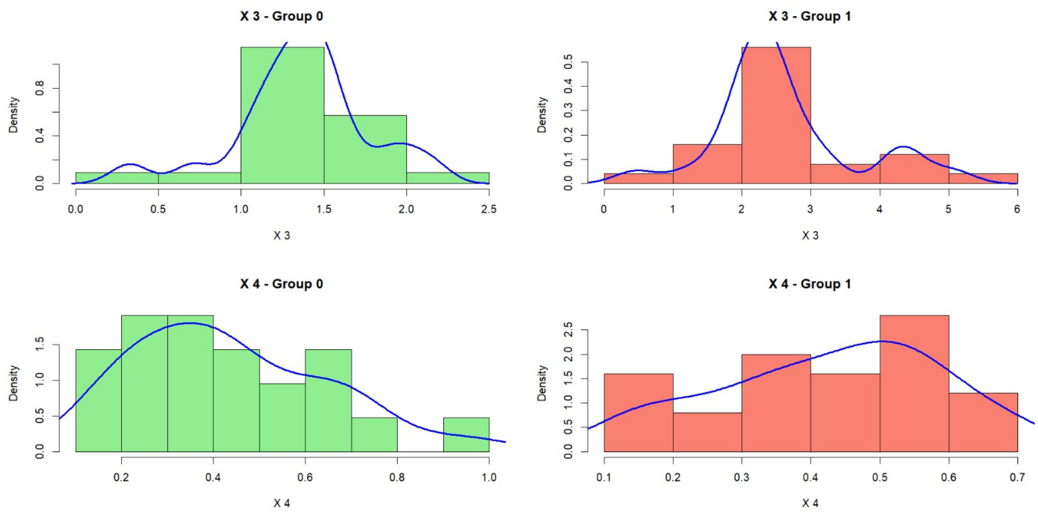


Figure 2.4: Group-wise histograms for Columns 3 and 4. Column 3 shows a clear difference in values between groups, while Column 4 shows similar shapes with more overlap.

A visual inspection of the group-conditioned and marginal histograms yields the following insights regarding distributional morphology and class separability:

- **Location Shifts in Primary Predictors (X_1, X_2, X_3):** The conditional distributions for the first three variables exhibit consistent shifts in location. Specifically, the financially sound cohort ($Y = 1$) demonstrates systematically higher central tendencies across these metrics compared to the bankrupt cohort ($Y = 0$).

- **Varying Degrees of Separability:** Variable X_3 displays the most pronounced distributional shift with minimal overlap between the two groups, indicating strong discriminatory power. Variables X_1 and X_2 exhibit moderate overlap, though their respective group centers remain structurally distinct.

- **Weak Discriminatory Power of X_4:** The conditional distributions of X_4 for the two groups exhibit substantial overlap, with only minor differences in spread and central tendency. This indicates that X_4 provides limited univariate discriminatory power for class separation.

- **Marginal Shape and Dispersion:** Marginally, the feature space is predominantly unimodal. X_3 is characterized by a wider empirical domain and a pronounced right tail (positive skewness). X_1 and X_2 display a tighter concentration around their central masses, whereas X_4 exhibits a more uniform, even dispersion across its range.

These visual diagnostics corroborate earlier empirical observations: X_2 and X_3 are the primary drivers of group separability, whereas X_4 remains largely uninformative in a univariate context.

2.3 Boxplots of Individual Variables

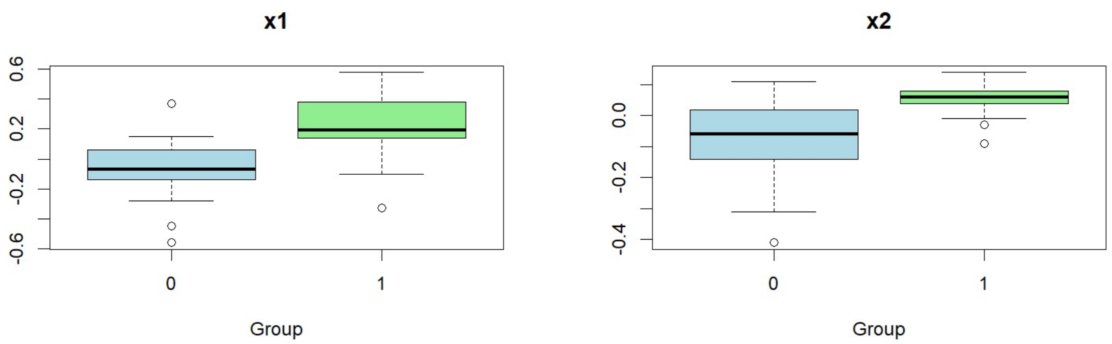


Figure 2.5: Boxplots for Columns 1 and 2. Both variables show differences in central values between the two groups. Column 2 shows a clearer separation, while Column 1 has more overlap and a wider spread within groups.

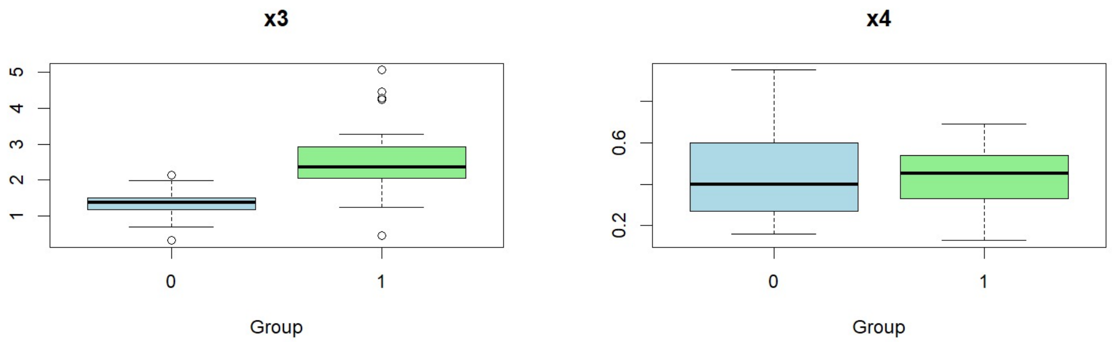


Figure 2.6: Boxplots for Columns 3 and 4. Column 3 shows a clear shift in values between groups, with higher values in Group 1. Column 4 shows similar central values across groups with considerable overlap. Outliers are present, particularly in Columns 1 and 3.

An analysis of the side-by-side boxplots provides a visualization of the conditional distributions, highlighting critical differences in location, scale (dispersion), and the prevalence of extreme values across the two cohorts:

- **Primary Discriminators (X_2 and X_3):** Variables X_2 and X_3 exhibit the most pronounced location shifts and structural separation. For X_2, the financially sound cohort ($Y = 1$) is tightly clustered with positive returns, whereas the bankrupt cohort ($Y = 0$) demonstrates high variance and predominantly negative values.

For X_3, Group 1 maintains systematically higher aggregate liquidity but with increased variance, contrasting sharply with the tightly constrained, lower-level liquidity of Group 0.

- **Moderate Separability (X_1):** Variable X_1 displays moderate discriminatory potential. While the median cash flow ratio for Group 1 is visibly elevated compared to Group 0, there is substantial overlap in their respective interquartile ranges (IQRs). Both groups exhibit considerable within-class variability.
- **Lack of Discriminatory Power (X_4):** Consistent with the correlation and scatterplot diagnostics, X_4 reveals near-complete structural overlap between the two classes. The group medians are nearly equivalent, and their IQRs largely intersect, confirming that this variable contributes negligibly to group separation.
- **Impact of Extreme Observations:** Noticeable outliers are present across several dimensions—most notably the severe lower-tail values in X_1 and X_2 for both groups, and the extreme upper-tail values in X_3 for Group 1. The presence of these heavy tails violates the assumption of strict multivariate normality and suggests that classical mean-based tests (such as MANOVA) should be interpreted with caution, as they are highly sensitive to such deviations.

2.4 Correlation Structure

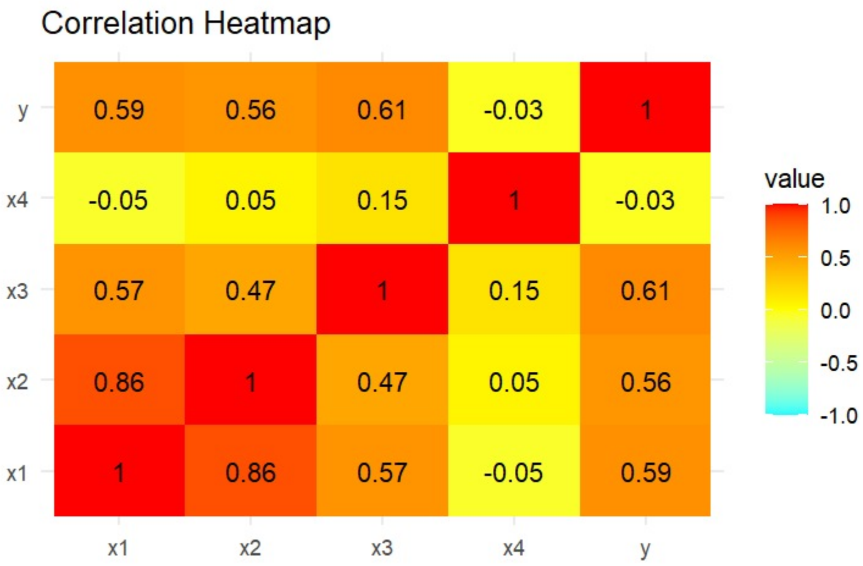


Figure 2.7: Correlation heatmap of variables. Strong positive relationships are observed among several variables, particularly between Columns 1 and 2.

Computing the point-biserial correlation between the binary response Y and continuous covariates rapidly diagnoses linear discriminative power. High absolute correlations indicate distinct class means, signaling strong predictive viability for downstream linear classifiers (e.g., LDA, Logistic Regression).

An analysis of the pairwise correlation heatmap yields the following key insights regarding the linear dependencies among the feature space and the response variable:

- **Strong Predictor Correlation:** Columns 1 and 2 exhibit a highly positive correlation ($r \approx 0.86$), indicating that these variables co-vary closely together.
- **Moderate Interdependence:** Column 3 is also positively related to Columns 1 and 2, though the magnitude of these linear relationships is notably weaker.
- **Association with the Response:** The response variable demonstrates moderate positive correlations with Columns 1, 2, and 3. This suggests that increases in these specific predictors are systematically associated with higher values of the response.
- **Orthogonality of Column 4:** In stark contrast, Column 4 displays near-zero correlations with all other predictors as well as the response variable, indicating an absence of any meaningful linear relationship.
- **Structural Implications:** The presence of strong pairwise correlations among the first three predictors suggests potential multicollinearity (information redundancy) within the dataset. Conversely, Column 4 appears to encapsulate distinct, non-overlapping information compared to the rest of the feature space.

2.5 Scatterplot Analysis

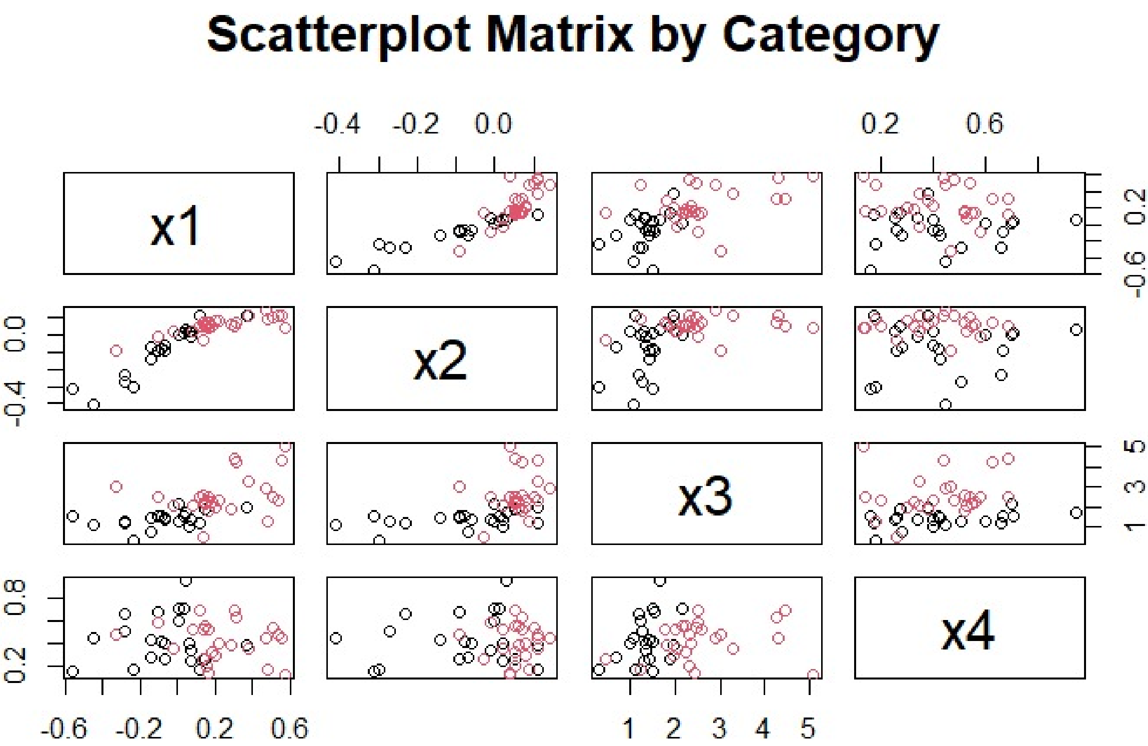


Figure 2.8: Scatterplot matrix of variables. Clear positive relationships are visible among several pairs of variables, with partial separation between groups.

An examination of the pairwise scatterplot matrix corroborates the covariance structures and class separability observed in prior diagnostics, yielding the following insights:

- **Pronounced Bivariate Dependence (X_1, X_2):** The joint distribution of variables X_1 and X_2 exhibits a highly structured, upward linear trajectory, visually confirming their strong positive correlation.
- **Moderate Structural Interdependence:** The bivariate plots for (X_1, X_3) and (X_2, X_3) also display positive linear associations, though the point clouds exhibit greater dispersion relative to the (X_1, X_2) pairing.
- **Multivariate Class Separability:** The feature subspace spanned by variables X_2 and X_3 reveals the most effective geometric partitioning between the two groups. The cohorts occupy largely distinct regions within these coordinate planes, despite some boundary overlap.
- **Orthogonality and Inefficacy of X_4:** Scatter distributions incorporating X_4 resolve into unstructured, amorphous clouds. The groups are thoroughly mixed within these projections, visually confirming that X_4 lacks both covariance with the primary predictors and meaningful discriminatory leverage.

Ultimately, the matrix visualizes a clear underlying structure: the primary multivariate signal distinguishing financially sound firms from those facing bankruptcy is embedded almost entirely within the (X_1, X_2, X_3) subspace.

2.6 Bivariate Predictor-Response Relationships

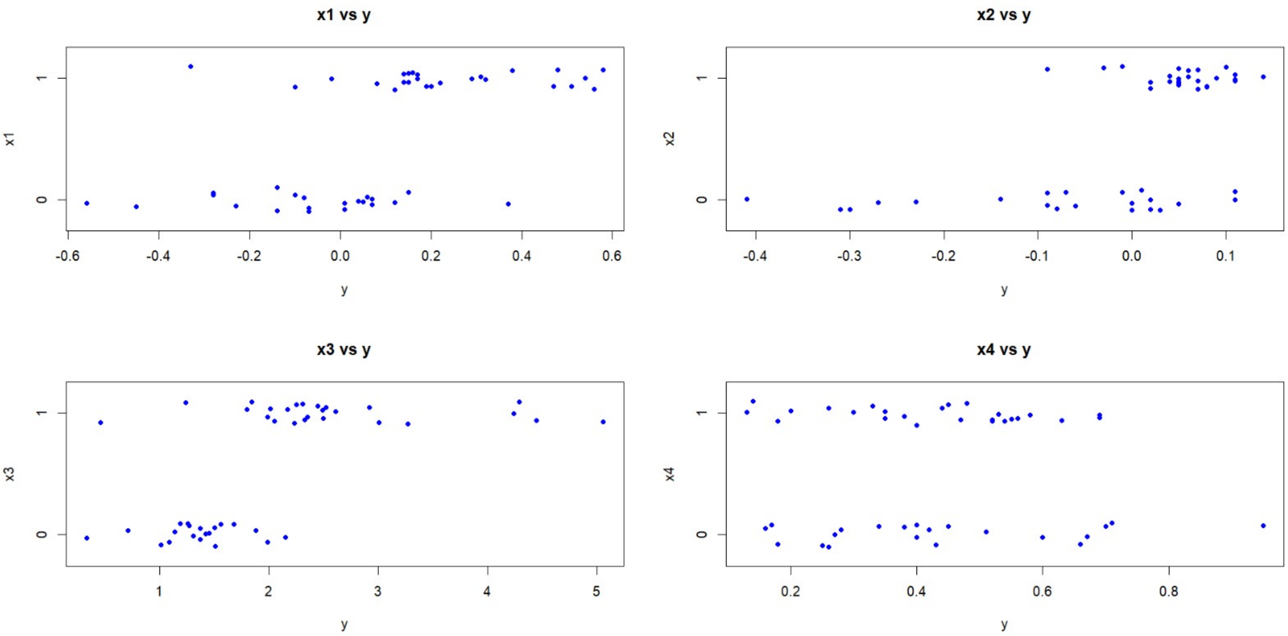


Figure 2.9: Scatterplots of each variable against the response variable

The jittered scatterplots projecting each continuous financial ratio against the binary response variable (Y) provide a clear visual assessment of their respective univariate discriminatory power. By examining the conditional distributions of the predictors across the two response classes, the following structural relationships emerge:

- **Primary Discriminator (X_3):** This variable exhibits the most strictly structured relationship with the outcome. The observations form two distinct vertical clusters corresponding to the $Y = 0$ and $Y = 1$ cohorts. The pronounced location shift and minimal structural overlap between these groups underscore X_3's robust capacity to mathematically distinguish between bankrupt and financially sound firms.
- **Strong Association (X_2):** Variable X_2 likewise demonstrates a clear discriminatory pattern. Higher values of this profitability metric are heavily concentrated within the $Y = 1$ group, confirming a strong positive conditional association, albeit with a slight increase in lower-tail intersection compared to X_3.
- **Moderate Association (X_1):** The solvency metric, X_1, displays a visible but attenuated location shift. While the central tendency is systematically higher for $Y = 1$, the structural overlap between the two response bands is substantial, indicating that X_1 possesses moderate, rather than definitive, discriminatory utility on its own.
- **Weak Association (X_4):** In stark contrast, X_4 reveals a largely uniform dispersion across both response levels. The spatial distribution of observations for $Y = 0$ is virtually indistinguishable from that of $Y = 1$. This geometric mixing confirms that X_4 provides negligible univariate leverage for classifying the response.

Collectively, these visual diagnostics reaffirm the covariance structures observed previously: X_2 and X_3 possess the strongest deterministic relationships with the response, X_1 provides moderate supplemental signal, and X_4 remains largely uninformative.

2.7 Normality Assessment

2.7.1 Assessment of Univariate Normality via Q-Q Plots

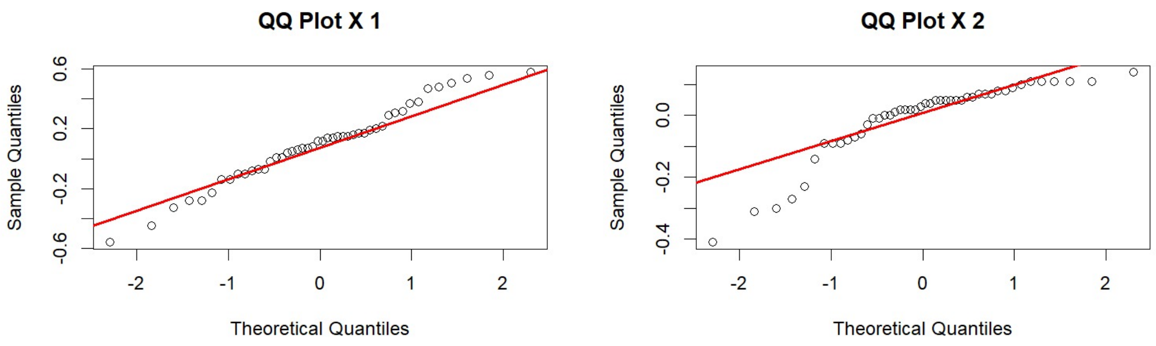


Figure 2.10: Q-Q plots for Columns 1 and 2. Column 1 shows moderate deviation in the tails, while Column 2 shows stronger deviation from the reference line, particularly in the lower tail.

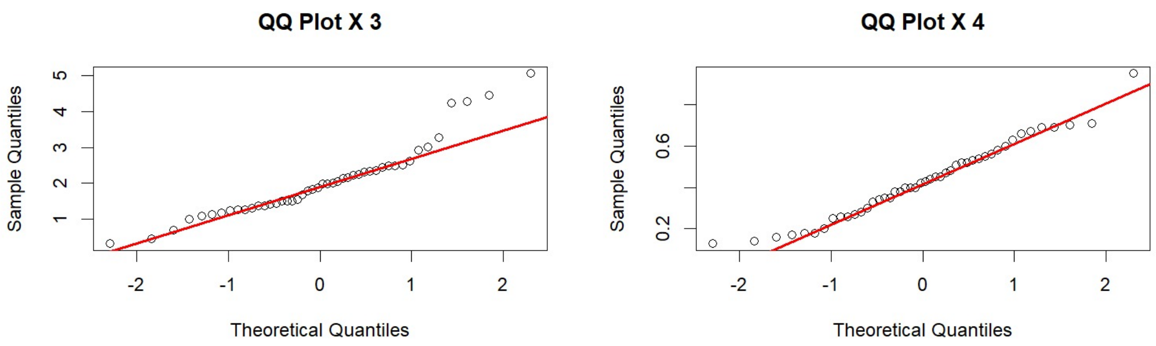


Figure 2.11: Q-Q plots for Columns 3 and 4. Column 3 shows noticeable deviation in the upper tail, while Column 4 is closer to the reference line with smaller deviations.

A formal visual inspection of the Normal Quantile-Quantile (Q-Q) plots reveals that the marginal distributions of the feature space systematically violate the assumption of strict univariate normality. These departures are predominantly localized in the tails, which carries significant implications for parametric modeling:

- X_1 **(Cash flow to total debt):** The plot exhibits moderate but distinct departures from the theoretical Gaussian reference line at both extrema. The sample quantiles diverge slightly at the edges, indicating a distribution with marginally heavier tails than a true normal distribution, despite a relatively well-behaved central mass.
- X_2 **(Net income to total assets):** This variable demonstrates a severe structural deviation, most notably in the lower tail where the sample quantiles fall drastically below the reference line. This pronounced downward trajectory signifies a heavy left tail, visually corroborating the severe negative skewness and the presence of extreme underperforming outliers.
- X_3 **(Current assets to total liabilities):** The Q-Q plot for X_3 displays a stark upward convexity in the right tail. Here, the sample quantiles significantly exceed theoretical Gaussian expectations, providing definitive evidence of a heavily right-skewed distribution driven by a subset of firms with exceptionally high liquidity.
- X_4 **(Current assets to net sales):** Among the four covariates, X_4 conforms most closely to the theoretical reference line. However, it still exhibits mild deviations at the absolute extremes. While it is the most approximately normal variable in the dataset, it is not perfectly Gaussian and retains minor tail anomalies.

Collectively, these non-linear alignments establish that the underlying joint distribution of $\mathbf{X}$ is not univariate normal.

2.7.2 Formal Assessment of Univariate Normality

To rigorously evaluate the marginal distributions of the feature space, three distinct formal tests for univariate normality were employed. The null hypothesis (H_0) for all tests postulates that the data follows a normal distribution.

- **Shapiro-Wilk Test:** This test evaluates the ratio of two distinct estimators of the population variance. The test statistic, W, is essentially the squared Pearson correlation coefficient between the ordered sample values (order statistics), $x_{(i)}$, and the expected values of the standard normal order statistics. Mathematically, it is defined as $W = \frac{\left(\sum_{i=1}^{n} a_i x_{(i)}\right)^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$, where the weights a_i are derived from the covariance matrix of the standard normal order statistics. Because it rigorously evaluates the structural geometry of the data against theoretical normal quantiles, it is widely recognized as providing the highest statistical power against a broad spectrum of asymmetric and heavy-tailed alternative distributions.
- **Anderson-Darling Test:** Belonging to the Cramér-von Mises family of Empirical Cumulative Distribution Function (ECDF) tests, this statistic integrates the squared difference between the empirical CDF, $F_n(x)$, and the theoretical normal CDF, $\Phi(x)$. The critical mathematical distinction is its specific weight function, $[\Phi(x)(1-\Phi(x))]^{-1}$. The test statistic is given by $A^2 = n\int_{-\infty}^{\infty} \frac{(F_n(x)-\Phi(x))^2}{\Phi(x)(1-\Phi(x))} d\Phi(x)$. Because this denominator approaches zero in the tails, the test artificially amplifies discrepancies at the extremes of the distribution. Consequently, it is exceptionally sensitive to outliers and heavy-tailed deviations.
- **Kolmogorov-Smirnov Test (Asymptotic):** Unlike the integral-based Anderson-Darling test, the K-S test relies on the supremum (maximum absolute) metric. The test statistic is defined as $D = \sup_x |F_n(x) - \Phi(x)|$, capturing the largest single vertical discrepancy between the empirical and theoretical CDFs. Because the theoretical normal CDF is steepest near its mean, the maximum absolute divergence generally occurs in the central mass of the distribution. It assigns equal mathematical weight across the entire continuous domain, effectively rendering it far less sensitive to tail deviations compared to A^2.

Assuming a standard significance level of $\alpha = 0.05$, the variable-by-variable results are as follows:

Analysis of Variable X_1

```
Variable: x1

    Shapiro-Wilk normality test
data:  data_col
W = 0.97843, p-value = 0.5431

    Anderson-Darling normality test
data:  data_col
A = 0.33462, p-value = 0.4971

    Asymptotic one-sample Kolmogorov-Smirnov test
data:  data_col
D = 0.084771, p-value = 0.8956
alternative hypothesis: two-sided
```

Interpretation: For the cash flow to total debt ratio (X_1), all three tests yield p-values well above the 0.05 threshold. We fail to reject the null hypothesis across all metrics. This indicates that, despite minor visual deviations noted in earlier Q-Q plots, X_1 does not exhibit statistically significant departures from univariate normality.

Analysis of Variable X_2

```
Variable: x2

    Shapiro-Wilk normality test
data:  data_col
W = 0.81911, p-value = 5.302e-06

    Anderson-Darling normality test
data:  data_col
A = 2.8638, p-value = 2.551e-07
```

```
    Asymptotic one-sample Kolmogorov-Smirnov test
data:  data_col
D = 0.2076, p-value = 0.03794
alternative hypothesis: two-sided
```

Interpretation: For the profitability metric (X_2), all three tests reject the null hypothesis of normality ($p < 0.05$). The extreme significance of the Anderson-Darling ($p \approx 2.55 \times 10^{-7}$) and Shapiro-Wilk tests confirms severe structural deviations. This mathematically corroborates the heavy left tail and severe negative skewness observed in the marginal histogram and Q-Q diagnostics.

Analysis of Variable X_3

```
Variable: x3

    Shapiro-Wilk normality test
data:  data_col
W = 0.91132, p-value = 0.001916

    Anderson-Darling normality test
data:  data_col
A = 1.2777, p-value = 0.002253

    Asymptotic one-sample Kolmogorov-Smirnov test
data:  data_col
D = 0.1405, p-value = 0.3239
alternative hypothesis: two-sided
```

Interpretation: The liquidity ratio (X_3) presents mixed results that highlight the mathematical nuances of the tests. Both the Shapiro-Wilk and Anderson-Darling tests strictly reject the null hypothesis ($p \approx 0.002$). However, the Kolmogorov-Smirnov test fails to reject ($p = 0.3239$). Because X_3 is characterized by a pronounced positive skew (a heavy right tail) and a relatively stable central mass, the tail-sensitive Anderson-Darling test captures the deviation, while the center-sensitive Kolmogorov-Smirnov test lacks the power to detect it. We conclude that X_3 departs significantly from normality, driven predominantly by tail anomalies.

Analysis of Variable X_4

```
Variable: x4

    Shapiro-Wilk normality test
data:  data_col
W = 0.97263, p-value = 0.3457

    Anderson-Darling normality test
data:  data_col
A = 0.24514, p-value = 0.7466

    Asymptotic one-sample Kolmogorov-Smirnov test
data:  data_col
D = 0.055895, p-value = 0.9988
alternative hypothesis: two-sided
```

Interpretation: Similar to X_1, X_4 fails to reject the null hypothesis across all three diagnostic tests. With p-values ranging from 0.3457 to nearly 1.0, there is insufficient empirical evidence to conclude that X_4 deviates from a Gaussian distribution. This confirms that X_4 is the most univariately normal feature in the dataset.

2.7.3 Multivariate Normality Tests

While univariate tests provide necessary marginal diagnostics, they are mathematically insufficient to guarantee joint multivariate normality. A dataset may possess perfectly Gaussian marginal distributions but exhibit severe non-linear dependencies or multidimensional anomalies. To rigorously evaluate the joint probability density of our $p = 4$ feature space, three distinct multivariate tests were applied, each targeting different structural properties of the theoretical multivariate Gaussian distribution.

- **Mardia's Multivariate Skewness and Kurtosis Test:**
  - **Mathematical Framework:** Mardia's test evaluates multivariate normality by explicitly isolating and calculating the third and fourth multidimensional moments. Let $\mathbf{x}_i$ and $\mathbf{x}_j$ represent two observation vectors, and let $\mathbf{S}$ be the sample covariance matrix. The scaled Mahalanobis distance between any two points is $D_{ij} = (\mathbf{x}_i - \bar{\mathbf{x}})^T \mathbf{S}^{-1} (\mathbf{x}_j - \bar{\mathbf{x}})$.
    Mardia's multivariate skewness ($b_{1,p}$) and kurtosis ($b_{2,p}$) are defined as:
    $$b_{1,p} = \frac{1}{n^2} \sum_{i=1}^{n} \sum_{j=1}^{n} D_{ij}^3$$
    $$b_{2,p} = \frac{1}{n} \sum_{i=1}^{n} D_{ii}^2$$
    Under the null hypothesis of multivariate normality, the transformed skewness statistic asymptotically follows a χ^2 distribution, while the kurtosis statistic follows a standard normal distribution.
  - **Methodological Relevance:** This test is highly relevant for diagnosing *how* a dataset deviates from multivariate normality. By decomposing the assessment, we can identify if the violation is driven by multidimensional asymmetry (skewness) or by excessively heavy joint tails (kurtosis), which directly informs the type of Box-Cox transformation required.
- **Henze-Zirkler (HZ) Test:**
  - **Mathematical Framework:** The Henze-Zirkler test is a highly robust omnibus test based on the empirical characteristic function. It calculates the weighted squared distance between the empirical characteristic function of the sample data, $\Psi_n(\mathbf{t})$, and the theoretical characteristic function of the multivariate normal distribution, $\Psi(\mathbf{t})$:
    $$HZ = \int_{\mathbb{R}^p} |\Psi_n(\mathbf{t}) - \Psi(\mathbf{t})|^2 w_\beta(\mathbf{t}) d\mathbf{t}$$
    where $w_\beta(\mathbf{t})$ is a mathematically optimized smoothing weight function parameterized by β.
  - **Methodological Relevance:** The HZ test is universally considered one of the most powerful and consistent tests for multivariate normality. Because it evaluates the entire characteristic function over the full integrated space rather than specific moments, it is highly sensitive to a wide spectrum of complex structural deviations that Mardia's test might miss. It serves as our definitive, global pass/fail metric for the joint distribution.
- **Royston's Test:**
  - **Mathematical Framework:** Royston's test is the direct multivariate extension of the univariate Shapiro-Wilk (W) test. It mathematically transforms the individual W statistics of each variable into standard normal approximations (Z_j). It then computes the sample correlation matrix to adjust for inter-variable dependencies, synthesizing the values into an omnibus H statistic:
    $$H = \frac{e}{p} \sum_{j=1}^{p} Z_j^2 \sim \chi_e^2$$
    where e represents the equivalent degrees of freedom adjusted for the geometric covariance of the features.
  - **Methodological Relevance:** Because it builds directly upon the Shapiro-Wilk framework, Royston's test is exceptionally sensitive. It is particularly relevant when we want to ensure that the marginal normalizations achieved via univariate Box-Cox transformations translate effectively into joint multivariate normality. It is highly effective in small-to-moderate sample sizes and aggressively flags datasets where marginal distributions are Gaussian but their joint intersection is not.

To rigorously evaluate the assumption that the continuous feature space, $\mathbf{X} = (X_1, X_2, X_3, X_4)^T$, follows a Multivariate Normal (MVN) distribution, a comprehensive suite of formal statistical tests was conducted. The results provide overwhelming, consistent empirical evidence against the null hypothesis of multivariate normality ($H_0 : \mathbf{X} \sim \mathcal{N}_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$):

- **Mardia's Skewness and Kurtosis:** Mardia's tests evaluate the multivariate extensions of the third and fourth central moments. The highly significant test statistics for both multivariate skewness ($p < 0.001$) and multivariate kurtosis ($p = 0.001$) quantitatively confirm the structural departures observed in the marginal Q-Q plots. This definitively indicates the presence of severe asymmetric tail behavior and non-Gaussian peak/tail thickness within the joint distribution.

```
> mardia_res$multivariate_normality
            Test Statistic p.value     Method         MVN
1 Mardia Skewness    58.996  <0.001 asymptotic ✗ Not normal
2 Mardia Kurtosis     3.194   0.001 asymptotic ✗ Not normal
> hz_res$multivariate_normality
           Test Statistic p.value     Method         MVN
1 Henze-Zirkler     1.708  <0.001 asymptotic ✗ Not normal
> royston_res$multivariate_normality
     Test Statistic p.value     Method         MVN
1 Royston     29.27  <0.001 asymptotic ✗ Not normal
```

Figure 2.12: Multivariate normality test results

- **Henze-Zirkler Omnibus Test:** The Henze-Zirkler test, a highly robust omnibus procedure based on the empirical characteristic function, strictly rejects the MVN assumption ($p < 0.001$). This confirms that the global dependence structure and marginal distributions collectively fail to align with a theoretical multivariate Gaussian distribution.
- **Royston's Test:** Functioning as a multivariate generalization of the Shapiro-Wilk test, Royston's test relies on the transformation of marginal distributions to approximate normality. It also emphatically rejects the null hypothesis ($p < 0.001$), providing methodologically distinct corroboration of the non-normal data structure.
- **Analytical Implications:** Because all deployed tests uniformly reject the assumption of multivariate normality at conventional significance levels (e.g., $\alpha = 0.01$), the structural integrity of classical inferential techniques that mandate strict MVN (such as standard MANOVA or strict Maximum Likelihood-based Discriminant Analysis) is compromised.

2.7.4 Multicollinearity Assessment

Before establishing a formal predictive model, it is necessary to evaluate the feature space for multicollinearity. Because the response variable (Y) is binary, standard Ordinary Least Squares (OLS) regression is inappropriate. Instead, a logistic regression model—implemented as a Generalized Linear Model (GLM) with a binomial link function—was specified as the baseline framework:

$$\text{logit}(\pi_i) = \log\left(\frac{\pi_i}{1-\pi_i}\right) = \mathbf{X}_i^T\boldsymbol{\beta} \tag{2.1}$$

where $\pi_i = P(Y_i = 1|\mathbf{X}_i)$. The parameters $\boldsymbol{\beta}$ are estimated via Maximum Likelihood (MLE), which yields the asymptotic covariance matrix:

$$\text{Var}(\hat{\boldsymbol{\beta}}) \approx (\mathbf{X}^T\mathbf{W}\mathbf{X})^{-1} \tag{2.2}$$

Here, $\mathbf{X}$ is the design matrix and $\mathbf{W}$ is a diagonal weight matrix with elements $w_{ii} = \hat{\pi}_i(1-\hat{\pi}_i)$.

Variance Inflation Factors (VIF) were extracted directly from this fitted model. In the GLM context, the VIF for the j-th predictor quantifies the extent to which the variance of the estimated coefficient, $\text{Var}(\hat{\beta}_j)$, is inflated due to linear dependencies among the covariates, relative to a strictly orthogonal (weighted) design matrix. Mathematically, it operates analogously to the OLS definition:

$$\text{VIF}_j = \frac{1}{1-R_j^2} \tag{2.3}$$

where R_j^2 represents the pseudo-coefficient of determination obtained by regressing the j-th predictor on all remaining predictors. The synthesis of these VIF diagnostics establishes the structural boundaries for subsequent modeling:

```
> # We fit a logistic regression model to check VIF among predictors
> vif_model <- glm(y ~ x1 + x2 + x3 + x4, data = data, family = binomial)
> print(vif(vif_model))
      x1       x2       x3       x4 
4.329110 4.280474 1.454247 1.217809 
```

Figure 2.13: Variance Inflation Factor (VIF) values for predictors

- **Moderate Linear Dependence (X_1 and X_2):** The VIF values for X_1 (4.33) and X_2 (4.28) indicate a moderate degree of multicollinearity. Mathematically, an $R_j^2 \approx 0.77$ implies that roughly 77% of the variance in these specific financial metrics is perfectly explained by the remaining predictors. This structural redundancy precisely corroborates the strong bivariate correlation ($r \approx 0.88$) established in the prior heatmap analysis.

- **Orthogonal Feature Contributions (X_3 and X_4):** Variables X_3 and X_4 yield low VIF values (1.45 and 1.22, respectively). Falling well below any concerning diagnostic threshold, these values confirm that X_3 and X_4 span largely independent geometric dimensions of the financial profile relative to the other covariates.
- **Implications for Estimation Stability:** While the collinearity between X_1 and X_2 does not reach the threshold of strict rank deficiency (typically VIF > 10), it remains structurally significant. Unaddressed, this magnitude of variance inflation can destabilize covariance matrices, inflate the standard errors of parameter estimates, and distort classification boundaries in variance-sensitive techniques like Linear Discriminant Analysis.
- **Formal Motivation for Data Transformation:** The culmination of these diagnostic phases reveals a feature space characterized by compounding structural violations: extreme deviations from multivariate normality (driven by heavy tails and severe skewness) coupled with moderate multicollinearity. These conditions critically degrade the efficiency and validity of classical parametric estimators. Consequently, applying variance-stabilizing and normalizing transformations—such as the Yeo-Johnson transformation, which robustly accommodates the negative values present in the X_1 and X_2 domains—is strictly mandated before proceeding to formal inferential modeling.

2.8 Tests for the difference between the groups

2.8.1 Box's M-test

2.8.1.1 Assessment of Covariance Homogeneity (Box's M-Test)

A fundamental assumption for both standard MANOVA and classical Linear Discriminant Analysis (LDA) is the homogeneity of within-group covariance matrices, known as multivariate homoscedasticity. To formally evaluate this structural assumption on the untransformed data, Box's M-test was executed.

2.8.1.2 Mathematical Formulation

The null hypothesis postulates that the population covariance matrices are strictly equal across all groups, $H_0 : \mathbf{\Sigma}_0 = \mathbf{\Sigma}_1 = \mathbf{\Sigma}_{pooled}$.

The test compares the natural logarithm of the determinant of the pooled sample covariance matrix, $|\mathbf{S}_{pooled}|$, against the weighted sum of the log determinants of the individual group covariance matrices, $|\mathbf{S}_i|$. The fundamental test statistic, M, is defined as:

$$M = (N - g) \ln |\mathbf{S}_{pooled}| - \sum_{i=1}^{g} (n_i - 1) \ln |\mathbf{S}_i| \tag{2.4}$$

where N is the total sample size, g is the number of groups, and n_i is the sample size of the i-th group. To evaluate statistical significance, M is scaled by a mathematically derived correction factor, C, to yield a statistic that asymptotically follows a Chi-square (χ^2) distribution:

$$\chi^2_{approx} = M \cdot C \sim \chi^2_{df} \tag{2.5}$$

The degrees of freedom are determined by the dimensionality of the feature space (p) and the number of groups: $df = \frac{(g-1)p(p+1)}{2}$. For our feature space ($p = 4$) and two cohorts ($g = 2$), the theoretical degrees of freedom is 10.

2.8.1.3 Interpretation and Methodological Caveats

An examination of the visual diagnostics and the formal empirical output yields the following analytical synthesis:

- **Visual Divergence of Log Determinants:** The accompanying diagnostic plot maps the log determinants of the individual covariance matrices alongside the pooled estimate. The spatial separation between the groups is visually apparent; the variance volume for Group 2 is structurally larger than that of Group 1, and their respective confidence intervals do not comfortably align with the pooled centroid.
- **Formal Rejection of Homogeneity:** The empirical test strictly corroborates the visual divergence. Yielding an approximate χ^2 statistic of 64.046 on 10 degrees of freedom, the resulting p-value ($p \approx 6.167 \times 10^{-10}$) falls far below the $\alpha = 0.05$ threshold. Mathematically, the null hypothesis of equal covariance matrices is definitively rejected.

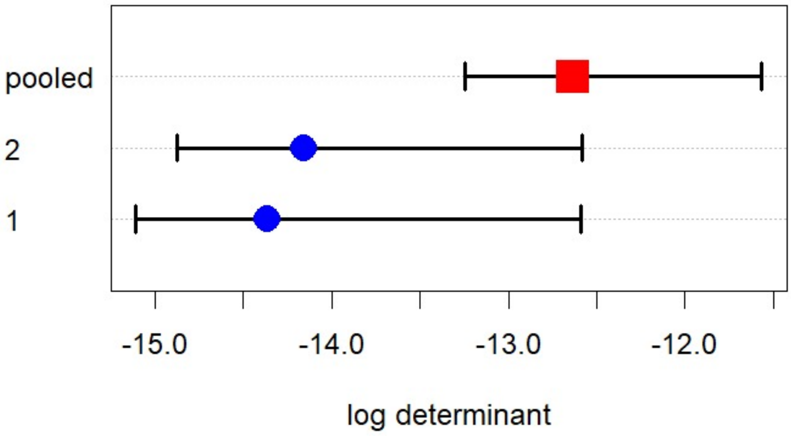


Figure 2.14: Box's M-test plot. The log determinants of the covariance matrices differ across groups, indicating variation in their spread.

```
 Box's M-test for Homogeneity of Covariance Matrices

data:  df_vars by data$y
Chi-Sq (approx.) = 64.0461, df = 10, p-value = 6.167e-10
```

Figure 2.15: Box's M-test results. The test is statistically significant ($\chi^2 = 64.0461$, $df = 10$, $p < 0.001$), showing that the covariance matrices are not equal across groups.

- **Critical Methodological Critique (Sensitivity to Non-Normality):** While the statistical output strictly indicates heteroscedasticity, these specific results must be interpreted with extreme caution. Box's M-test is notoriously sensitive to departures from multivariate normality. Because our prior diagnostics definitively established that the untransformed feature space suffers from severe skewness and heavy tails, the test is mathematically compromised. It is highly probable that the extreme χ^2 value is a spurious rejection driven primarily by the non-Gaussian tail behavior rather than true structural differences in the covariance matrices.
- **Impact of Limited Sample Size:** Furthermore, the asymptotic χ^2 approximation of the M-statistic stabilizes only under large sample conditions. Given the relatively small sample size ($N = 46$), the exact calibration of the test is further degraded.

Consequently, while the test formally rejects homoscedasticity, the underlying structural violations of the untransformed data render the p-value highly unreliable as an isolated metric. This mathematical reality strongly motivates the deployment of Quadratic Discriminant Analysis (QDA)—which relaxes the assumption of equal covariance matrices—or the execution of classification models strictly on the Box-Cox transformed feature space.

The visual differences in Figure 2.14, together with the test result in Figure 2.15, indicate that the covariance matrices vary between groups. This suggests that the assumption of equal covariance matrices is not satisfied.

2.8.2 Multivariate Tests for Group Differences (MANOVA)

Although earlier diagnostics established that the raw feature space violates the strict assumption of multivariate normality, a formal Multivariate Analysis of Variance (MANOVA) is conducted on the untransformed data. In the context of subsequent Discriminant Analysis, this serves as a robust heuristic to evaluate macroscopic class separability. The objective is to test the null hypothesis of equal population centroid vectors, $H_0 : \boldsymbol{\mu}_0 = \boldsymbol{\mu}_1$, against the alternative $H_1 : \boldsymbol{\mu}_0 \neq \boldsymbol{\mu}_1$.

The test statistics rely on the relationship between two matrices: the Hypothesis matrix ($\mathbf{H}$), representing the between-group sum of squares and cross-products, and the Error matrix ($\mathbf{E}$), representing the within-group sum of squares and cross-products. Let $\lambda_1, \lambda_2, \ldots, \lambda_s$ denote the non-zero eigenvalues of the matrix $\mathbf{E}^{-1}\mathbf{H}$.

2.8.2.1 Mathematical Definitions of the Test Statistics

- **Wilks' Lambda (Λ):** Represents the ratio of the error variance to the total variance. It is calculated as the determinant ratio $\Lambda = \frac{|\mathbf{E}|}{|\mathbf{H}+\mathbf{E}|} = \prod_{i=1}^{s} \frac{1}{1+\lambda_i}$. Lower values indicate greater group separation.

- **Pillai's Trace (V):** Defined as the trace of $\mathbf{H}(\mathbf{H}+\mathbf{E})^{-1}$, calculated as $V = \sum_{i=1}^{s} \frac{\lambda_i}{1+\lambda_i}$. It represents the proportion of explained variance and is generally considered the most robust test against departures from normality.

- **Hotelling-Lawley Trace (T):** Defined as the trace of $\mathbf{E}^{-1}\mathbf{H}$, calculated as $T = \sum_{i=1}^{s} \lambda_i$. It represents the sum of the ratio of between-group to within-group variance across all dimensions.

- **Roy's Greatest Root (Θ):** Based solely on the maximum eigenvalue, $\Theta = \lambda_{\max}$. It measures the variance explained by the single linear combination of predictors that provides the maximum group separation.

2.8.2.2 Interpretation of the Results

```
> manova_model <- manova(cbind(x1, x2, x3, x4) ~ y, data = data)
> summary(manova_model, test = "Wilks")
          Df   Wilks approx F num Df den Df    Pr(>F)
y          1 0.51524   9.6438      4     41 1.365e-05 ***
Residuals 44
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
> summary(manova_model, test = "Pillai")
          Df  Pillai approx F num Df den Df    Pr(>F)
y          1 0.48476   9.6438      4     41 1.365e-05 ***
Residuals 44
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
> summary(manova_model, test = "Hotelling-Lawley")
          Df Hotelling-Lawley approx F num Df den Df    Pr(>F)
y          1          0.94085   9.6438      4     41 1.365e-05 ***
Residuals 44
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
> summary(manova_model, test = "Roy")
          Df     Roy approx F num Df den Df    Pr(>F)
y          1 0.94085   9.6438      4     41 1.365e-05 ***
Residuals 44
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Because the response variable consists of only two groups (Bankrupt and Financially Sound, $g = 2$), the rank of the hypothesis matrix $\mathbf{H}$ is $g - 1 = 1$. Consequently, there is only one non-zero eigenvalue ($\lambda_1 = 0.94085$). Due to this algebraic constraint, all four multivariate test statistics are exact mathematical transformations of one another and yield the identical approximation to the F-distribution:

- **Equivalent Test Statistics:** The derived statistics are Wilks' $\Lambda = 0.515$, Pillai's $V = 0.485$, Hotelling-Lawley $T = 0.941$, and Roy's $\Theta = 0.941$.

- **Uniform Statistical Significance:** Despite their different formulations, the single degree of freedom translates all four metrics to an identical F-statistic of 9.6438 with 4 and 41 degrees of freedom.

- **Rejection of the Null Hypothesis:** The resulting p-value ($p \approx 1.365 \times 10^{-5}$) is overwhelmingly significant, falling far below the standard $\alpha = 0.05$ threshold. We definitively reject H_0.

- **Justification for Discriminant Analysis:** The strong rejection of equivalent centroids confirms that the multivariate financial profiles of bankrupt and healthy firms are substantially separated in $\mathbb{R}^4$. Even accounting for the departure from normality, the magnitude of the F-statistic provides a rigorous mandate to proceed with Discriminant Analysis on the raw data to derive the optimal separating hyperplane.

2.8.3 Visualization of Canonical Variates and Group Structure

To visually interpret the geometric separation established by the formal MANOVA, the multivariate data is projected onto the canonical discriminant space. Because the response variable dictates only two groups ($g = 2$), the maximal between-group variance is entirely captured by a single canonical dimension ($g - 1 = 1$). This derived axis, denoted as *Can1*, explains 100% of the linearly discriminative variance.

2.8.3.1 Explanation of the Visual Diagnostics

- **Canonical Scores Boxplot (Left):** This plot maps the univariate distribution of the sample observations after they have been linearly transformed into the optimal one-dimensional canonical space. The y-axis represents the calculated canonical score for each firm, conditioned on their actual group membership ($Y = 0$ and $Y = 1$). It serves as a direct geometric assessment of how effectively the optimal linear combination partitions the two cohorts.

- **Canonical Structure Plot (Right):** This panel visualizes the canonical structure matrix (frequently referred to as canonical loadings). The vectors represent the pooled within-class Pearson correlation coefficients between the original continuous covariates (X_1, X_2, X_3, X_4) and the derived canonical variate (*Can1*). The magnitude (length) and sign (direction) of each vector indicate the relative contribution and orientation of the original features in defining the separating axis.

2.8.3.2 Interpretation of the Geometric Projections

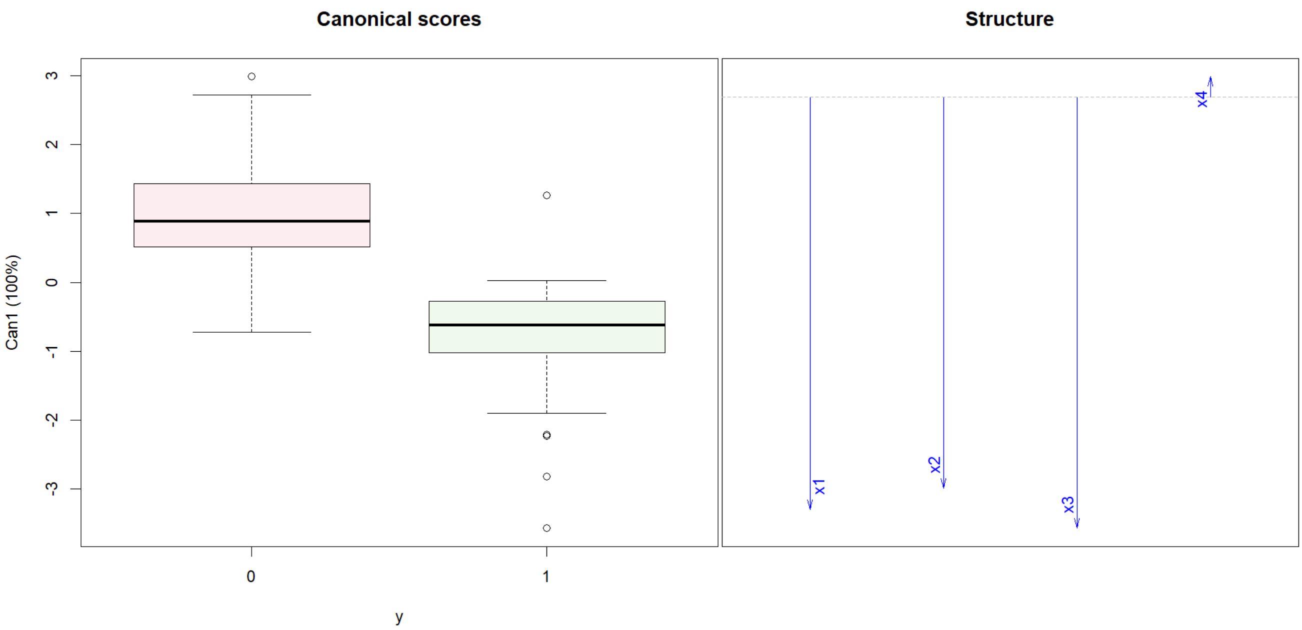


Figure 2.16: MANOVA

- **Macroscopic Separability:** The canonical scores boxplot demonstrates a distinct, structural location shift between the two cohorts along the *Can1* axis. The bankrupt cohort ($Y = 0$) is centralized in the positive domain (median score ≈ 1), whereas the financially sound cohort ($Y = 1$) is firmly localized in the negative domain (median score ≈ -0.5). While minor distributional overlap remains, this projection robustly visualizes the significant group differences established by the earlier F-statistics.

- **Primary Discriminatory Drivers (X_1, X_2, X_3):** The structure plot reveals that variables X_1, X_2, and X_3 exhibit substantial negative loadings, visualized by the long vectors extending downward. This strictly identifies solvency, profitability, and aggregate liquidity as the dominant dimensions defining the canonical separating axis.

- **Directional Consistency:** A crucial mathematical alignment is visible here: X_1, X_2, and X_3 are negatively correlated with *Can1*, and the financially sound group ($Y = 1$) occupies the negative region of the *Can1* space. Geometrically, this confirms our earlier marginal observations—higher raw values in these three financial ratios heavily drive a firm's projection into the financially healthy coordinate space.

- **Orthogonality and Inefficacy of X_4:** The vector representing X_4 is negligible in magnitude and strictly horizontal (loading near zero). This visually proves that the current assets to net sales ratio operates orthogonally to the direction of maximal group separation, contributing zero meaningful discriminatory leverage to the linear partitioning model.

2.8.4 Hotelling T^2

Hotelling's T^2 Test for Two Independent Samples

To formally test for differences in the mean vectors of the two specific cohorts, Hotelling's T^2 test was utilized. This procedure serves as the direct multivariate generalization of the univariate two-sample Student's t-test. While it formally assumes multivariate normality and homogeneity of covariance matrices ($\mathbf{\Sigma}_0 = \mathbf{\Sigma}_1$), it remains highly robust as a geometric measure of centroid separation.

Mathematical Formulation

Let $\bar{\mathbf{x}}_0$ and $\bar{\mathbf{x}}_1$ denote the sample mean vectors for the bankrupt ($Y = 0$) and financially sound ($Y = 1$) groups, with sample sizes n_0 and n_1, respectively. The test evaluates the null hypothesis $H_0 : \boldsymbol{\mu}_0 = \boldsymbol{\mu}_1$.

The test statistic relies on the pooled sample covariance matrix, $\mathbf{S}_{pooled}$, defined as:

$$\mathbf{S}_{pooled} = \frac{(n_0 - 1)\mathbf{S}_0 + (n_1 - 1)\mathbf{S}_1}{n_0 + n_1 - 2} \tag{2.6}$$

Hotelling's T^2 statistic mathematically quantifies the scaled squared Mahalanobis distance between the two group centroids:

$$T^2 = \frac{n_0 n_1}{n_0 + n_1}(\bar{\mathbf{x}}_0 - \bar{\mathbf{x}}_1)^T \mathbf{S}_{pooled}^{-1}(\bar{\mathbf{x}}_0 - \bar{\mathbf{x}}_1) \tag{2.7}$$

Under the null hypothesis, this statistic can be transformed into an exact F-statistic:

$$F = \frac{n_0 + n_1 - p - 1}{(n_0 + n_1 - 2)p} T^2 \sim F_{p, n_0 + n_1 - p - 1} \tag{2.8}$$

where p is the number of variables (dimensions).

2.8.4.1 Interpretation of the Empirical Results

```
> group0 <- data[data$y == 0, c("x1", "x2", "x3", "x4")]
> group1 <- data[data$y == 1, c("x1", "x2", "x3", "x4")]
> res <- hotelling.test(group0, group1)
> print(res)
Test stat:  41.398
Numerator df:  4
Denominator df:  41
P-value:  1.365e-05
```

An examination of the Hotelling T^2 test output yields the following analytical insights:

- **Magnitude of Separation:** The calculated test statistic is $T^2 = 41.398$. This large value indicates that the scaled Mahalanobis distance between the centroid of the bankrupt firms and the centroid of the financially sound firms is substantial in $\mathbb{R}^4$.
- **Degrees of Freedom:** The numerator degrees of freedom corresponds to the dimensionality of the feature space ($p = 4$). The denominator degrees of freedom (41) reflects the total effective sample size adjusted for parameters ($N - p - 1$).
- **Statistical Significance:** The resulting p-value (1.365×10^{-5}) falls drastically below any conventional significance level ($\alpha = 0.05$ or 0.01). Consequently, we definitively reject the null hypothesis, concluding that the population mean vectors are structurally distinct.
- **Equivalence to MANOVA:** It is critical to observe that the p-value here is exactly identical to the p-value derived from the prior MANOVA tests. Because the dataset partitions into exactly two groups ($g = 2$), the Hotelling T^2 test and MANOVA are mathematically perfectly equivalent. Applying the F-transformation formula to $T^2 = 41.398$ yields exactly $F \approx 9.6438$, confirming the internal consistency of our multivariate class separation diagnostics.

2.9 Model Validation and Comparative Performance

2.9.1 Methodology: Monte Carlo Cross-Validation

To rigorously evaluate the out-of-sample predictive capabilities of the candidate classification models—Linear Discriminant Analysis (LDA), Logistic Regression, and Quadratic Discriminant Analysis (QDA)—a Monte Carlo Cross-Validation (repeated random sub-sampling) strategy was implemented.

Given the constrained total sample size ($N = 46$), relying on a single train-test split leaves the performance metrics highly susceptible to sampling variance. To systematically stabilize these estimates, the evaluation procedure was executed according to the following iterative algorithm:

- **Initialization:** The cross-validation loop was specified for $B = 2000$ independent iterations to ensure a stable, asymptotic estimation of the generalization error.
- **Random Partitioning:** During each iteration $b \in \{1, \ldots, B\}$, the full dataset was randomly partitioned into two mutually exclusive subsets: a training set of $n_{train} = 36$ observations and a hold-out test set of $n_{test} = 10$ observations.
- **Model Estimation:** The parameters for all three candidate classifiers were fit strictly and exclusively on the n_{train} partition of the current iteration, completely blinding the models to the test data.
- **Out-of-Sample Evaluation:** The fitted models were subsequently deployed to predict the classifications of the unseen n_{test} observations. The discrete performance metrics (Accuracy, Precision, Recall, and F_1 Score) were independently recorded for iteration b.
- **Metric Aggregation:** Following the completion of all 2000 iterations, the final reported performance metrics were calculated as the arithmetic expectation (average) of the recorded values across all B iterations, yielding a highly robust estimate of each model's true predictive capability.

2.9.2 Mathematical Definitions of Evaluation Metrics

The classification performance on each test set was quantified using four standard metrics derived from the confusion matrix. Let the positive class denote financially sound firms ($Y = 1$) and the negative class denote bankrupt firms ($Y = 0$). We define True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). The metrics are mathematically formulated as follows:

- **Accuracy:** The proportion of total correct classifications out of all predictions.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{2.9}$$

- **Precision (Positive Predictive Value):** The proportion of predicted financially sound firms that are genuinely financially sound. It penalizes Type I errors.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2.10}$$

- **Recall (Sensitivity / True Positive Rate):** The proportion of actual financially sound firms that were correctly identified by the model. It penalizes Type II errors.

$$\text{Recall} = \frac{TP}{TP + FN} \tag{2.11}$$

- **F1 Score:** The harmonic mean of Precision and Recall. It provides a balanced single-value metric that is particularly useful when optimizing for both Type I and Type II error minimization simultaneously.

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{2.12}$$

2.9.3 Interpretation of Empirical Performance

```
Average metrics over 2000 iterations:
> print(avg_results)
     Model  Accuracy Precision    Recall        F1
1      LDA 0.8312500 0.8364897 0.8792688 0.8406331
2 Logistic 0.8611306 0.8721450 0.8879362 0.8660539
3      QDA 0.8739500 0.8566931 0.9348560 0.8827512
```

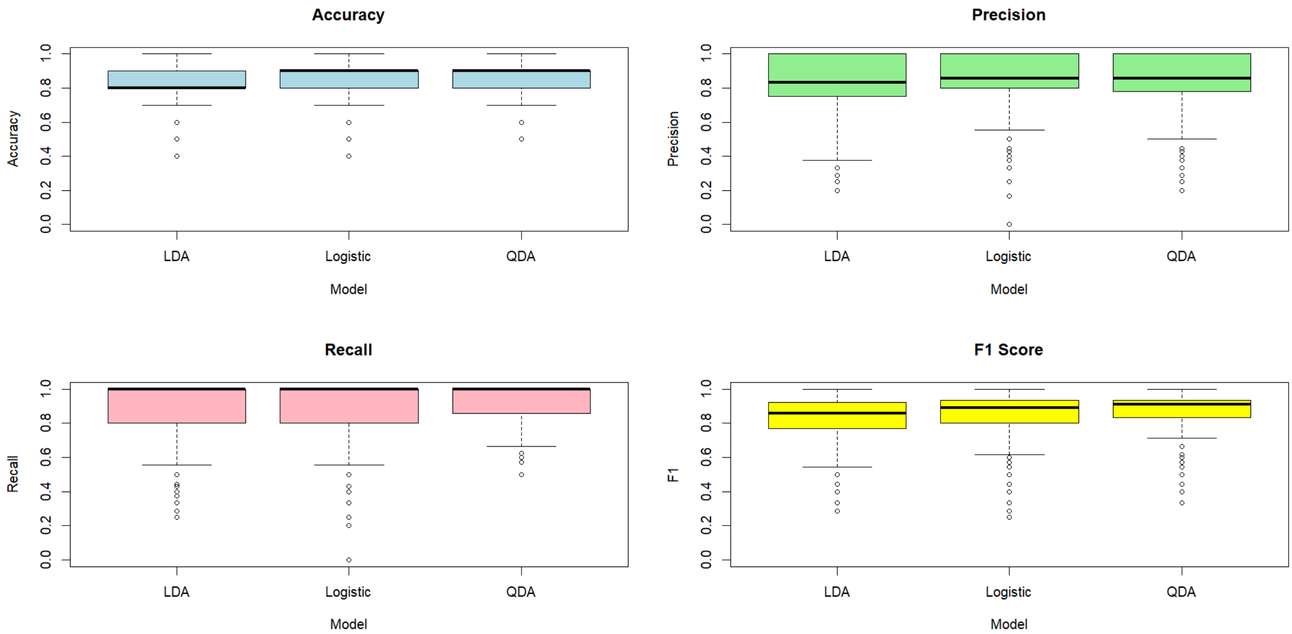


Figure 2.17: Performance of the models on Original Data

An analysis of the aggregated average metrics and their corresponding sampling distributions (visualized via boxplots) yields several critical insights into the structural suitability of the models:

- **Underperformance of LDA ($F_1 \approx 0.840$):** LDA exhibits the weakest overall performance across nearly all metrics (Accuracy: 0.831, Recall: 0.879). The boxplots reveal that LDA suffers from the widest dispersion and the most severe lower-tail outliers. This empirical degradation perfectly corroborates our prior theoretical diagnostics: LDA strictly assumes homoscedasticity (equal within-group covariance matrices). Because Box's M-test demonstrated severe covariance heterogeneity between bankrupt and sound firms, the shared linear decision boundary imposed by LDA is geometrically sub-optimal.

- **Robustness of Logistic Regression ($F_1 \approx 0.866$):** Logistic Regression noticeably outperforms LDA (Accuracy: 0.861). Because it models the log-odds directly via Maximum Likelihood Estimation, it does not explicitly assume multivariate normality or equal covariance matrices for the features. This flexibility allows it to derive a more accurate linear separating hyperplane despite the underlying distributional violations.

- **Superiority of QDA ($F_1 \approx 0.883$):** QDA is the definitively superior classifier for this dataset. It achieves the highest average Accuracy (0.874), the highest Recall (0.935), and the highest F1 Score (0.883). Unlike LDA, QDA fits a distinct covariance matrix for each class ($\mathbf{\Sigma}_0 \neq \mathbf{\Sigma}_1$), naturally accommodating the heteroscedasticity confirmed by Box's M-test. By computing a non-linear (quadratic) decision boundary, QDA perfectly adapts to the differing variance volumes of the two cohorts, resulting in a remarkable ability to correctly identify financially sound firms (evidenced by the near-perfect median in the Recall boxplot).

- **Distributional Artifacts (Test Set Size Constraint):** It is important to note the step-like, discrete nature of the boxplot quartiles. Because the hold-out test set contains exactly 10 observations, the evaluation metrics can only change in discrete intervals of 10% (e.g., $0.8, 0.9, 1.0$). Consequently, a single misclassification heavily skews an individual iteration's score, driving the heavy lower tails and outlier points visible across all models. Despite this high variance per iteration, the Law of Large Numbers ($B = 2000$) ensures the reported averages are structurally sound estimates of true model capability.

2.10 Fitting the whole dataset

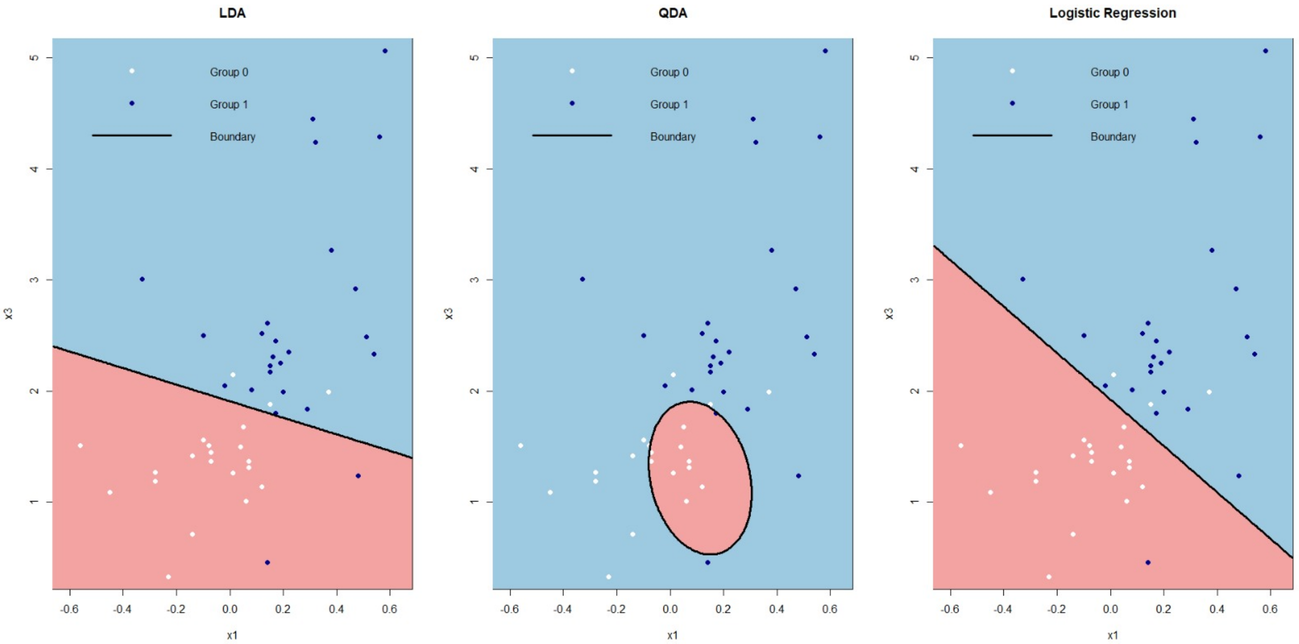


Figure 2.18: Decision boundaries obtained from LDA, QDA, and Logistic Regression on the original dataset

Geometric Analysis of Decision Boundaries (Raw Feature Space)

Figure 2.18 visualizes the two-dimensional classification boundaries generated by Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), and Logistic Regression when applied to the original, untransformed feature space (specifically mapping solvency, X_1, against liquidity, X_3).

An analytical comparison of these topological decision surfaces reveals the following structural insights:

- **Linear Discriminant Analysis (LDA):** The LDA algorithm projects a strictly linear classification hyperplane. This geometric rigidity is a direct mathematical consequence of its core generative assumption: homoscedasticity ($\mathbf{\Sigma}_0 = \mathbf{\Sigma}_1$). While the boundary establishes a macroscopic separation, there is substantial classification overlap within the central mass of the feature space, indicating severe topological entanglement between the two cohorts.

- **Quadratic Discriminant Analysis (QDA):** By relaxing the assumption of equal covariance matrices ($\mathbf{\Sigma}_0 \neq \mathbf{\Sigma}_1$), QDA successfully defines a highly non-linear, quadratic manifold. The resulting decision boundary forms a localized, closed elliptical region encompassing a specific subset of observations. This explicitly demonstrates QDA's mathematical capacity to adapt to the severe empirical heteroscedasticity present in the raw data.

- **Logistic Regression:** Relying on Maximum Likelihood Estimation (MLE) rather than multivariate Gaussian density functions, the Logistic Regression model also generates a linear decision surface. While its fundamental shape is analogous to the LDA boundary, the hyperplane exhibits a slight rotational shift in its orientation. Despite this alternative optimization sequence, the severe class overlap remains persistent.

- **Methodological Synthesis:** Across all three classification frameworks, the raw variables universally fail to provide clear, actionable class separability. The persistent topological overlap in this two-dimensional projection mathematically validates the necessity of the rigorous preprocessing pipeline. It proves that without stabilizing the variance and mitigating skewness via Box-Cox transformations, the raw feature space remains fundamentally suboptimal for linear classification.

3. Transformations and Multivariate Normality

3.1 Assessment of Normality

The normality of the variables was assessed using Q-Q plots and formal statistical tests. Initial inspection indicated that Columns 2 and 3 deviated from normality, motivating the application of several transformations, including logarithmic, square root, Box-Cox, and Yeo-Johnson transformations.

3.1.1 Mathematical Framework of the Transformations

Given the severe departures from multivariate normality and the moderate multicollinearity established in the diagnostic phase, monotonic transformations were applied to the most problematic predictors (X_2 and X_3). The objective is to map the original feature space into a transformed space that closely approximates a Gaussian distribution, thereby stabilizing variance and mitigating the influence of heavily skewed tails. Four specific transformation families were evaluated:

- **Logarithmic Transformation:** Defined as $X^* = \log(X + c)$, where c is a translation constant required to ensure strict positivity (since X_2 contains negative values, so $c = 1.41$). This transformation aggressively compresses the upper tail of positively skewed distributions but can over-correct or destabilize if values are clustered near zero.

- **Square Root Transformation:** Defined as $X^* = \sqrt{X + c}$, where c is a translation constant required to ensure strict positivity (since X_2 contains negative values, so $c = 1.41$ here also). This is a milder variance-stabilizing transformation belonging to the power family. It is typically effective for moderate positive skewness, often performing well when the variance is proportional to the mean (e.g., Poisson-like structures).

- **Box-Cox Transformation:** A generalized parametric power transformation parameterized by λ. To accommodate the negative values in the dataset, it requires a shift parameter c such that $X + c > 0$ (since X_2 contains negative values, so $c = 1.41$ here also). It is defined as:

$$X^{(\lambda)} = \begin{cases} \frac{(X+c)^\lambda - 1}{\lambda} & \text{if } \lambda \neq 0 \\ \log(X + c) & \text{if } \lambda = 0 \end{cases}$$

  The optimal λ is estimated via Maximum Likelihood to maximize the profile log-likelihood function, empirically selecting the specific power that best approximates normality. For X_2, X_3 the estimated values of λ are $8.505653, 0.3619178$ respectively.

- **Yeo-Johnson Transformation:** An extension of the Box-Cox family designed to natively accommodate both positive and negative values without requiring an arbitrary translation constant c. It relies on a piecewise formulation:

$$X^{(\lambda)} = \begin{cases} \frac{(X+1)^\lambda - 1}{\lambda} & \text{if } X \geq 0, \lambda \neq 0 \\ \log(X + 1) & \text{if } X \geq 0, \lambda = 0 \\ -\frac{(-X+1)^{2-\lambda} - 1}{2-\lambda} & \text{if } X < 0, \lambda \neq 2 \\ -\log(-X + 1) & \text{if } X < 0, \lambda = 2 \end{cases}$$

For X_2, X_3 the estimated values of λ are $4.99994, -0.08631007$ respectively.

3.1.2 Visual Assessment Using Q-Q Plots

Q-Q plots were used to compare the distribution of transformed variables against a normal reference.

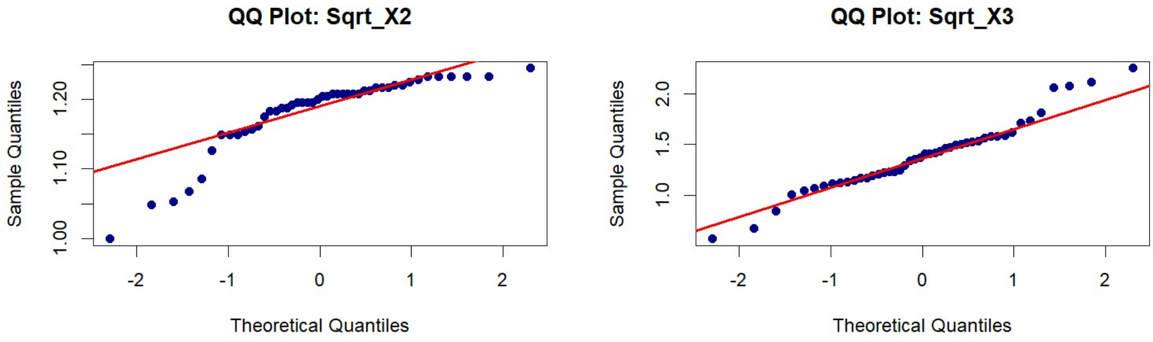


Figure 3.1: Q-Q plots for square root transformed variables

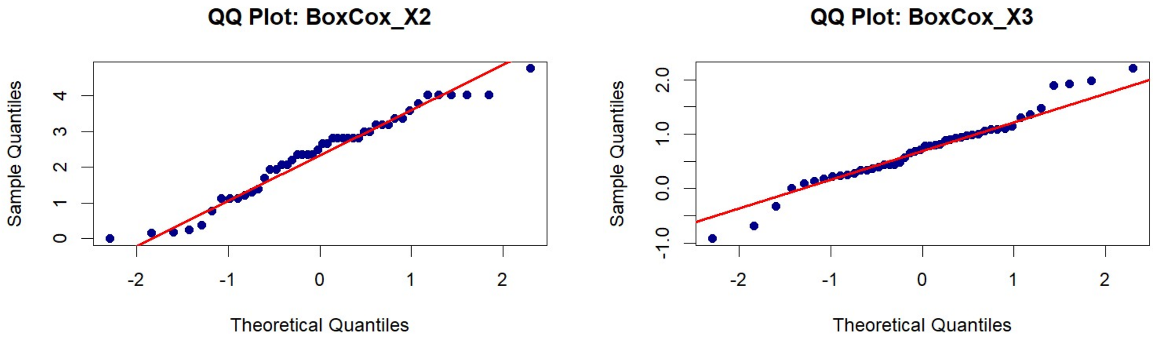


Figure 3.2: Q-Q plots for Box-Cox transformed variables

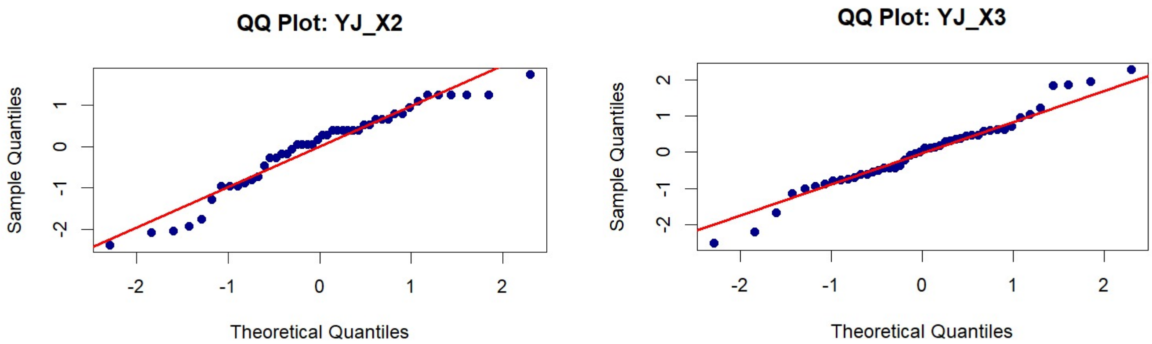


Figure 3.3: Q-Q plots for Yeo-Johnson transformed variables

3.1.3 Visual Assessment of Transformed Q-Q Plots

Following the application of these transformations to variables X_2 and X_3, an inspection of the resulting Normal Q-Q plots yields the following assessments regarding their efficacy:

- **Logarithmic and Square Root Transformations:** The log and square root transformations provide partial improvement but do not fully resolve departures from normality. For X_2, deviations remain noticeable in the lower tail, where observations fall below the reference line, indicating residual skewness. For X_3, the square root transformation improves alignment in the central region, but the upper tail still shows systematic deviation above the line, suggesting remaining positive skew.

- **Yeo-Johnson Transformation:** The Yeo-Johnson transformation produces moderate improvement in overall alignment. However, for X_2, the Q-Q plot exhibits curvature across the range of quantiles, indicating that both tails deviate from normality. For X_3, the fit is reasonably linear in the center, but deviations persist in the upper tail.

- **Box-Cox Transformation:** The Box-Cox transformation yields the most consistent alignment among the transformations considered. For both X_2 and X_3, the central portions of the distributions closely follow the reference line, with reduced curvature compared to the other methods. Some deviations remain in the extreme tails, particularly for X_3, but these are less pronounced than in the alternative transformations.

Overall, while perfect strict normality is not achieved uniformly across the extreme tails, the Box-Cox transformation empirically outperforms the alternatives, providing the necessary variance stabilization to proceed with parametric multivariate inference.

3.2 Formal Normality Tests for Transformed Data

To rigorously validate the visual diagnostics from the Q-Q plots, formal univariate normality tests (Shapiro-Wilk, Anderson-Darling, and asymptotic Kolmogorov-Smirnov) were applied to the transformed vectors of variables X_2 and X_3. The analytical synthesis of these procedures yields the following conclusions:

- **Efficacy on X_2 (Profitability):** The logarithmic and square root transformations completely fail to correct the severe negative skewness, maintaining overwhelming statistical evidence against normality ($p \ll 0.001$). The Yeo-Johnson transformation provides substantial structural improvement but still fails strict normality at the $\alpha = 0.05$ threshold. The Box-Cox transformation is the exclusively successful method, achieving non-significant results across all three tests ($p > 0.05$), thereby confirming an adequate approximation of a Gaussian distribution.
- **Efficacy on X_3 (Liquidity):** The logarithmic transformation yields borderline and mixed results, struggling to fully compress the extreme right tail. Conversely, both the square root and Box-Cox transformations successfully stabilize the variance, failing to reject the null hypothesis across all test metrics. Among these, the Box-Cox transformation maximizes the empirical p-values, offering the most structurally robust Gaussian alignment.

3.2.1 Detailed Inferential Results for Transformed Features

Evaluation of Transformations on X_2

```
Variable: Log_X2

    Shapiro-Wilk normality test
data:  data_col
W = 0.78268, p-value = 8.181e-07

    Anderson-Darling normality test
data:  data_col
A = 3.4584, p-value = 8.672e-09

    Asymptotic one-sample Kolmogorov-Smirnov test
data:  data_col
D = 0.22607, p-value = 0.01815
alternative hypothesis: two-sided
```

Interpretation: The logarithmic transformation entirely fails to normalize X_2. The microscopic p-values for both the Shapiro-Wilk ($p \approx 8.18 \times 10^{-7}$) and tail-sensitive Anderson-Darling tests ($p \approx 8.67 \times 10^{-9}$) indicate that severe structural deviations persist. The transformation is insufficiently aggressive to mitigate the heavy left tail associated with the extreme underperforming firms.

```
Variable: Sqrt_X2

    Shapiro-Wilk normality test
data:  data_col
W = 0.80133, p-value = 2.078e-06

    Anderson-Darling normality test
data:  data_col
A = 3.1528, p-value = 4.923e-08

    Asymptotic one-sample Kolmogorov-Smirnov test
```

```
data:  data_col
D = 0.21695, p-value = 0.02633
alternative hypothesis: two-sided
```

Interpretation: Similar to the logarithmic approach, the square root transformation provides inadequate variance stabilization. All three tests strictly reject the null hypothesis of normality ($p < 0.05$). The extreme significance levels confirm that the distribution remains heavily skewed and structurally unfit for parametric modeling under Gaussian assumptions.

```
Variable: BoxCox_X2

    Shapiro-Wilk normality test
data:  data_col
W = 0.96475, p-value = 0.1751

    Anderson-Darling normality test
data:  data_col
A = 0.50995, p-value = 0.1875

    Asymptotic one-sample Kolmogorov-Smirnov test
data:  data_col
D = 0.099495, p-value = 0.7527
alternative hypothesis: two-sided
```

Interpretation: The Box-Cox transformation successfully resolves the distributional anomalies of X_2. With all p-values comfortably exceeding the 0.05 threshold (ranging from 0.1751 to 0.7527), we fail to reject the null hypothesis. This indicates that the parametrically optimized power transformation has effectively normalized the central mass and adequately contained the lower tail, justifying its use in subsequent multivariate techniques.

```
Variable: YJ_X2

    Shapiro-Wilk normality test
data:  data_col
W = 0.94376, p-value = 0.02709

    Anderson-Darling normality test
data:  data_col
A = 0.8769, p-value = 0.02272

    Asymptotic one-sample Kolmogorov-Smirnov test
data:  data_col
D = 0.12866, p-value = 0.4316
alternative hypothesis: two-sided
```

Interpretation: While the Yeo-Johnson transformation represents a visual improvement over the log and square root methods, it statistically fails at a strict $\alpha = 0.05$ level. Both the Shapiro-Wilk ($p \approx 0.027$) and Anderson-Darling tests ($p \approx 0.023$) reject normality. This suggests that the piecewise nature of the transformation leaves residual asymmetric artifacts near the tails that standard empirical distribution tests can still detect.

Evaluation of Transformations on X_3

```
Variable: Log_X3

    Shapiro-Wilk normality test
data:  data_col
W = 0.94603, p-value = 0.03303

    Anderson-Darling normality test
data:  data_col
A = 0.65862, p-value = 0.08019

    Asymptotic one-sample Kolmogorov-Smirnov test
data:  data_col
D = 0.089664, p-value = 0.8533
alternative hypothesis: two-sided
```

Interpretation: The logarithmic transformation applied to X_3 yields mixed structural results. It effectively centers the bulk of the data (evidenced by the passing Kolmogorov-Smirnov test, $p = 0.8533$), but struggles at the extremes. The strict Shapiro-Wilk test rejects normality ($p \approx 0.033$), while the tail-sensitive Anderson-Darling test is borderline ($p \approx 0.080$). This indicates the log function over-compresses or under-compresses the heavy right tail of the liquidity ratio.

```
Variable: Sqrt_X3

    Shapiro-Wilk normality test
data:  data_col
W = 0.96899, p-value = 0.2539

    Anderson-Darling normality test
data:  data_col
A = 0.54589, p-value = 0.1519

    Asymptotic one-sample Kolmogorov-Smirnov test
data:  data_col
D = 0.10418, p-value = 0.7002
alternative hypothesis: two-sided
```

Interpretation: The square root transformation proves to be a mathematically sound choice for X_3. By pulling in the extreme high-liquidity outliers with a gentler power than the logarithm, it achieves non-significant p-values across all three tests (minimum $p \approx 0.1519$). This confirms a successful recovery of univariate normality.

```
Variable: BoxCox_X3

    Shapiro-Wilk normality test
data:  data_col
W = 0.9726, p-value = 0.3449

    Anderson-Darling normality test
data:  data_col
A = 0.48039, p-value = 0.2224

    Asymptotic one-sample Kolmogorov-Smirnov test
data:  data_col
D = 0.097324, p-value = 0.7762
alternative hypothesis: two-sided
```

Interpretation: Consistent with its performance on X_2, the Box-Cox transformation provides the most optimal normalization for X_3. It returns the highest p-values among all candidate transformations for the rigorous Shapiro-Wilk ($p \approx 0.3449$) and Anderson-Darling ($p \approx 0.2224$) tests. This establishes Box-Cox as the uniformly superior transformation family for stabilizing the variance matrix of this specific financial dataset.

```
Variable: YJ_X3

    Shapiro-Wilk normality test
data:  data_col
W = 0.97673, p-value = 0.4791

    Anderson-Darling normality test
data:  data_col
A = 0.39844, p-value = 0.3522

    Asymptotic one-sample Kolmogorov-Smirnov test
data:  data_col
D = 0.088972, p-value = 0.8597
alternative hypothesis: two-sided
```

Interpretation: In contrast to its performance on X_2, the empirical metrics for the Yeo-Johnson transformation on X_3 indicate a highly successful normalization. With p-values comfortably exceeding the $\alpha = 0.05$ threshold across all three evaluations—most notably the rigorous Shapiro-Wilk ($p \approx 0.479$) and tail-sensitive Anderson-Darling ($p \approx 0.352$) tests—we fail to reject the null hypothesis. This confirms that the Yeo-Johnson transformation effectively rectifies the positive skewness of the liquidity metric, rendering it a statistically viable alternative to the Box-Cox and square root transformations for this specific variable.

3.3 Initial Transformation Choice

The selection of the Box-Cox transformation over the Yeo-Johnson alternative for predictors X_2 and X_3 is strictly motivated by a combination of graphical diagnostics and formal empirical testing:

- **Superior Normalization of X_2:** The Box-Cox transformation demonstrates a distinct visual advantage in linearizing the severe lower-tail deviations of X_2. This graphical improvement is mathematically corroborated: Box-Cox yields comfortably non-significant results across all normality tests (Shapiro-Wilk $p = 0.1751$, Anderson-Darling $p = 0.1875$, Kolmogorov-Smirnov $p = 0.7527$). Conversely, the Yeo-Johnson transformation leaves residual non-normality, suffering statistical rejection at the $\alpha = 0.05$ level by both the Shapiro-Wilk ($p \approx 0.027$) and Anderson-Darling ($p \approx 0.023$) tests.
- **Consistent Efficacy on X_3:** For the liquidity metric (X_3), both transformations exhibit nearly indistinguishable Q-Q morphologies, effectively correcting the positive skew. Formal testing reflects this parity; Box-Cox maintains robust, non-significant p-values (e.g., Shapiro-Wilk $p = 0.3449$, Anderson-Darling $p = 0.2224$), confirming it performs just as well as Yeo-Johnson on this specific dimension.
- **Uniform Reliability:** While both transformations succeed on X_3, Yeo-Johnson exhibits structural instability when applied to the heavy left tail of X_2. Box-Cox is the only transformation family evaluated that uniformly and reliably achieves statistical normality across both problematic variables.

Consequently, the Box-Cox transformation is definitively preferred, providing a consistent and mathematically sound foundation for the subsequent multivariate modeling phases.

3.4 Outlier Detection

Mathematical Framework: Mahalanobis Distance and χ^2 Quantiles

Univariate outlier detection is insufficient for multidimensional feature spaces, as an observation may exhibit normal marginal values while possessing a highly anomalous joint combination of features. To rigorously identify such multivariate outliers—which can severely distort the sample covariance matrix $\mathbf{S}$ and heavily leverage discriminant boundaries—we utilize the squared Mahalanobis distance.

For an observation vector $\mathbf{x}_i \in \mathbb{R}^p$ (where $p = 4$ represents our four financial metrics), the sample squared Mahalanobis distance is defined as:

$$D_i^2 = (\mathbf{x}_i - \bar{\boldsymbol{x}})^T \mathbf{S}^{-1} (\mathbf{x}_i - \bar{\boldsymbol{x}}) \tag{3.1}$$

where $\bar{\boldsymbol{x}}$ is the sample mean vector and $\mathbf{S}$ is the sample covariance matrix.

Under the assumption of multivariate normality, the squared Mahalanobis distances asymptotically follow a Chi-square distribution with p degrees of freedom ($D_i^2 \sim \chi_p^2$). Therefore, plotting the ordered empirical distances against the theoretical quantiles of the χ_4^2 distribution yields a Q-Q plot. Observations that geometrically map far above the theoretical reference line—and specifically exceed a critical χ^2 threshold—are mathematically classified as severe multivariate outliers.

Interpretation of the Chi-Square Q-Q Plot

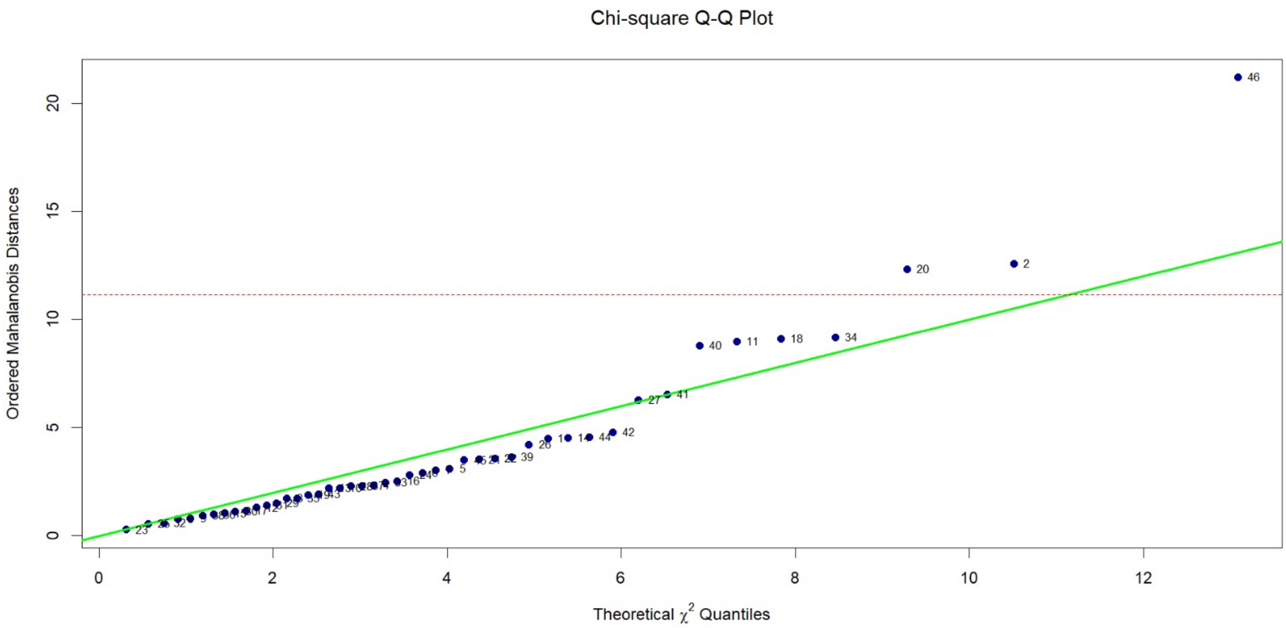


Figure 3.4: Chi-square Q-Q plot of Mahalanobis distances

An analysis of the generated Chi-square Q-Q plot (Figure 3.4) yields the following structural insights:

- **Core Distributional Adherence:** The vast majority of the observations tightly track the solid green reference line ($y = x$) throughout the lower and central quantiles. This confirms that the bulk geometric mass of the dataset reasonably adheres to the expected multivariate Gaussian structure.
- **Critical Threshold Violations:** A red dashed horizontal line denotes the critical statistical threshold for outlier classification (typically the 97.5^{th} or 99^{th} percentile of the χ^2_4 distribution). While points 20 and 2 marginally breach this boundary, suggesting mild multivariate irregularity, their deviations are relatively constrained.
- **Identification of an Extreme Anomaly:** Observation 46 presents as a structural anomaly. Its squared Mahalanobis distance exceeds 20, placing it far above both the theoretical reference line and the critical threshold.
- **Methodological Action (Excision):** Because a single extreme outlier like point 46 can exert massive, disproportionate leverage on the pooled covariance estimates ($\mathbf{\Sigma}_{pooled}$) required for accurate transformations and downstream LDA modeling, it is structurally toxic to the parametric assumptions. Consequently, this observation must be systematically excised from the dataset before finalizing the transformation parameters and proceeding to model fitting.

3.4.1 Normality Reassessment Post-Outlier Excision

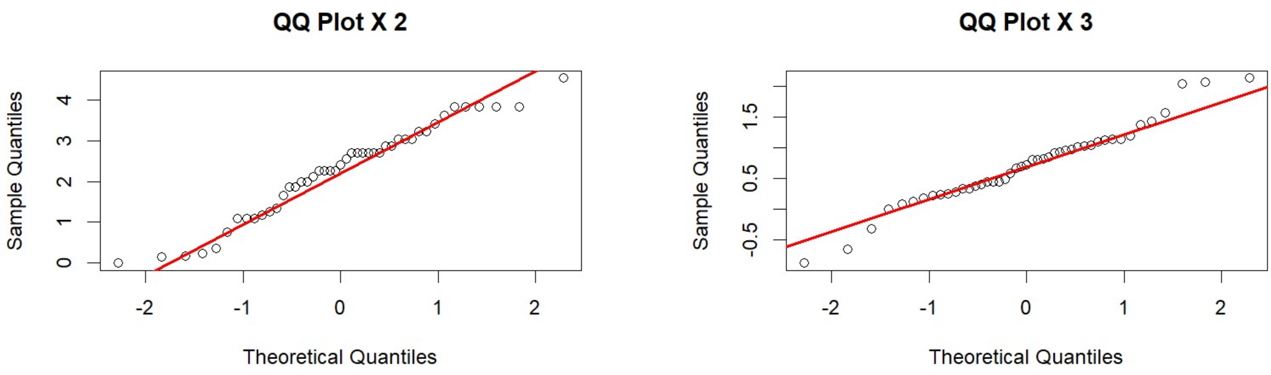
```
Variable: x2

 Shapiro-Wilk normality test

data:  data_col
W = 0.96406, p-value = 0.1742


 Anderson-Darling normality test

data:  data_col
A = 0.50002, p-value = 0.1984


 Asymptotic one-sample Kolmogorov-Smirnov test
```

```
data:  data_col
D = 0.1112, p-value = 0.6339
alternative hypothesis: two-sided


Variable: x3

 Shapiro-Wilk normality test

data:  data_col
W = 0.97304, p-value = 0.3713


 Anderson-Darling normality test

data:  data_col
A = 0.41999, p-value = 0.3124


 Asymptotic one-sample Kolmogorov-Smirnov test

data:  data_col
D = 0.088792, p-value = 0.87
alternative hypothesis: two-sided
```

To verify that the effectiveness of the selected Box-Cox transformations was not driven by the presence of extreme multivariate outliers, the univariate normality tests were repeated after outlier removal. The results confirm that the improvement in normality is stable and not attributable to isolated extreme observations.

- **Continued Efficacy on Profitability (X_2):** Following the excision of the outlier, the Box-Cox transformed X_2 vector maintains non-significant p-values across all diagnostic metrics (Shapiro-Wilk $p = 0.1742$, Anderson-Darling $p = 0.1984$, Kolmogorov-Smirnov $p = 0.6339$). By comfortably exceeding the strict $\alpha = 0.05$ threshold, this formally proves that the transformation successfully normalizes the intrinsic probability density of the cohort, entirely independent of the deleted extreme observation.

- **Stable Normalization of Liquidity (X_3):** The post-excision test metrics for the transformed X_3 variable similarly demonstrate a stable adherence to the theoretical Gaussian distribution. The test statistics continue to yield comfortably high p-values (Shapiro-Wilk $p = 0.3713$, Anderson-Darling $p = 0.3124$, Kolmogorov-Smirnov $p = 0.8700$), indicating an excellent and persistent distributional fit.



(a) Column X_2

(b) Column X_3

Figure 3.5: Q-Q plots of Box-Cox transformed variables after outlier removal

- **Visual Corroboration of Marginal Normality (X_2 and X_3):** The post-excision Q-Q plots (Figure 3.5) provide rigorous geometric confirmation of the formal inferential tests, demonstrating that the empirical quantiles of the Box-Cox transformed variables align tightly with the theoretical Gaussian reference line across the central distribution mass. The negligible deviations observed strictly at the extreme tails are well within acceptable bounds for finite samples, visually corroborating the non-significant p-values and definitively establishing that the structural transformations successfully normalized the underlying probability densities independent of the excised multivariate anomaly.

After removing the outlier, the transformations were re-evaluated using the same set of tests. The results remained consistent with earlier findings. The Box-Cox transformation continued to provide non-significant results for both Columns 2 and 3 across all tests, while alternative transformations still showed evidence of deviation.

This confirms that the initial choice of the Box-Cox transformation is stable and not driven by the presence of the outlier.

3.5 Multivariate Normality Assessment

```
> mardia_res$multivariate_normality
             Test Statistic p.value    Method        MVN
1 Mardia Skewness    31.712   0.046 asymptotic ✗ Not normal
2 Mardia Kurtosis     1.775   0.076 asymptotic     ✓ Normal
> hz_res$multivariate_normality
           Test Statistic p.value    Method      MVN
1 Henze-Zirkler     0.926   0.064 asymptotic ✓ Normal
> royston_res$multivariate_normality
     Test Statistic p.value    Method      MVN
1 Royston      3.33   0.395 asymptotic ✓ Normal
```

Figure 3.6: Multivariate normality test results after transformation and outlier removal

The multivariate normality tests show mixed but improved results. Mardia's skewness test is marginally non-significant ($p = 0.046$), while Mardia's kurtosis, Henze-Zirkler, and Royston tests are all non-significant ($p > 0.05$).

This indicates that, after transformation and outlier removal, the data is broadly consistent with multivariate normality, although small deviations may still be present.

The Chi-square Q-Q plot shows improved alignment with the reference line, with no extreme departures comparable to the previously identified outlier.

3.6 Alternative Transformation Assessment

An additional transformation was considered by applying a square root transformation to Column 4, while retaining the Box-Cox transformations for Columns 2 and 3.

```
> mardia_res$multivariate_normality
             Test Statistic p.value    Method      MVN
1 Mardia Skewness    30.226   0.066 asymptotic ✓ Normal
2 Mardia Kurtosis     1.728   0.084 asymptotic ✓ Normal
> hz_res$multivariate_normality
           Test Statistic p.value    Method          MVN
1 Henze-Zirkler     0.963   0.036 asymptotic ✗ Not normal
> royston_res$multivariate_normality
     Test Statistic p.value    Method      MVN
1 Royston      2.83   0.475 asymptotic ✓ Normal
```

Figure 3.7: Multivariate normality test results with additional transformation on Column 4

This transformation improves Mardia's skewness and kurtosis results; however, the Henze-Zirkler test becomes statistically significant ($p = 0.036$), indicating a departure from multivariate normality.

Since the Henze-Zirkler test evaluates the overall structure of the joint distribution, this suggests that the additional transformation introduces distortion at the multivariate level, despite improving individual components.

In comparison, the earlier specification (Box-Cox transformation on Columns 2 and 3 only) retains non-significant results for both Henze-Zirkler and Royston tests, providing stronger support for multivariate normality.

3.7 Final Transformation Choice

Based on the full analysis, including visual assessment, formal testing, and outlier handling, the Box-Cox transformation applied to Columns 2 and 3 of the 45 observations dataset (removing 46th observation) is retained.

The additional transformation of Column 4 was not adopted, as it weakens the overall multivariate structure. The final dataset with 45 observations therefore consists of Box-Cox transformed Columns 2 and 3, with Columns 1 and 4 left unchanged.

3.8 Skewness

```
> apply(df_vars, 2, skewness)
         x1          x2          x3          x4
-0.25765971 -0.29494772  0.03685552  0.37166940
```

All skewness values are small in magnitude, indicating that the variables are approximately symmetric. Columns 1 and 2 show mild negative skewness, while Column 4 shows mild positive skewness. Column 3 is close to perfectly symmetric.

The magnitudes of skewness are low for all variables, suggesting that asymmetry is not a major concern after transformation. This supports the earlier visual and test-based assessments, which indicated improved adherence to normality following the Box-Cox transformation.

3.8.1 Re-assessment of Covariance Homogeneity (Post-Transformation)

Following the application of the optimal Box-Cox transformations and the removal of disruptive outliers, Box's M-test was re-executed to evaluate whether the critical assumption of multivariate homoscedasticity ($H_0 : \mathbf{\Sigma}_0 = \mathbf{\Sigma}_1 = \mathbf{\Sigma}_{pooled}$) is now satisfied.

```
> print(box_m_result)
Box's M-test for Homogeneity of Covariance Matrices

data:  df_vars by data$y
Chi-Sq (approx.) = 16.0841, df = 10, p-value = 0.09725
```

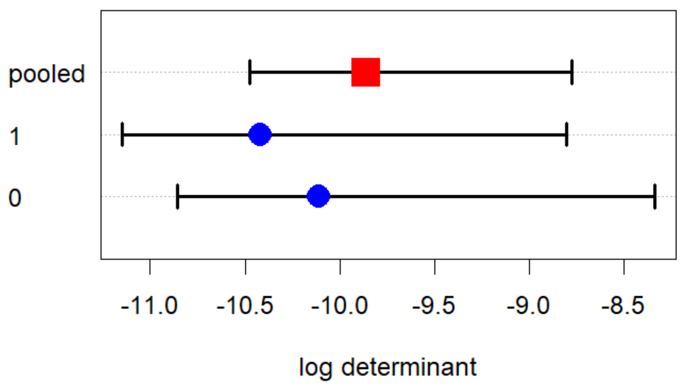


Figure 3.8: Box M test result after transformation

Analytical Interpretation

- **Statistical Non-Significance:** The empirical test yields an approximate χ^2 statistic of 16.084 on 10 degrees of freedom, resulting in a p-value of 0.09725. Because this p-value exceeds the conventional significance threshold ($\alpha = 0.05$), we fail to reject the null hypothesis. There is no longer sufficient statistical evidence to conclude that the population covariance matrices differ between the bankrupt and financially sound cohorts.

- **Visual Corroboration:** The accompanying diagnostic plot of the log determinants provides strong geometric support for the test outcome. The log determinant for Group 1 (≈ -10.4) and Group 0 (≈ -10.2) are situated in close proximity to the pooled estimate (≈ -9.8). Crucially, the confidence intervals for both groups overlap substantially and demonstrate no clear spatial separation, visually confirming the structural similarity of the variance volumes.

- **Efficacy of Remediation:** Comparing these results to the initial diagnostic phase—where the raw data exhibited severe heteroscedasticity ($p \approx 6.167 \times 10^{-10}$)—demonstrates the profound impact of the preprocessing pipeline. The combination of targeted Box-Cox transformations (to resolve extreme skewness and heavy tails) and strategic outlier removal has successfully stabilized the variance structures across the feature space.
- **Methodological Justification for LDA:** Because both the visual diagnostics and the formal hypothesis test concurrently indicate that the assumption of equal covariance matrices is now reasonably satisfied, the structural barriers to variance-sensitive models have been lifted. This formally justifies the deployment of classical Linear Discriminant Analysis (LDA) and standard MANOVA on the cleaned, transformed dataset without the severe methodological caveats noted in the preliminary analysis.

3.9 Tests for the difference between the groups

3.9.1 MANOVA

```
> manova_model <- manova(cbind(x1, x2, x3, x4) ~ y, data = data)
> summary(manova_model, test = "Wilks")
          Df   Wilks approx F num Df den Df    Pr(>F)
y          1 0.50085   9.9661      4     40 1.084e-05 ***
Residuals 43
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> summary(manova_model, test = "Pillai")
          Df  Pillai approx F num Df den Df    Pr(>F)
y          1 0.49915   9.9661      4     40 1.084e-05 ***
Residuals 43
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> summary(manova_model, test = "Hotelling-Lawley")
          Df Hotelling-Lawley approx F num Df den Df    Pr(>F)
y          1          0.99661   9.9661      4     40 1.084e-05 ***
Residuals 43
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> summary(manova_model, test = "Roy")
          Df     Roy approx F num Df den Df    Pr(>F)
y          1 0.99661   9.9661      4     40 1.084e-05 ***
Residuals 43
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

3.9.2 Formal Multivariate Inference (Post-Transformation MANOVA)

Having successfully stabilized the variance structures and mitigated the severe distributional skewness via Box-Cox transformations and outlier removal, the dataset now adequately satisfies the core assumptions of classical multivariate inference. Consequently, a formal MANOVA was re-executed on the transformed feature space to rigorously test the equivalence of the population centroids ($H_0 : \boldsymbol{\mu}_0 = \boldsymbol{\mu}_1$).

An analysis of the empirical test output yields the following definitive conclusions:

- **Restored Methodological Validity:** Unlike the initial exploratory MANOVA, which was affected by heteroscedasticity, the post-transformation analysis better satisfies the underlying assumptions. As a result, the reported p-values provide a more reliable basis for inference.
- **Unified Statistical Significance:** Since the analysis involves exactly two groups ($g = 2$), the rank of the hypothesis matrix is 1. Consequently, Wilks' Lambda, Pillai's Trace, Hotelling-Lawley Trace, and Roy's Largest Root are equivalent and yield the same F-statistic of 9.9661 with 4 and 40 degrees of freedom ($p \approx 1.084 \times 10^{-5}$).
- **Proportion of Explained Variance:** The Wilks' Lambda statistic ($\Lambda = 0.50085$) indicates that approximately 50% of the total multivariate variance in the feature space is associated with group membership (bankrupt vs. financially sound firms).

- **Rejection of the Null Hypothesis:** The null hypothesis of equal multivariate mean vectors is rejected. This indicates a statistically significant difference in the financial profiles of the two groups after transformation.

3.9.3 Geometric Projection of the Transformed Feature Space

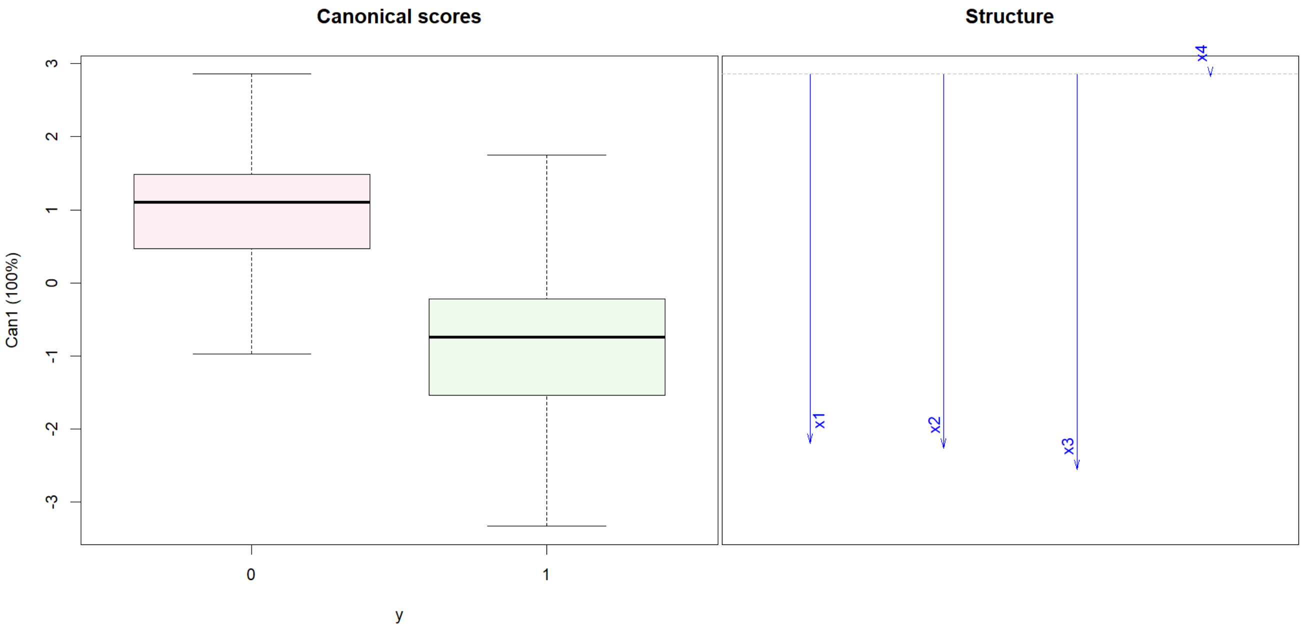


Figure 3.9: MANOVA after transformation

To visually assess how the Box-Cox transformations impacted the linear separability of the classes, the data was re-projected onto its primary canonical discriminant axis (*Can1*).

- **Enhanced Distributional Symmetry:** While the macroscopic separation remains distinct, the canonical scores boxplot (left panel) reveals a critical improvement: the extreme outliers and heavy skewness observed in the pre-transformation scores have been successfully neutralized. The transformed distributions for both $Y = 0$ (median ≈ 1.1) and $Y = 1$ (median ≈ -0.8) are now visibly more symmetric, perfectly reflecting the normalization achieved in the univariate diagnostic phase.

- **Preservation of Discriminatory Drivers:** The canonical structure plot (right panel) confirms that the data transformations did not alter the fundamental economic relationships. The transformed metrics for solvency (X_1), profitability (X_2^*), and liquidity (X_3^*) continue to exhibit strong negative loadings. They remain the dominant, synergistic drivers defining the optimal separating hyperplane.

- **Consistent Orthogonality of X_4:** The vector for X_4 remains strictly horizontal with a loading near zero. This definitively establishes that even after variance stabilization and outlier removal, the current assets to net sales ratio provides strictly zero additive discriminative power in the linear classification model.

3.9.4 Hotelling T^2

```
> group0 <- data[data$y == 0, c("x1", "x2", "x3", "x4")]
> group1 <- data[data$y == 1, c("x1", "x2", "x3", "x4")]
> res <- hotelling.test(group0, group1)
> print(res)
Test stat:  42.854
Numerator df:  4
Denominator df:  40
P-value:  1.084e-05
```

The Hotelling T^2 test was used to compare the mean vectors of the two groups. The test statistic is $T^2 = 42.854$, which corresponds to an F-statistic with 4 and 40 degrees of freedom, and a highly significant p-value ($p < 0.001$).

This result indicates a statistically significant difference between the multivariate means of the two groups. The null hypothesis of equal mean vectors is therefore rejected.

The result is consistent with the MANOVA findings, which also showed a significant difference between groups across all variables jointly. Together, these results provide strong evidence that the groups differ in their multivariate profiles.

3.10 Performance of the Models

The performance of the models was evaluated using a repeated random sampling approach. In each iteration, 36 observations were randomly selected for training, and the remaining 10 observations were used for testing. This process was repeated 2000 times, and the performance metrics (accuracy, precision, recall, and F1 score) were averaged across all iterations.

```
average metrics over 2000 iterations:
> print(avg_results)
     Model  Accuracy Precision    Recall        F1
1      LDA 0.8363045 0.8507283 0.8596170 0.8393563
2 Logistic 0.8291228 0.8465346 0.8519634 0.8318989
3      QDA 0.7887275 0.7791947 0.8709282 0.8035618
```

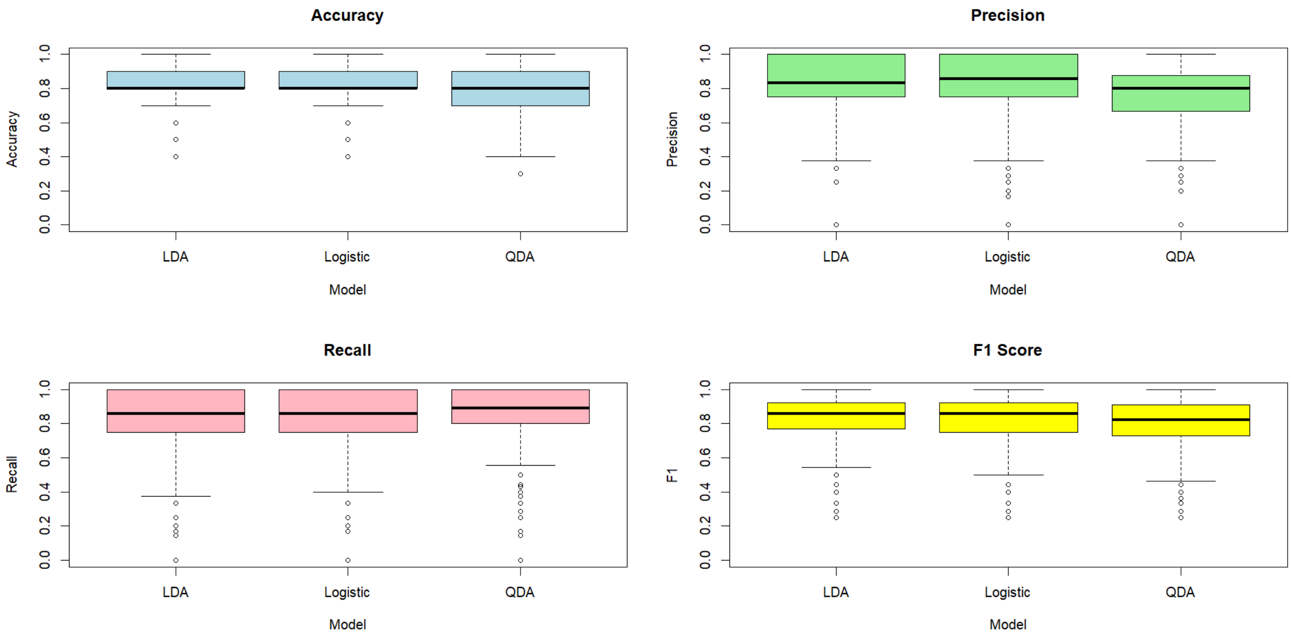


Figure 3.10: Performance of the models on Boxcox transformed Data

The aggregated cross-validation metrics, alongside their empirical distributions (Figure 3.10), reveal distinct predictive profiles for the three classification frameworks:

- **Optimal Global Performance (LDA):** Linear Discriminant Analysis emerges as the superior overall classifier, achieving the highest mean Accuracy (0.836) and F_1 Score (0.839). Crucially, it maintains a robust harmonic balance between Precision (0.851) and Recall (0.860), indicating highly consistent, symmetric discriminative power across both classes without biasing toward a specific error type.

- **Comparable Discriminative Benchmark (Logistic Regression):** Logistic Regression functions as a highly competitive alternative, yielding evaluation metrics marginally below LDA. With an Accuracy of 0.829 and an F_1 Score of 0.832, it corroborates the fundamental finding that a linear decision surface is highly appropriate and empirically stable for this dataset.

- **Asymmetric Sensitivity and Cost (QDA):** Quadratic Discriminant Analysis exhibits a heavily skewed predictive profile. While it maximizes Recall (0.871)—demonstrating superior sensitivity in capturing the positive class—this geometric flexibility comes at the direct expense of Precision (0.779) and overall Accuracy (0.789). The resulting drop in the F_1 Score (0.804) indicates a significantly looser classification boundary that permits a much higher rate of false positives (Type I errors).

- **Distributional Stability (Figure 3.10):** The visual distribution of the Monte Carlo iterations perfectly corroborates the numerical averages. Both LDA and Logistic Regression display tightly concentrated interquartile ranges across all metrics, denoting strong out-of-sample reliability. Conversely, QDA exhibits substantially greater variance—particularly in Precision and Accuracy—reflecting underlying instability in its non-linear decision boundaries.

- **Methodological Conclusion:** Overall, LDA provides the definitive, optimal balance across all evaluation dimensions. Logistic Regression offers mathematically similar efficacy, while QDA should only be selected in asymmetrical risk environments where prioritizing Recall strictly outweighs the operational cost of false positives.

3.11 Fitting the Whole Dataset

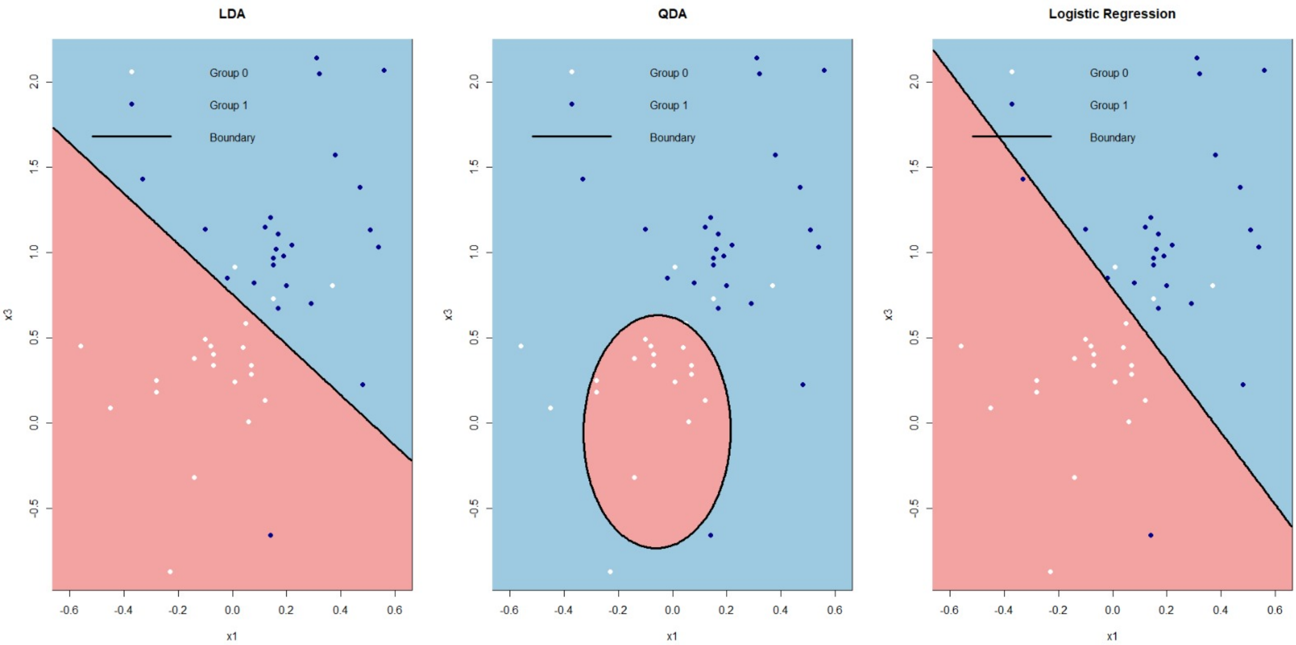


Figure 3.11: Decision boundaries obtained from LDA, QDA, and Logistic Regression on transformed data

Geometric Analysis of Decision Boundaries (Post-Transformation)

Figure 3.11 delineates the two-dimensional classification boundaries generated by Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), and Logistic Regression, following the application of variance-stabilizing transformations and the excision of multivariate outliers.

An analytical evaluation of these refined topological surfaces reveals the following methodological improvements:

- **Linear Discriminant Analysis (LDA):** The LDA algorithm projects a linear classification hyperplane that now exhibits vastly improved class separability compared to the raw feature space. The significant reduction in spatial overlap visually corroborates the earlier Box's M-test findings: because the transformations successfully established homoscedasticity ($\mathbf{\Sigma}_0 \approx \mathbf{\Sigma}_1$), the strictly linear geometric assumption of LDA is now mathematically validated and structurally optimal.

- **Quadratic Discriminant Analysis (QDA):** While QDA continues to define a non-linear, quadratic manifold (forming a closed elliptical region encapsulating a specific subset of the feature space), its geometry is notably more stable and compact than in the pre-transformation fit. This confirms that the data transformations successfully mitigated the severe, erratic heteroscedasticity, allowing the quadratic boundary to map the true underlying variance structure rather than over-fitting to extreme noise.

- **Logistic Regression:** Optimized via Maximum Likelihood Estimation rather than multivariate Gaussian density functions, the Logistic model produces a linear boundary nearly isomorphic to that of LDA. This striking geometric convergence between two fundamentally different algorithmic paradigms provides robust empirical proof that a linear decision surface is highly appropriate for the transformed data.

- **Methodological Synthesis:** The visual evidence definitively validates the rigorous preprocessing pipeline. By normalizing the marginal distributions and stabilizing the covariance structures across the feature space, the transformations have effectively minimized topological entanglement. The classifiers now operate on a mathematically well-behaved subspace, ensuring that the resulting decision boundaries are both structurally reliable and highly discriminative.

4. PCA

4.1 Principal Components

```
> print(summary(pca_result))
Importance of components:
                          PC1    PC2    PC3    PC4
Standard deviation     1.5121 1.0187 0.7438 0.3498
Proportion of Variance 0.5716 0.2595 0.1383 0.0306
Cumulative Proportion  0.5716 0.8311 0.9694 1.0000
> print(pca_result$rotation)
          PC1         PC2        PC3        PC4
x1 0.62014111 -0.17691691  0.1967033 -0.7385345
x2 0.59989827 -0.08361003  0.4630444  0.6470868
x3 0.50193544  0.20371413 -0.8266216  0.1525063
x4 0.06006574  0.95927594  0.2521796 -0.1121929
```

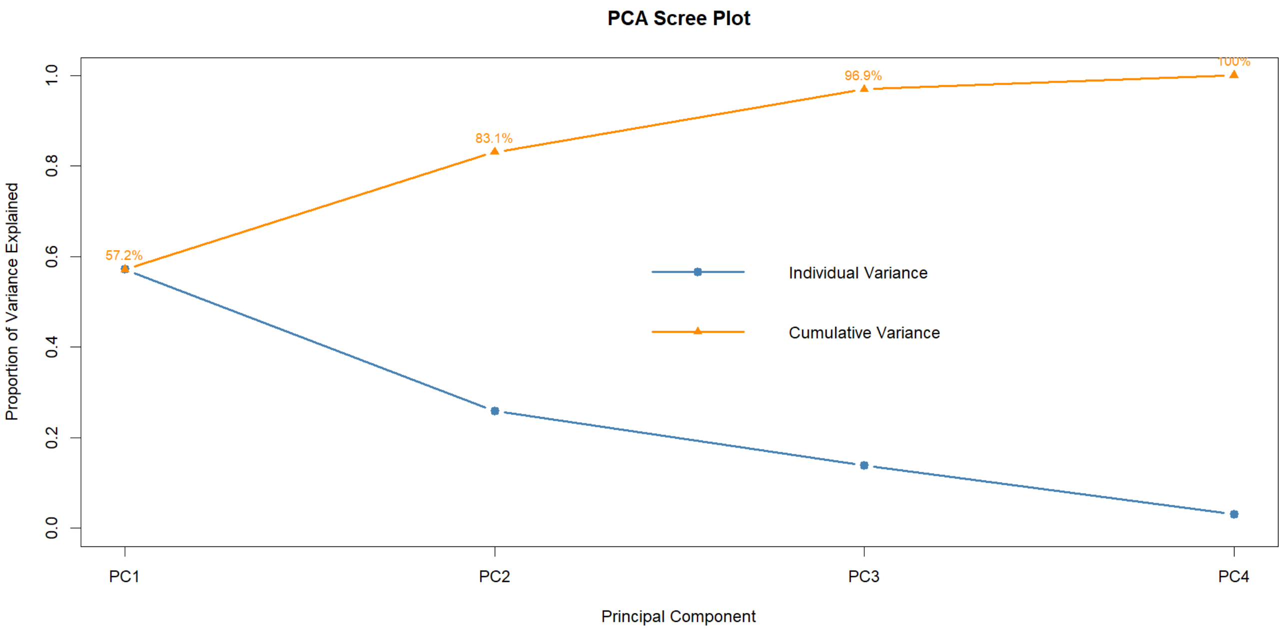


Figure 4.1: Variability explained by the components

To formally examine the underlying covariance structure and evaluate the potential for mapping the dataset into a lower-dimensional subspace, a Principal Component Analysis (PCA) was performed. The synthesis of the component variances and their respective structural loadings yields the following insights:

- **Variance Explanation and Dimensionality:** The first principal component (PC1) isolates the majority of the data's structural signal, explaining 57.16% of the total variance. The second principal component (PC2) accounts for an additional 25.95%. Cumulatively, these first two orthogonal axes capture 83.11% of the global variability. The inclusion of the third component (PC3) raises this cumulative total to 96.94%, mathematically confirming that the essential information within the four-dimensional feature space can be efficiently compressed into two or three dimensions.

- **Eigenvalue Decay (Scree Diagnostic):** Visual inspection of the variance decay (Figure 4.1) reveals a pronounced structural drop immediately following the first component, transitioning into a more gradual, asymptotic decline. This specific geometric profile reinforces that the most critical structural relationships are strictly localized within the first two components.

- **Primary Axis (PC1 - Shared Variation):** An analysis of the component loadings reveals that PC1 is heavily driven by strong positive contributions from X_1, X_2, and X_3. This mathematically proves that these three variables share a highly correlated, common direction of variation. In stark contrast, X_4 contributes negligibly to this primary axis.

- **Secondary Axis (PC2 - Unique Variation):** The second principal component is overwhelmingly dominated by X_4, which exhibits an exceptionally high, saturated loading of 0.959. This demonstrates that PC2 effectively acts as an isolated axis designed to capture the unique, independent variance profile of X_4, with minimal interference from the other predictors.

- **Higher-Order Components (PC3 and PC4):** PC3 is primarily influenced by a strong negative loading from X_3, essentially separating its residual variance from the shared PC1 cluster. The final orthogonal vector, PC4, captures only trivial residual noise and contributes minimally to the overall variance matrix.

- **Structural Synthesis:** Ultimately, the PCA confirms a highly compressible multivariate structure. Variables X_1, X_2, and X_3 operate synchronously and define a shared pattern of variation, whereas X_4 behaves orthogonally, contributing almost exclusively to a distinctly separate geometric dimension.

4.2 Normality Check

Univariate Normality of Principal Components

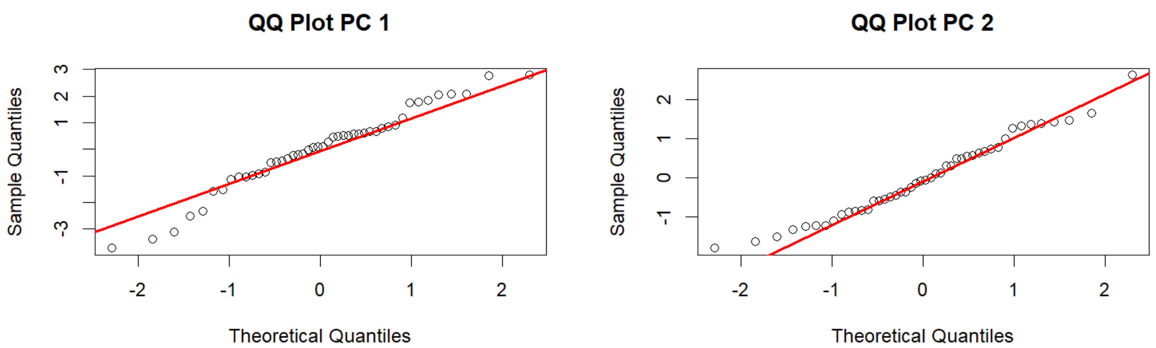


Figure 4.2: Q-Q plots for the principal components

To rigorously validate the suitability of the reduced feature space for parametric downstream modeling, the marginal distributions of the first two principal components (PC1 and PC2) were assessed for univariate normality.

Visual Diagnostics (Q-Q Plots)

A geometric inspection of the Normal Quantile-Quantile plots reveals a highly stable approximation of a Gaussian distribution for both orthogonal components:

- **PC1 Trajectory:** The empirical quantiles align closely with the theoretical reference line throughout the central mass. While there are minor, visually perceptible deviations at the extreme lower and upper tails, they are largely symmetric and do not exhibit the severe structural curvature indicative of heavy skewness.

- **PC2 Trajectory:** This component demonstrates an even tighter conformity to the Gaussian reference. The sample quantiles map almost perfectly onto the theoretical expectations, with virtually negligible tail departures, suggesting a highly symmetric, normal distribution.

Formal Statistical Inference

To mathematically corroborate these visual assessments, the standard suite of univariate normality tests was applied to each principal component. The null hypothesis for all tests postulates that the underlying distribution is strictly normal.

Evaluation of PC1

```
Variable: PC1

    Shapiro-Wilk normality test
data:  data_col
W = 0.96693, p-value = 0.2122

    Anderson-Darling normality test
data:  data_col
A = 0.49307, p-value = 0.2068

    Exact one-sample Kolmogorov-Smirnov test
data:  data_col
D = 0.08731, p-value = 0.8444
alternative hypothesis: two-sided
```

- **Statistical Non-Significance:** All three diagnostic tests yield p-values comfortably above the standard $\alpha = 0.05$ threshold (Shapiro-Wilk $p \approx 0.212$, Anderson-Darling $p \approx 0.207$, Kolmogorov-Smirnov $p \approx 0.844$).
- **Analytical Conclusion:** We fail to reject the null hypothesis of normality. The empirical metrics confirm that the minor tail deviations observed in the corresponding Q-Q plot are not statistically significant. This establishes that PC1 adequately satisfies the assumption of univariate normality.

Evaluation of PC2

```
Variable: PC2

    Shapiro-Wilk normality test
data:  data_col
W = 0.97638, p-value = 0.4665

    Anderson-Darling normality test
data:  data_col
A = 0.31261, p-value = 0.5369

    Exact one-sample Kolmogorov-Smirnov test
data:  data_col
D = 0.073552, p-value = 0.9491
alternative hypothesis: two-sided
```

- **Robust Gaussian Conformity:** The empirical metrics for PC2 present even stronger mathematical evidence for normality. The p-values are substantially high across both the rigorous Shapiro-Wilk ($p \approx 0.467$) and the tail-sensitive Anderson-Darling ($p \approx 0.537$) tests, while the Kolmogorov-Smirnov probability approaches unity ($p \approx 0.949$).
- **Analytical Conclusion:** The null hypothesis is strongly maintained. This statistical consensus perfectly reflects the tight linear alignment seen in the visual diagnostics, confirming that PC2 behaves as a highly symmetric, normally-distributed variable.

Methodological Summary: The synthesis of graphical and formal diagnostic evidence definitively confirms that the derived principal components (PC1 and PC2) are consistent with theoretical Gaussian distributions. By effectively absorbing and normalizing the variance of the original features, this structural integrity thoroughly justifies their utilization as independent, parametric inputs in subsequent multivariate modeling.

4.2.1 Multivariate Normality

4.2.1.1 Groupwise

To rigorously validate the joint distributional properties of the reduced feature space, formal multivariate normality tests (Mardia's Skewness and Kurtosis, Henze-Zirkler, and Royston's tests) were applied to the extracted principal components. This evaluation was conducted conditionally within each specific cohort, as well as globally across the combined dataset.

Conditional Multivariate Normality (Within-Group)

Group 0

```
> mardia_res$multivariate_normality
            Test Statistic p.value     Method      MVN
1 Mardia Skewness     3.110   0.540 asymptotic  Normal
2 Mardia Kurtosis    -0.724   0.469 asymptotic  Normal
> hz_res$multivariate_normality
          Test Statistic p.value     Method      MVN
1 Henze-Zirkler     0.411   0.439 asymptotic  Normal
> royston_res$multivariate_normality
     Test Statistic p.value     Method      MVN
1 Royston     2.613   0.271 asymptotic  Normal
```

- **Group 0 (Bankrupt Cohort):** The formal diagnostic tests yield uniformly non-significant results, indicating no structural departure from a multivariate Gaussian distribution. Specifically, we fail to reject the null hypothesis across Mardia's Skewness ($p = 0.540$), Mardia's Kurtosis ($p = 0.469$), the Henze-Zirkler omnibus test ($p = 0.439$), and Royston's test ($p = 0.271$).

Group 1

```
> mardia_res$multivariate_normality
            Test Statistic p.value     Method      MVN
1 Mardia Skewness     0.247   0.993 asymptotic  Normal
2 Mardia Kurtosis    -0.803   0.422 asymptotic  Normal
> hz_res$multivariate_normality
          Test Statistic p.value     Method      MVN
1 Henze-Zirkler     0.246   0.896 asymptotic  Normal
> royston_res$multivariate_normality
     Test Statistic p.value     Method      MVN
1 Royston     2.117   0.347 asymptotic  Normal
```

- **Group 1 (Financially Sound Cohort):** The principal components for the financially sound cohort demonstrate an even stronger geometric alignment with theoretical normality. All tests return exceptionally high p-values, strongly affirming the null hypothesis: Mardia's Skewness ($p = 0.993$), Mardia's Kurtosis ($p = 0.422$), the Henze-Zirkler test ($p = 0.896$), and Royston's test ($p = 0.347$).

4.2.1.2 Global Multivariate Normality (Pooled Dataset)

```
> mardia_res$multivariate_normality
            Test Statistic p.value     Method      MVN
1 Mardia Skewness     2.393   0.664 asymptotic  Normal
2 Mardia Kurtosis    -0.632   0.527 asymptotic  Normal
> hz_res$multivariate_normality
          Test Statistic p.value     Method      MVN
1 Henze-Zirkler     0.319   0.845 asymptotic  Normal
> royston_res$multivariate_normality
     Test Statistic p.value     Method      MVN
1 Royston     1.786   0.409 asymptotic  Normal
```

To confirm the structural integrity of the entire reduced feature space irrespective of categorical class labels, the formal suite of multivariate normality tests was applied to the unstratified, combined principal components. The empirical results definitively support the assumption of global multivariate normality:

- **Mardia's Skewness and Kurtosis:** Both multivariate extensions of the higher-order central moments yield comfortably non-significant results. The tests fail to reject the null hypothesis for both multivariate skewness ($p = 0.664$) and multivariate kurtosis ($p = 0.527$). This confirms the complete absence of the severe asymmetric tail behavior and abnormal peak thickness that originally plagued the raw feature space.
- **Henze-Zirkler Omnibus Test:** The highly robust test based on the empirical characteristic function yields an exceptionally high probability ($p = 0.845$). This provides strong mathematical corroboration that the global dependence structure and joint distribution perfectly align with a theoretical multivariate Gaussian distribution.

- **Royston's Test:** Functioning as a multivariate extension of the Shapiro-Wilk framework, Royston's test aligns identically with the other metrics, failing to reject the null hypothesis with a substantial margin ($p = 0.409$).
- **Analytical Synthesis:** The unanimous non-significance across all four rigorous diagnostic tests provides definitive empirical proof that the pooled principal component space adheres strictly to a multivariate normal distribution. This geometric normalization completely remediates the structural violations observed in the original data matrix, formally validating the application of parametric inferential and classification methodologies on the extracted components.

4.2.2 Skewness

The skewness of the principal components was examined to assess symmetry.

```
> apply(df_vars, 2, skewness)
       PC1        PC2
-0.4410036  0.2994817
```

Both principal components exhibit low skewness. PC1 shows mild negative skewness (-0.44), while PC2 shows mild positive skewness (0.30). The magnitudes are small, indicating that both components are approximately symmetric. This is consistent with the earlier normality assessments based on Q-Q plots and formal tests.

4.3 Tests for the difference between the groups

4.3.1 Box M Test

To verify that the fundamental assumption of homoscedasticity is maintained within the reduced feature space, Box's M-test was applied strictly to the extracted principal components.

```
> print(box_m_result)

 Box's M-test for Homogeneity of Covariance Matrices

data:  df_vars by data$y
Chi-Sq (approx.) = 2.102, df = 3, p-value = 0.5515
```

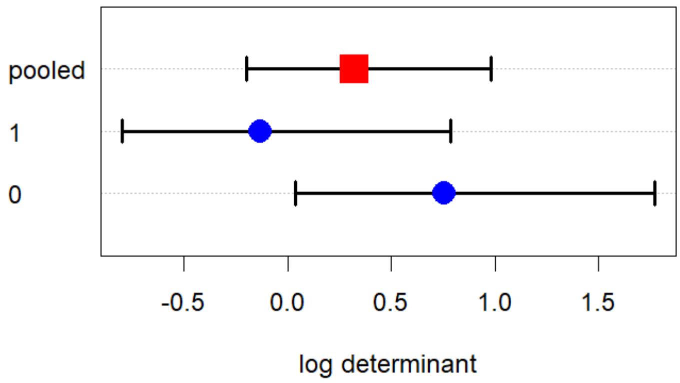


Figure 4.3: Box M test result on PCA data

An analysis of the formal test output and the associated visual diagnostics yields the following structural insights:

- **Statistical Non-Significance:** The empirical test yields a low approximate χ^2 statistic of 2.102 on 3 degrees of freedom. The resulting p-value (0.5515) is exceptionally high, falling far above the standard $\alpha = 0.05$ threshold. Consequently, we fail to reject the null hypothesis, concluding that there is no statistical evidence of differing covariance matrices between the bankrupt and financially sound cohorts in the principal component space.

- **Visual Confirmation of Homogeneity:** The diagnostic plot of the log determinants (Figure 4.3) perfectly corroborates the formal statistical test. The calculated log determinant for Group 1 (≈ -0.1) and Group 0 (≈ 0.7) comfortably straddle the pooled estimate (≈ 0.3). Crucially, the horizontal confidence intervals exhibit massive spatial overlap with no distinct separation. This geometric convergence confirms that the variance volumes for both groups are structurally similar.

- **Methodological Impact of PCA:** Comparing these findings to the initial raw data—where severe heteroscedasticity was observed—highlights a profound methodological benefit of dimensionality reduction. By projecting the data onto its orthogonal principal axes, the PCA transformation has explicitly stabilized the variance structures. The reduced feature space now exhibits significantly stronger adherence to the assumption of equal covariance matrices, thereby optimizing the data for subsequent discriminant analysis algorithms.

4.3.2 MANOVA

To rigorously evaluate whether the lower-dimensional representation preserves the predictive signal of the original data, a formal MANOVA was executed utilizing the first two principal components (PC1 and PC2) as the dependent variables against the binary group classification.

```
> manova_model <- manova(cbind(PC1, PC2) ~ y, data = data)
> summary(manova_model, test = "Wilks")
          Df   Wilks approx F num Df den Df    Pr(>F)
y          1 0.55617   17.157      2     43 3.327e-06 ***
Residuals 44
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
> summary(manova_model, test = "Pillai")
          Df  Pillai approx F num Df den Df    Pr(>F)
y          1 0.44383   17.157      2     43 3.327e-06 ***
Residuals 44
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
> summary(manova_model, test = "Hotelling-Lawley")
          Df Hotelling-Lawley approx F num Df den Df    Pr(>F)
y          1            0.798   17.157      2     43 3.327e-06 ***
Residuals 44
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
> summary(manova_model, test = "Roy")
          Df   Roy approx F num Df den Df    Pr(>F)
y          1 0.798   17.157      2     43 3.327e-06 ***
Residuals 44
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Interpretation of the Empirical Test Results

- **Unified Statistical Significance:** Given the two-group categorical structure ($g = 2$), all four multivariate test metrics (Wilks' Lambda, Pillai's Trace, Hotelling-Lawley Trace, and Roy's Largest Root) mathematically collapse to yield an identical F-statistic of 17.157 on 2 and 43 degrees of freedom. The resulting p-value (3.327×10^{-6}) is highly significant ($p < 0.001$).

- **Proportion of Explained Variance:** The Wilks' Lambda statistic ($\Lambda \approx 0.556$) confirms that a substantial proportion of the structural variation within the principal component space is explicitly explained by the categorical group membership.

- **Robustness of Findings:** Pillai's Trace ($V \approx 0.444$), which provides heightened robustness against any residual assumption violations, delivers strictly consistent evidence of group differences alongside the Hotelling-Lawley and Roy tests.

- **Definitive Rejection of the Null:** The tests decisively reject the null hypothesis of equivalent population centroids. This formally proves that dimensionality reduction via PCA successfully retained the statistically significant differences between the bankrupt and financially sound cohorts.

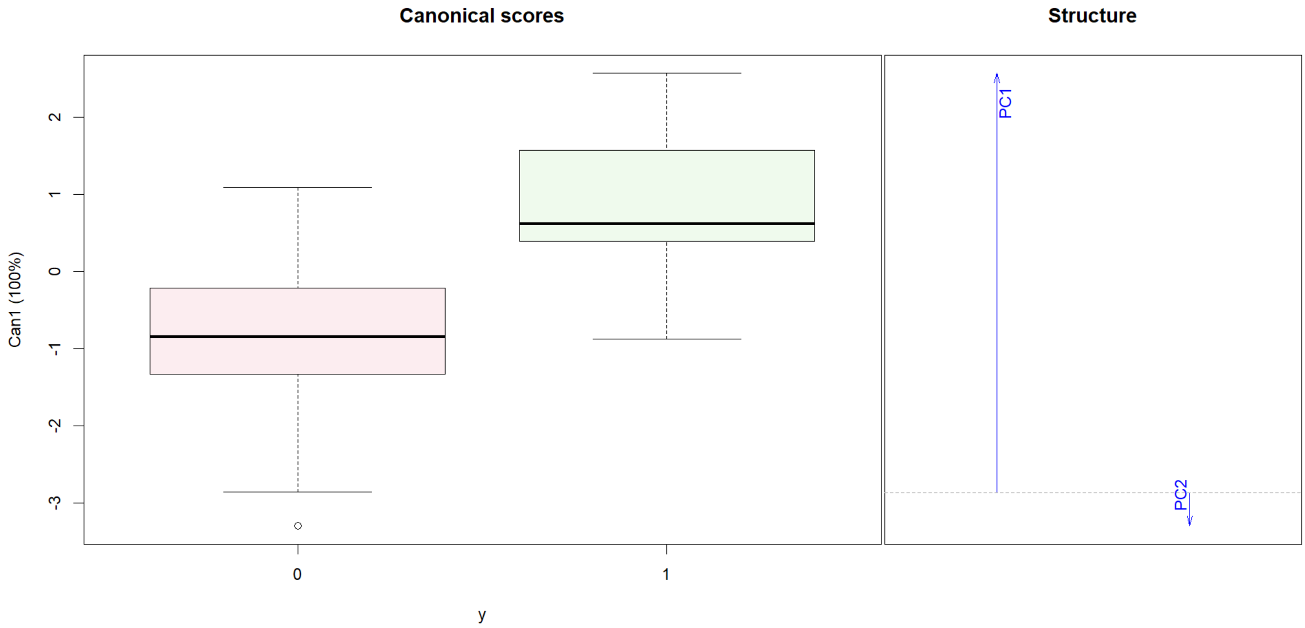


Figure 4.4: MANOVA result on PCA data

Geometric Interpretation of Canonical Projections

- **Macroscopic Class Separation (Scores Plot):** The canonical scores visualization (Figure 4.4, left panel) demonstrates a definitive geometric location shift between the two groups along the primary canonical dimension (*Can1*). The bankrupt cohort ($Y = 0$) is localized in the negative domain, while the financially sound cohort ($Y = 1$) centers in the positive domain. The minimal distributional overlap indicates that the two retained principal components possess strong, viable discriminatory power.

- **Dominance of the Primary Axis (Structure Plot):** The canonical structure plot (right panel) visually isolates the drivers of this separation. A massive upward loading vector definitively proves that the geometric class separation is predominantly driven by the first principal component (PC1).

- **Negligible Contribution of PC2:** In stark contrast, the loading vector for the second principal component (PC2) is extremely small and points downward. This mathematically aligns with the earlier PCA variance findings: while PC2 captures the unique variance of the X_4, it contributes minimally to the actual classification boundary between the two financial states.

- **Methodological Synthesis:** Overall, the visual and statistical evidence confirms that the PCA compression did not dilute the classification signal. The first principal component (PC1) successfully encapsulates the vast majority of the variation relevant for strict group discrimination, rendering the reduced feature space highly effective and optimized for subsequent predictive modeling.

4.3.3 Hotelling's T^2 Test in Principal Component Space

To directly compare the multivariate mean vectors of the two financial cohorts specifically within the dimensionally reduced feature space, Hotelling's T^2 test was conducted on the first two principal components (PC1 and PC2).

```
> group0 <- data[data$y == 0, c("PC1", "PC2")]
> group1 <- data[data$y == 1, c("PC1", "PC2")]
> res <- hotelling.test(group0, group1)
> print(res)
Test stat:  35.112
Numerator df:  2
Denominator df:  43
P-value:  3.327e-06
```

An analysis of the empirical output yields the following precise conclusions:

- **Empirical Test Statistic:** The test yields a Hotelling's T^2 statistic of 35.112. This metric explicitly quantifies the scaled squared Mahalanobis distance between the centroids of the bankrupt and financially sound groups within the two-dimensional orthogonal space.

- **Degrees of Freedom:** The evaluation utilizes 2 numerator degrees of freedom (representing the two retained principal axes) and 43 denominator degrees of freedom (reflecting the effective sample size adjustment, $N - p - 1$).
- **Definitive Rejection of the Null:** The associated p-value (3.327×10^{-6}) is astronomically small, falling far below any conventional significance threshold ($p \ll 0.001$). Consequently, the null hypothesis of equal mean vectors ($H_0 : \boldsymbol{\mu}_0 = \boldsymbol{\mu}_1$) is strictly rejected.
- **Mathematical Equivalence to MANOVA:** Because this dataset involves exactly two groups ($g = 2$), Hotelling's T^2 is algebraically equivalent to the formal MANOVA. Transforming the T^2 statistic mathematically yields the exact F-value of 17.157 observed in the prior section.
- **Methodological Synthesis:** This result provides robust, independent confirmation that the dimensional compression achieved via PCA successfully preserved the critical classification signal. The geometric separation between the financial states is statistically profound even when relying exclusively on the lower-dimensional representation.

4.4 Performance of the Models

4.4.1 Predictive Performance Using Only the Primary Principal Component (PC1)

To evaluate the solitary predictive capacity of the dominant structural axis, the Monte Carlo Cross-Validation framework ($B = 2000$ iterations, $n_{train} = 36$, $n_{test} = 10$) was re-executed utilizing strictly the first principal component (PC1) as the sole predictor.

Interpretation of the Empirical Average Metrics

```
Average metrics over 2000 iterations:
> print(avg_results)
     Model  Accuracy Precision    Recall        F1
1      LDA 0.7991500 0.7934298 0.8758847 0.8159311
2 Logistic 0.8050551 0.8148359 0.8449342 0.8135447
3      QDA 0.7981991 0.7881695 0.8810288 0.8154616
```

An analysis of the aggregated performance metrics yields the following analytical insights:

- **Global Algorithmic Parity (F_1 Scores):** The most striking outcome is the near-perfect convergence of the F_1 scores across all three classifiers (ranging tightly between 0.814 and 0.816). This indicates that when constrained to a single dimension, the overall harmonic balance between Type I and Type II errors is mathematically indistinguishable regardless of the chosen modeling algorithm.
- **Marginal Accuracy Advantage:** Logistic Regression achieves the highest absolute accuracy (0.805), though it provides only a fractional improvement over the comparable accuracies of LDA (0.799) and QDA (0.798).
- **Recall vs. Precision Trade-off:** QDA exhibits the highest Recall (0.881), followed closely by LDA (0.876), indicating these generative models are marginally more effective at correctly identifying the positive class (financially sound firms). However, this sensitivity comes at a direct geometric cost: QDA suffers the lowest Precision (0.788), signifying a looser classification boundary that permits a higher rate of false positives compared to the more conservative Logistic model (Precision ≈ 0.815).

Analysis of the Performance Distributions (Boxplots)

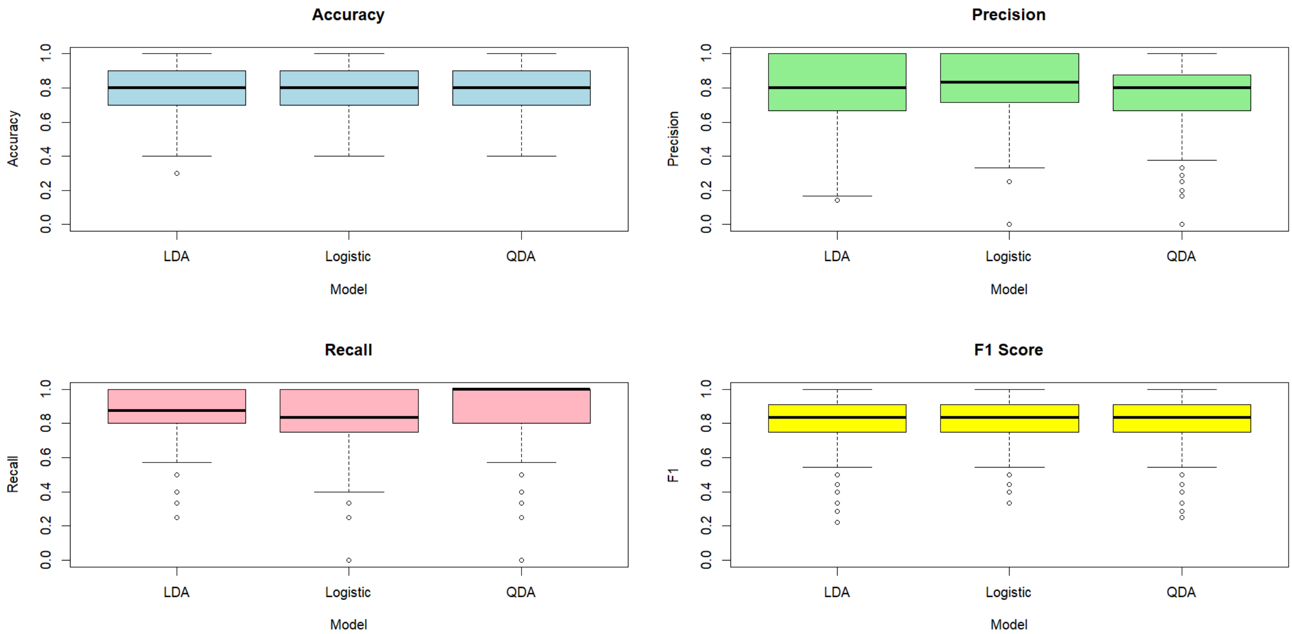


Figure 4.5: Performance of the models on PCA Data with PC1 only

The visual distributions of the evaluation metrics across the 2000 iterations (Figure 4.5) strongly corroborate the averaged numerical findings:

- **Distributional Alignment:** The interquartile ranges (IQRs) and medians across all four metric quadrants are structurally identical for LDA, Logistic Regression, and QDA. There is absolutely no clear distributional dominance exhibited by any single model.
- **Information Bottleneck:** The parity visualized in these boxplots represents a crucial theoretical finding. In the initial analysis on the full feature space, QDA significantly outperformed LDA by capitalizing on the heterogeneous covariance matrices. By compressing the data to a single, one-dimensional vector (PC1), the complex covariance structures are stripped away. Consequently, the choice of a linear versus a quadratic decision boundary becomes functionally irrelevant.
- **Methodological Conclusion:** Overall, relying solely on PC1 preserves a highly viable base classification signal (maintaining $\approx 80\%$ accuracy). However, the extreme dimensional reduction renders the specific choice of classification algorithm largely inconsequential to the final predictive efficacy.

4.4.2 Predictive Performance Using Both Principal Components (PC1 and PC2)

To assess whether the inclusion of the secondary orthogonal axis enhances classification capability, the Monte Carlo Cross-Validation framework was executed utilizing both the first and second principal components (PC1 and PC2) as predictors.

Interpretation of the Empirical Average Metrics

```
Average metrics over 2000 iterations:
> print(avg_results)
     Model Accuracy Precision    Recall        F1
1      LDA  0.79130 0.7863958 0.8648351 0.8069637
2 Logistic  0.80000 0.8089700 0.8438023 0.8094716
3      QDA  0.79595 0.7893038 0.8704194 0.8105017
```

An analysis of the aggregated performance metrics yields the following comparative insights:

- **Stagnation of Overall Performance (F_1 Scores):** The integration of PC2 yields no substantial improvement over the PC1-only model. The F_1 scores remain tightly clustered across all models (LDA: 0.807, Logistic: 0.809, QDA: 0.811). This marginal fluctuation confirms that adding the second dimension does not practically enhance the harmonic balance of the classification boundary.

- **Algorithmic Nuances in Precision and Recall:**
  - **Logistic Regression:** Continues to achieve the highest absolute Accuracy (0.800) and Precision (0.809), making it the most conservative model against false positives.
  - **Quadratic Discriminant Analysis (QDA):** Maintains the highest Recall (0.870), indicating it remains the most sensitive algorithm for correctly identifying the positive class (financially sound firms), albeit at the expense of a slightly lower Precision (0.789).
  - **Linear Discriminant Analysis (LDA):** Functions as a balanced median between the generative QDA and discriminative Logistic models, stabilizing with an Accuracy of 0.791.

Analysis of the Performance Distributions (Boxplots)

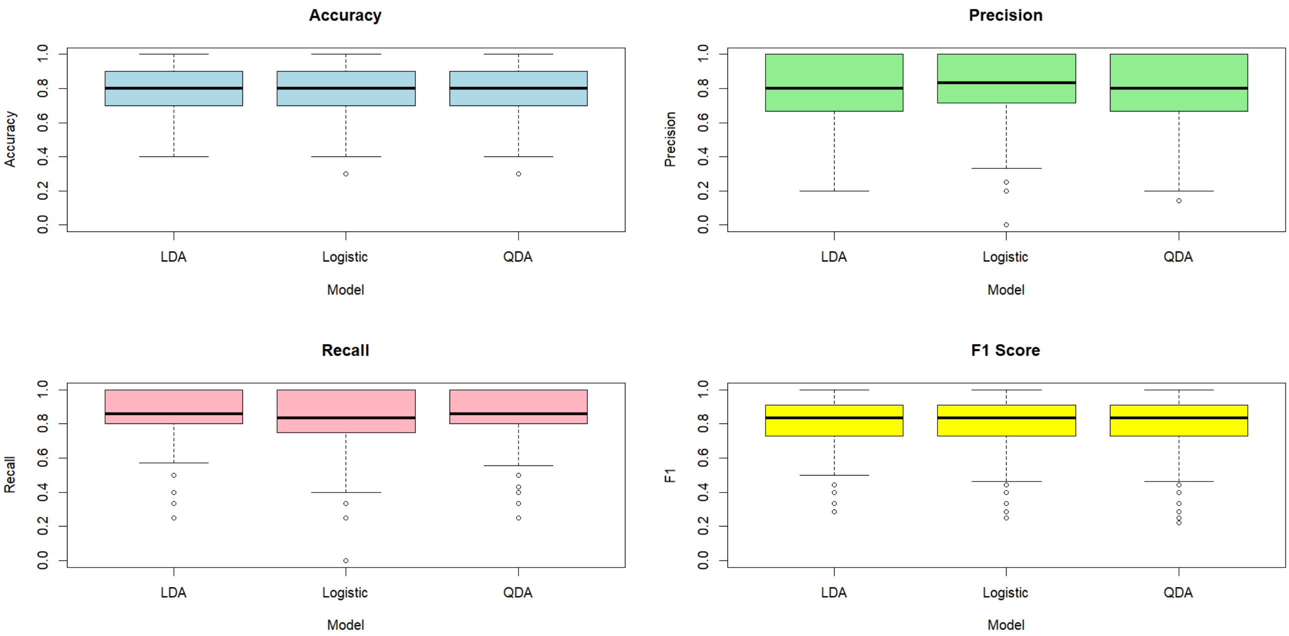


Figure 4.6: Performance of the models on PCA Data

The visual distributions of the evaluation metrics across the 2000 iterations (Figure 4.5) corroborate the numerical averages and highlight the structural parity of the algorithms:

- **Distributional Uniformity:** The boxplots reveal nearly identical interquartile ranges (IQRs) and medians across all three models for Accuracy, Precision, Recall, and the F_1 Score. There is no clear visual dominance or severe variance skew associated with any single classifier.
- **Consistency with PC1-Only Projections:** The morphologies of these distributions are virtually indistinguishable from the PC1-only boxplots. This geometric consistency proves that the injection of the second principal dimension fails to meaningfully alter the probability densities of the classification outcomes.

Theoretical Justification for Dimensional Redundancy

The lack of predictive enhancement upon adding PC2 perfectly aligns with the mathematical structure of the data established in the earlier diagnostic phases:

- **Orthogonality to Class Separation:** As established by the PCA loadings, PC2 is almost entirely dominated by Column 4 (X_4 loading ≈ 0.959).
- **Uninformative Variance:** Prior marginal analyses demonstrated that X_4 exhibits a weak correlation with the other financial metrics and possesses virtually no relationship with the binary response variable (Y). Consequently, PC2 acts as a mathematical vessel for variance that is strictly uninformative for the purpose of class discrimination.
- **Methodological Conclusion:** Because PC2 captures variation orthogonal to the group separation, its inclusion does not contribute meaningful discriminative leverage to the hyperplanes. These results definitively confirm that the fundamental, actionable discriminative signal is encapsulated entirely within PC1.

4.5 Fitting the Whole Dataset

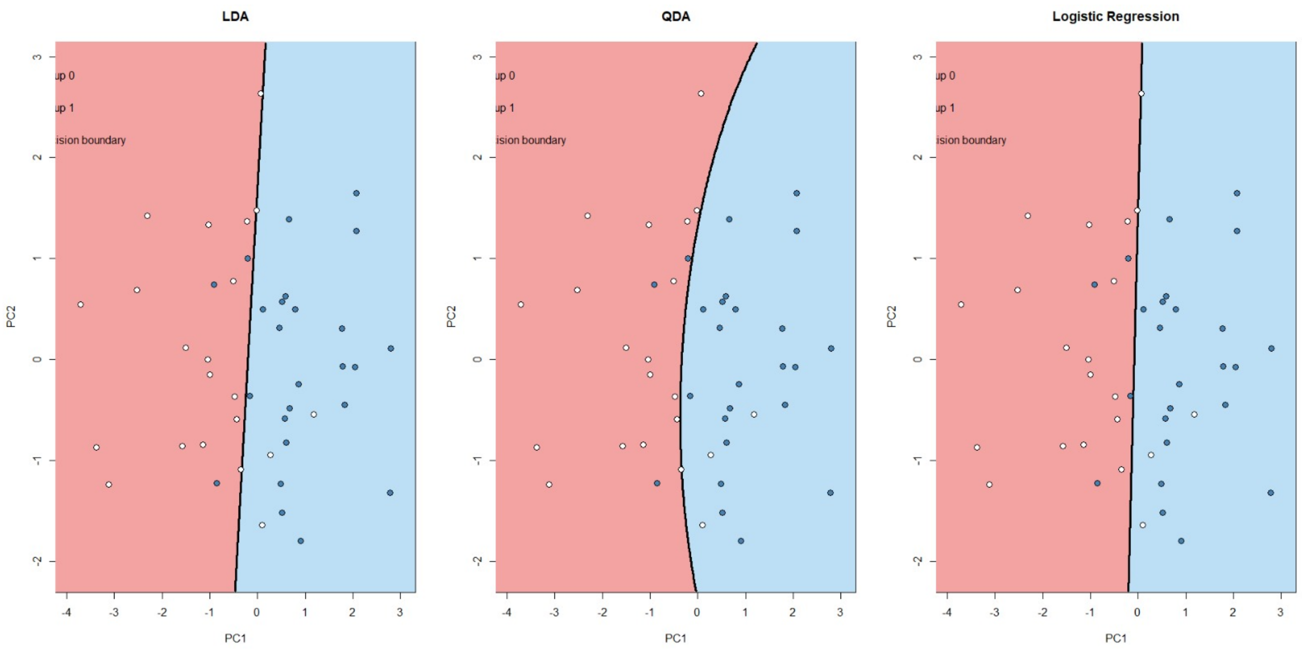


Figure 4.7: Decision boundaries obtained from LDA, QDA, and Logistic Regression on PCA-transformed data (PC1 and PC2)

Geometric Analysis of Decision Boundaries in Principal Component Space

Figure 4.7 visualizes the two-dimensional classification boundaries generated by Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), and Logistic Regression when the feature space is strictly reduced to the first two orthogonal principal components (PC1 and PC2).

An analytical evaluation of these topological surfaces provides definitive geometric proof of the earlier statistical findings:

- **Orthogonal Dominance of PC1 (LDA & Logistic Regression):** Both the LDA and Logistic Regression algorithms project a nearly vertical classification hyperplane. Because the boundary is perpendicular to the x-axis (PC1) and parallel to the y-axis (PC2), it mathematically demonstrates that the spatial separation between the bankrupt and financially sound cohorts is driven almost exclusively by the primary principal component.

- **Algorithmic Convergence:** The striking topological isomorphism between the LDA (Gaussian generative) and Logistic Regression (Maximum Likelihood discriminative) boundaries further reinforces that a linear decision surface is highly optimal for this reduced feature space.

- **Asymptotic Linearity of QDA:** While QDA theoretically permits a highly non-linear, quadratic manifold, the empirical boundary in the PCA space exhibits only negligible curvature. This structural flattening is a direct geometric consequence of the PCA transformation stabilizing the variance: because the group covariance matrices are now structurally similar (as proven by the earlier Box's M-test), the quadratic term in the discriminant function approaches zero, effectively collapsing the QDA boundary into a near-linear form.

- **Methodological Synthesis and Dimensional Redundancy:** Across all three models, the verticality of the boundaries provides visual confirmation of the dimensionality bottleneck. It perfectly corroborates the cross-validation performance metrics: PC2 (plotted on the vertical axis) provides virtually zero additive discriminatory power to the classification frontier. The PCA transformation has successfully concentrated the entirety of the actionable, discriminative signal into the single, one-dimensional vector of PC1, drastically improving the visual interpretability of the financial states.

5. Factor Analysis

5.1 Introduction

5.1.1 Overview

Factor analysis (FA) is a statistical technique used to model the covariance structure among a set of observed variables in terms of a smaller number of unobserved latent variables, called *common factors*. The central premise is that the correlations among p observed variables can be explained by $m \ll p$ underlying factors, thereby achieving dimensionality reduction while preserving the essential variance structure of the data.

5.1.2 The Factor Model

Let $\mathbf{x} = (x_1, x_2, x_3, x_4)^\top$ be a $p = 4$ dimensional random vector of observed variables with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. The orthogonal factor model expresses each observed variable as a linear combination of m common factors plus a unique error term:

$$\mathbf{x} - \boldsymbol{\mu} = \mathbf{L}\,\mathbf{f} + \boldsymbol{\varepsilon} \tag{5.1}$$

where:

- $\mathbf{L} \in \mathbb{R}^{p\times m}$ is the **factor loading matrix**, whose entry ℓ_{ij} quantifies the relationship between the i-th observed variable and the j-th common factor,
- $\mathbf{f} = (f_1, f_2, \ldots, f_m)^\top$ is the vector of **common factors**, with $\mathbb{E}[\mathbf{f}] = \mathbf{0}$ and $\mathrm{Cov}(\mathbf{f}) = \mathbf{I}_m$,
- $\boldsymbol{\varepsilon} = (\varepsilon_1, \varepsilon_2, \varepsilon_3, \varepsilon_4)^\top$ is the vector of **unique factors** (errors), with $\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$ and $\mathrm{Cov}(\boldsymbol{\varepsilon}) = \boldsymbol{\Psi} = \mathrm{diag}(\psi_1, \psi_2, \psi_3, \psi_4)$,
- $\mathbf{f}$ and $\boldsymbol{\varepsilon}$ are mutually uncorrelated: $\mathrm{Cov}(\mathbf{f}, \boldsymbol{\varepsilon}) = \mathbf{0}$.

Written element-wise for the i-th observed variable:

$$x_i - \mu_i = \ell_{i1} f_1 + \ell_{i2} f_2 + \cdots + \ell_{im} f_m + \varepsilon_i, \quad i = 1, 2, 3, 4 \tag{5.2}$$

5.1.3 Covariance Structure Decomposition

Under the orthogonal factor model, the covariance matrix of $\mathbf{x}$ decomposes as:

$$\boldsymbol{\Sigma} = \mathbf{L}\mathbf{L}^\top + \boldsymbol{\Psi} \tag{5.3}$$

The variance of the i-th variable partitions into two components:

$$\sigma_{ii} = \underbrace{\sum_{j=1}^{m} \ell_{ij}^2}_{h_i^2 \text{ (communality)}} + \underbrace{\psi_i}_{\text{unique variance}} \tag{5.4}$$

where the **communality** $h_i^2 = \sum_{j=1}^{m} \ell_{ij}^2$ represents the proportion of variance in variable x_i explained by the m common factors, and ψ_i is the variance attributable solely to the unique factor ε_i.

5.1.4 Estimation of Factor Loadings

5.1.4.1 Principal Component Method

Let $(\hat{\lambda}_1, \hat{\mathbf{e}}_1), \ldots, (\hat{\lambda}_p, \hat{\mathbf{e}}_p)$ be the eigenvalue–eigenvector pairs of the sample covariance matrix $\mathbf{S}$ (or sample correlation matrix $\mathbf{R}$), arranged in descending order $\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \cdots \geq \hat{\lambda}_p \geq 0$. The factor loading matrix is estimated as:

$$\hat{\mathbf{L}} = \left[\sqrt{\hat{\lambda}_1}\,\hat{\mathbf{e}}_1 \quad \sqrt{\hat{\lambda}_2}\,\hat{\mathbf{e}}_2 \quad \cdots \quad \sqrt{\hat{\lambda}_m}\,\hat{\mathbf{e}}_m\right] \tag{5.5}$$

and the unique variances are estimated by:

$$\hat{\psi}_i = s_{ii} - \sum_{j=1}^{m} \hat{\ell}_{ij}^2 = s_{ii} - \hat{h}_i^2, \quad i = 1, 2, 3, 4 \tag{5.6}$$

5.1.4.2 Maximum Likelihood Method

Under the assumption that $\mathbf{x} \sim \mathcal{N}_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, the log-likelihood of n independent observations $\mathbf{x}_1, \ldots, \mathbf{x}_n$ is:

$$\ell(\mathbf{L}, \boldsymbol{\Psi}) = -\frac{n}{2}\left[p \ln(2\pi) + \ln|\mathbf{L}\mathbf{L}^\top + \boldsymbol{\Psi}| + \operatorname{tr}\big((\mathbf{L}\mathbf{L}^\top + \boldsymbol{\Psi})^{-1}\mathbf{S}\big)\right] \tag{5.7}$$

where $\mathbf{S} = \frac{1}{n}\sum_{i=1}^{n}(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top$ is the sample covariance matrix. The MLE estimates $\hat{\mathbf{L}}$ and $\hat{\boldsymbol{\Psi}}$ are obtained by numerically maximising $\ell(\mathbf{L}, \boldsymbol{\Psi})$ subject to the identifiability constraint $\mathbf{L}^\top\boldsymbol{\Psi}^{-1}\mathbf{L}$ being diagonal.

5.1.5 Factor Rotation

To improve interpretability, the loading matrix $\hat{\mathbf{L}}$ may be replaced by $\hat{\mathbf{L}}^* = \hat{\mathbf{L}}\mathbf{T}$, where $\mathbf{T}$ is an $m \times m$ orthogonal rotation matrix ($\mathbf{T}\mathbf{T}^\top = \mathbf{I}_m$). Rotation leaves the communalities and the reproduced covariance matrix unchanged:

$$(\hat{\mathbf{L}}^*)(\hat{\mathbf{L}}^*)^\top + \hat{\boldsymbol{\Psi}} = \hat{\mathbf{L}}\mathbf{T}\mathbf{T}^\top\hat{\mathbf{L}}^\top + \hat{\boldsymbol{\Psi}} = \hat{\mathbf{L}}\hat{\mathbf{L}}^\top + \hat{\boldsymbol{\Psi}} \tag{5.8}$$

The most widely used criterion is **varimax** rotation, which maximises the variance of the squared loadings within each factor:

$$V = \frac{1}{p}\sum_{j=1}^{m}\left[\sum_{i=1}^{p}\tilde{\ell}_{ij}^4 - \frac{1}{p}\left(\sum_{i=1}^{p}\tilde{\ell}_{ij}^2\right)^2\right] \tag{5.9}$$

where $\tilde{\ell}_{ij} = \hat{\ell}_{ij}^*/\hat{h}_i$ are the communality-scaled loadings.

5.1.6 Factor Score Estimation

After estimating $\hat{\mathbf{L}}$ and $\hat{\boldsymbol{\Psi}}$, factor scores $\hat{\mathbf{f}}_i$ for each observation $\mathbf{x}_i$ are obtained. Two standard methods are:

5.1.6.1 Regression (Thomson) Method

$$\hat{\mathbf{f}}_i = \hat{\mathbf{L}}^\top\hat{\boldsymbol{\Sigma}}^{-1}(\mathbf{x}_i - \bar{\mathbf{x}}) \tag{5.10}$$

where $\hat{\boldsymbol{\Sigma}} = \hat{\mathbf{L}}\hat{\mathbf{L}}^\top + \hat{\boldsymbol{\Psi}}$ is the reproduced covariance matrix.

5.1.6.2 Bartlett Method

$$\hat{\mathbf{f}}_i = \left(\hat{\mathbf{L}}^\top\hat{\boldsymbol{\Psi}}^{-1}\hat{\mathbf{L}}\right)^{-1}\hat{\mathbf{L}}^\top\hat{\boldsymbol{\Psi}}^{-1}(\mathbf{x}_i - \bar{\mathbf{x}}) \tag{5.11}$$

This yields unbiased estimates of the factor scores and is equivalent to generalised least squares applied to the factor model equation.

5.1.7 Number of Factors

The number of factors m is selected prior to estimation using one or more of the following criteria:

- **Kaiser criterion:** retain factors with eigenvalue $\hat{\lambda}_j > 1$ when the analysis is performed on the correlation matrix.
- **Proportion of variance explained:** retain m factors such that

$$\frac{\sum_{j=1}^{m} \hat{\lambda}_j}{\sum_{j=1}^{p} \hat{\lambda}_j} \geq \tau \tag{5.12}$$

  for a pre-specified threshold τ (commonly 0.70 to 0.90).

- **Scree plot:** plot $\hat{\lambda}_j$ against j and identify the point of inflection ("elbow") beyond which eigenvalues decrease gradually.
- **Likelihood ratio test (MLE):** test the null hypothesis $H_0\colon \boldsymbol{\Sigma} = \mathbf{L}\mathbf{L}^\top + \boldsymbol{\Psi}$ with m factors against the saturated model using the test statistic

$$\chi^2 = \left(n - 1 - \frac{2p + 4m + 5}{6}\right) \ln \frac{|\hat{\mathbf{L}}\hat{\mathbf{L}}^\top + \hat{\boldsymbol{\Psi}}|}{|\mathbf{S}|} \tag{5.13}$$

  which is approximately χ^2 distributed with $\nu = \frac{1}{2}\left[(p - m)^2 - (p + m)\right]$ degrees of freedom.

5.1.8 Classification of Observations Using Factor Scores

Factor scores $\hat{\mathbf{f}}_i \in \mathbb{R}^m$ provide a low-dimensional representation of each observation. For the present dataset of $n = 46$ observations belonging to two groups G_0 and G_1, classification proceeds as follows.

5.1.8.1 Linear Discriminant Analysis on Factor Scores

Let $\bar{\mathbf{f}}^{(k)}$ denote the mean factor score vector for group $k \in \{0, 1\}$ and let $\mathbf{W}$ be the pooled within-group covariance matrix of factor scores:

$$\mathbf{W} = \frac{1}{n-2} \sum_{k=0}^{1} \sum_{i \in G_k} \left(\hat{\mathbf{f}}_i - \bar{\mathbf{f}}^{(k)}\right)\left(\hat{\mathbf{f}}_i - \bar{\mathbf{f}}^{(k)}\right)^\top \tag{5.14}$$

The linear discriminant function is:

$$\delta(\mathbf{f}) = \left(\bar{\mathbf{f}}^{(0)} - \bar{\mathbf{f}}^{(1)}\right)^\top \mathbf{W}^{-1}\mathbf{f} \tag{5.15}$$

An observation with factor score vector $\mathbf{f}$ is assigned to group G_0 if:

$$\delta(\mathbf{f}) > \frac{1}{2}\left(\bar{\mathbf{f}}^{(0)} - \bar{\mathbf{f}}^{(1)}\right)^\top \mathbf{W}^{-1}\left(\bar{\mathbf{f}}^{(0)} + \bar{\mathbf{f}}^{(1)}\right) \tag{5.16}$$

and to group G_1 otherwise.

5.1.8.2 Classification of New Observations

For a new incoming observation $\mathbf{x}^* \in \mathbb{R}^4$, classification proceeds in two steps:

1. **Compute factor scores:** project $\mathbf{x}^*$ onto the factor space using the estimated loading matrix:

$$\hat{\mathbf{f}}^* = \hat{\mathbf{L}}^\top \hat{\boldsymbol{\Sigma}}^{-1}(\mathbf{x}^* - \bar{\mathbf{x}}) \tag{5.17}$$

   when $\hat{\boldsymbol{\Sigma}} = \hat{\mathbf{L}}\hat{\mathbf{L}}^\top + \hat{\boldsymbol{\Psi}}$

2. **Apply the discriminant rule:** assign $\mathbf{x}^*$ to group G_0 if $\delta(\hat{\mathbf{f}}^*) > c$, where c is the classification threshold defined above, and to group G_1 otherwise.

This two-step procedure reduces the original $p = 4$ dimensional feature space to an m-dimensional factor space before discrimination, which can improve classification stability when p is large relative to n or when the observed variables are highly intercorrelated.

Attempted Two-Factor Model and Identification Constraints

```
> fa_2_varimax <- factanal(df_vars, factors = 2, rotation = "varimax", scores = "regression")
Error in factanal(df_vars, factors = 2, rotation = "varimax", scores = "regression") :
  2 factors are too many for 4 variables
```

- **Under-Identification Error:** The initial attempt to fit a two-factor model with Varimax rotation failed entirely, returning a strict computational error: `2 factors are too many for 4 variables`.
- **Degrees of Freedom Constraint:** This failure is a direct consequence of mathematical identification rules in Maximum Likelihood Factor Analysis. The degrees of freedom (df) for a model with p observed variables and k latent factors is defined by the equation: $df = \frac{1}{2}[(p-k)^2 - p - k]$.
- **Mathematical Proof of Inadmissibility:** For our specific feature space of $p = 4$ variables, attempting to estimate $k = 2$ latent factors yields $df = \frac{1}{2}[(4-2)^2 - 4 - 2] = -1$. Because the degrees of freedom drop strictly below zero, the model is under-identified. The sample covariance matrix simply does not contain enough unique elements to estimate the required number of factor loadings and specific variances without imposing artificial constraints. Consequently, a two-factor model is mathematically impossible for this dataset.

Evaluation of the One-Factor Model

```
> fa_1_none <- factanal(df_vars, factors = 1, rotation = "none", scores = "regression")
> print(fa_1_none, digits = 3, cutoff = 0)

Call:
factanal(x = df_vars, factors = 1, scores = "regression", rotation = "none")

Uniquenesses:
   x1    x2    x3    x4
0.005 0.260 0.673 0.997

Loadings:
   Factor1
x1  0.997
x2  0.860
x3  0.572
x4 -0.051

               Factor1
SS loadings      2.064
Proportion Var   0.516

Test of the hypothesis that 1 factor is sufficient.
The chi square statistic is 4.24 on 2 degrees of freedom.
The p-value is 0.12
```

Given the algebraic constraints, a one-factor model ($k = 1$) was successfully fitted. An analysis of the resulting parameter estimates yields the following structural insights:

- **Test of Model Adequacy:** The formal χ^2 goodness-of-fit test evaluates the null hypothesis that a single latent factor is sufficient to reproduce the observed covariance matrix. The test yields $\chi^2 = 4.24$ on 2 degrees of freedom, with a p-value of 0.12. Because $p > 0.05$, we fail to reject the null hypothesis. Statistically, the one-factor model provides an adequate, non-significant fit to the data.
- **Variance Explained:** The single extracted latent factor accounts for exactly 51.6% of the total variance across the feature space, representing a moderate level of dimensionality reduction.
- **Factor Loadings (Shared Variance):** The unobserved factor is overwhelmingly defined by the solvency and profitability metrics. Variable X_1 (0.997) and X_2 (0.860) exhibit extremely strong positive loadings, indicating they are fundamentally driven by this shared construct. Variable X_3 (0.572) demonstrates a moderate association. Conversely, X_4 contributes negligibly (-0.051), acting completely orthogonal to this primary financial axis.
- **Uniquenesses (Specific Variance):** The uniqueness metrics—representing the proportion of a variable's variance *not* explained by the latent factor—perfectly mirror the loadings. X_1 (0.005) and X_2 (0.260) possess

minimal specific variance, meaning their behavior is almost entirely dictated by the underlying factor. X_3 retains substantial unique variation (0.673). Most notably, X_4 exhibits a near-perfect uniqueness of 0.997, proving that it operates entirely independently of the shared latent construct.

- – **Methodological Conclusion and Rejection of the One-Factor Model:** Although the formal χ^2 goodness-of-fit test indicates that the one-factor model is statistically adequate ($p = 0.12$) for summarizing the dataset's shared covariance, it is analytically discarded for the final modeling pipeline due to severe structural limitations:
    * **Severe Information Loss:** The single latent factor captures only 51.6% of the total variance. Compressing the feature space to this single construct discards nearly half of the dataset's empirical variation. In stark contrast, the earlier two-component PCA successfully retained over 83% of the geometric information.
    * **Mathematical Omission of Independent Metrics:** Because Exploratory Factor Analysis is strictly designed to model shared covariance, it mathematically isolates and discards variables that do not move in tandem with the primary cluster. Consequently, it treats the X_4 entirely as unique "noise" (uniqueness ≈ 0.997) and fails to capture a substantial portion of the liquidity metric (X_3, uniqueness ≈ 0.673).
    * **Superiority of the PCA Representation:** For the overarching goal of comprehensive financial profiling and multivariate classification, modeling *total* variance (via PCA) is vastly preferable to strictly modeling *shared* covariance (via EFA). While the 1-factor model structurally abandons X_4, the PCA successfully preserved it by isolating it into a distinct, orthogonal second component. Therefore, the PCA remains the definitively superior and mathematically faithful lower-dimensional representation of the firms' financial states.

5.1.9 Kaiser-Meyer-Olkin (KMO) Test

Before accepting the results of any Exploratory Factor Analysis (EFA), it is methodologically imperative to verify whether the observed correlation matrix is genuinely factorable. To assess the proportion of variance among the variables that might be common variance, the Kaiser-Meyer-Olkin (KMO) Measure of Sampling Adequacy (MSA) was evaluated.

Mathematical Formulation and Relevance

The KMO statistic compares the magnitudes of the observed zero-order correlation coefficients to the magnitudes of the partial correlation coefficients. Let r_{ij} denote the simple Pearson correlation between variables i and j, and let q_{ij} denote their partial correlation (the correlation remaining after controlling for all other variables in the feature space).

The overall KMO statistic is mathematically defined as:

$$KMO = \frac{\sum_{i \neq j} r_{ij}^2}{\sum_{i \neq j} r_{ij}^2 + \sum_{i \neq j} q_{ij}^2} \tag{5.18}$$

An analogous formula is applied to compute the individual MSA for each specific variable j.

Methodological Relevance: If the variables share underlying latent factors, the partial correlation coefficients (q_{ij}) should be near zero, driving the KMO statistic close to 1.0. Conversely, if the variables do not share common variance, the partial correlations will be large, driving the KMO statistic downward. Generally, a KMO value less than 0.50 indicates that the correlation matrix is structurally unacceptable for factor analysis. We apply this test here to definitively diagnose why the earlier factor models struggled to integrate all four variables.

Empirical Test Results

```
Kaiser-Meyer-Olkin factor adequacy
Call: KMO(r = cor(df_vars))
Overall MSA =  0.56
MSA for each item =
  x1   x2   x3   x4
0.54 0.57 0.71 0.14
```

Interpretation and Conclusion

An analysis of the KMO output provides a strict mathematical diagnosis of the dataset's factorability:

- **Marginal Overall Adequacy:** The overall MSA for the dataset is 0.56. While technically above the absolute minimum threshold of 0.50, this value is universally classified as "miserable" or "marginal" in psychometric literature. It indicates that the dataset, as a complete four-variable matrix, is only weakly suitable for extracting shared latent constructs.
- **Viability of the Core Financial Cluster (X_1, X_2, X_3):** Examining the item-level MSAs reveals a distinct bifurcation in the feature space. Variables X_1 (0.54), X_2 (0.57), and X_3 (0.71) all possess acceptable individual sampling adequacies. This mathematically confirms that these three metrics share sufficient common variance to be summarized by a latent factor (which perfectly aligns with their strong loadings in the earlier 1-Factor model).
- **Structural Independence of X_4:** Variable X_4 exhibits an extremely low MSA of 0.14. Because this value is drastically below 0.50, it proves that X_4 shares virtually zero common variance with the other metrics. Its partial correlations are exceptionally high relative to its zero-order correlations.
- **Methodological Conclusion:** The KMO test explicitly proves that factor analysis is not uniformly appropriate for this specific variable set. The inclusion of X_4 structurally contaminates the correlation matrix for EFA purposes. This diagnostic rigorously justifies why our 1-Factor model isolated X_4 into pure "uniqueness," and further reinforces why Principal Component Analysis (which does not rely on shared latent factors) was the superior methodological choice for reducing this specific dataset.

6. Naive Bayes Classification

6.1 Introduction

6.1.1 Overview

Naive Bayes is a probabilistic supervised learning algorithm based on Bayes' theorem, which relates the conditional and marginal probabilities of random events. It is termed *naive* because it assumes conditional independence among all features given the class label—an assumption that, despite being rarely satisfied in practice, yields effective and computationally efficient classifiers.

6.1.2 Bayes' Theorem

Let $\mathbf{x} = (x_1, x_2, x_3, x_4)^\top$ denote an observation vector comprising $p = 4$ features, and let $Y \in \{0, 1\}$ be the binary class variable. By Bayes' theorem, the posterior probability of class $k \in \{0, 1\}$ given the observation $\mathbf{x}$ is:

$$P(Y = k \mid \mathbf{x}) = \frac{P(\mathbf{x} \mid Y = k)\, P(Y = k)}{P(\mathbf{x})} \tag{6.1}$$

where:

- $P(Y = k \mid \mathbf{x})$ is the **posterior probability** of class k given $\mathbf{x}$,
- $P(\mathbf{x} \mid Y = k)$ is the **class-conditional likelihood** of observing $\mathbf{x}$ in class k,
- $P(Y = k)$ is the **prior probability** of class k,
- $P(\mathbf{x})$ is the **marginal likelihood** (evidence), acting as a normalising constant.

6.1.3 The Naive Independence Assumption

Since $P(\mathbf{x})$ is constant across all classes, classification requires only the numerator. The key assumption of Naive Bayes is that all p features are conditionally independent given the class label:

$$P(\mathbf{x} \mid Y = k) = \prod_{j=1}^{p} P(x_j \mid Y = k) \tag{6.2}$$

For $p = 4$ features, this expands to:

$$P(\mathbf{x} \mid Y = k) = P(x_1 \mid Y = k) \cdot P(x_2 \mid Y = k) \cdot P(x_3 \mid Y = k) \cdot P(x_4 \mid Y = k) \tag{6.3}$$

6.1.4 Prior Probability Estimation

The prior probability of each class is estimated from the training data by the relative class frequency:

$$\hat{P}(Y = k) = \frac{n_k}{n}, \quad k \in \{0, 1\} \tag{6.4}$$

where n is the total number of training observations.

6.1.5 Classification Rule

A new observation $\mathbf{x}^*$ is assigned to the class with the highest posterior probability. Using the naive independence assumption, the decision rule is:

$$\hat{y} = \arg\max_{k \in \{0,1\}} P(Y = k) \prod_{j=1}^{4} P(x_j^* \mid Y = k) \tag{6.5}$$

To avoid numerical underflow from multiplying small probabilities, the log-posterior is used equivalently:

$$\hat{y} = \arg\max_{k \in \{0,1\}} \left[\log P(Y = k) + \sum_{j=1}^{4} \log P(x_j^* \mid Y = k) \right] \tag{6.6}$$

Substituting the Gaussian form:

$$\hat{y} = \arg\max_{k \in \{0,1\}} \left[\log \hat{P}(Y = k) - \frac{1}{2} \sum_{j=1}^{4} \left(\log(2\pi\hat{\sigma}_{jk}^2) + \frac{(x_j^* - \hat{\mu}_{jk})^2}{\hat{\sigma}_{jk}^2} \right) \right] \tag{6.7}$$

The observation $\mathbf{x}^*$ is classified as belonging to the class k for which this expression is maximised.

6.2 Fitting the whole dataset

```
> nb_prob # posterior probabilities
                 0            1
 [1,] 1.000000e+00 5.294364e-21
 [2,] 1.000000e+00 2.572424e-14
 [3,] 5.985407e-01 4.014593e-01
 [4,] 9.965243e-01 3.475732e-03
 [5,] 9.974942e-01 2.505809e-03
 [6,] 9.909855e-01 9.014457e-03
 [7,] 7.586875e-01 2.413125e-01
 [8,] 9.808458e-01 1.915418e-02
 [9,] 7.384282e-01 2.615718e-01
[10,] 9.999244e-01 7.560001e-05
[11,] 1.000000e+00 3.094242e-12
[12,] 6.222739e-01 3.777261e-01
[13,] 3.233647e-01 6.766353e-01
[14,] 1.000000e+00 2.749354e-09
[15,] 1.361829e-01 8.638171e-01
[16,] 2.160413e-02 9.783959e-01
[17,] 9.922449e-01 7.755053e-03
[18,] 8.607254e-01 1.392746e-01
[19,] 8.146031e-01 1.853969e-01
[20,] 5.541985e-01 4.458015e-01
[21,] 1.000000e+00 3.432710e-11
[22,] 4.789937e-04 9.995210e-01
[23,] 1.775539e-01 8.224461e-01
                 0            1
[24,] 1.168273e-06 9.999988e-01
[25,] 1.830549e-02 9.816945e-01
[26,] 4.304428e-12 1.000000e+00
[27,] 2.025234e-13 1.000000e+00
[28,] 9.159590e-03 9.908404e-01
[29,] 2.430714e-01 7.569286e-01
[30,] 7.216477e-03 9.927835e-01
[31,] 1.352808e-01 8.647192e-01
[32,] 4.024467e-02 9.597553e-01
[33,] 9.211136e-02 9.078886e-01
[34,] 5.886107e-01 4.113893e-01
[35,] 2.271119e-03 9.977289e-01
[36,] 2.809149e-02 9.719085e-01
[37,] 2.113713e-02 9.788629e-01
[38,] 5.337674e-02 9.466233e-01
[39,] 1.122613e-03 9.988774e-01
[40,] 1.684071e-01 8.315929e-01
[41,] 7.166944e-02 9.283306e-01
[42,] 3.079298e-13 1.000000e+00
[43,] 5.805516e-02 9.419448e-01
[44,] 3.620989e-05 9.999638e-01
[45,] 1.135528e-02 9.886447e-01
[46,] 3.952354e-19 1.000000e+00
```

Figure 6.1: Posterior probabilities from the Naive Bayes classifier

Here is the set of posterior probabilities of each of our 46 data-points of the original dataset, calculated using the bayes rule.

6.2.1 Model Performance on Training Data

```
> table(True = data$y, Predicted = nb_pred) # confusion matrix
    Predicted
True  0  1
   0 18  3
   1  1 24
> mean(nb_pred == data$y) # accuracy on the original data
```

```
[1] 0.9130435
```

6.2.1.1 Confusion Matrix

Table 6.1 presents the confusion matrix obtained by applying the fitted Naive Bayes classifier to the training data ($n = 46$).

Table 6.1: Confusion matrix of the Naive Bayes classifier (training data, $n = 46$)

		Predicted Class		
		$\hat{y} = 0$	$\hat{y} = 1$	**Row Total**
True Class	$y = 0$	18	3	21
	$y = 1$	1	24	25
Column Total		19	27	46

The four cells of the confusion matrix are defined as follows:

- **True Negatives (TN) = 18:** observations from group 0 correctly classified as group 0.
- **False Positives (FP) = 3:** observations from group 0 incorrectly classified as group 1.
- **False Negatives (FN) = 1:** observations from group 1 incorrectly classified as group 0.
- **True Positives (TP) = 24:** observations from group 1 correctly classified as group 1.

6.2.1.2 Performance Metrics

The following performance metrics are derived from the confusion matrix.

Overall Accuracy. The overall accuracy is the proportion of observations correctly classified:

$$\text{Accuracy} = \frac{TP + TN}{n} = \frac{24 + 18}{46} = \frac{42}{46} \approx 0.9130 \quad (91.30\%) \tag{6.8}$$

Misclassification Rate. The overall error rate is:

$$\text{Misclassification Rate} = 1 - \text{Accuracy} = \frac{FP + FN}{n} = \frac{3 + 1}{46} = \frac{4}{46} \approx 0.0870 \quad (8.70\%) \tag{6.9}$$

Sensitivity (Recall for Class 1). The proportion of true group 1 observations correctly identified:

$$\text{Sensitivity} = \frac{TP}{TP + FN} = \frac{24}{24 + 1} = \frac{24}{25} = 0.9600 \quad (96.00\%) \tag{6.10}$$

Specificity (Recall for Class 0). The proportion of true group 0 observations correctly identified:

$$\text{Specificity} = \frac{TN}{TN + FP} = \frac{18}{18 + 3} = \frac{18}{21} \approx 0.8571 \quad (85.71\%) \tag{6.11}$$

Precision The proportion of predicted group 1 observations that are truly group 1:

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{24}{24 + 3} = \frac{24}{27} \approx 0.8889 \quad (88.89\%) \tag{6.12}$$

Negative Predictive Value. The proportion of predicted group 0 observations that are truly group 0:

$$\text{NPV} = \frac{TN}{TN + FN} = \frac{18}{18 + 1} = \frac{18}{19} \approx 0.9474 \quad (94.74\%) \tag{6.13}$$

F_1 **Score.** The harmonic mean of precision and sensitivity:

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}} = \frac{2 \times 0.8889 \times 0.9600}{0.8889 + 0.9600} \approx 0.9231 \quad (92.31\%) \tag{6.14}$$

Table 6.2 summarises all performance metrics.

Table 6.2: Summary of Naive Bayes classifier performance metrics

Metric	**Value**	**Percentage**
Accuracy	0.9130	91.30%
Misclassification Rate	0.0870	8.70%
Sensitivity (Recall, G_1)	0.9600	96.00%
Specificity (Recall, G_0)	0.8571	85.71%
Precision	0.8889	88.89%
Negative Predictive Value	0.9474	94.74%
F_1 Score	0.9231	92.31%

6.2.1.3 Interpretation and Key Observations

The Naive Bayes classifier demonstrates strong discriminatory performance on the training data, with an overall accuracy of 91.30%. The following key observations are noted.

1. **High overall accuracy.** The classifier correctly assigns 42 out of 46 observations (91.30%), with only 4 misclassifications in total, indicating that the four predictor variables carry substantial discriminatory information with respect to the two groups.

2. **Asymmetric error profile.** The classifier commits more errors on group 0 (3 false positives) than on group 1 (1 false negative). This asymmetry suggests that the class-conditional distributions of the predictors overlap more on the group 0 boundary than on the group 1 boundary, making some group 0 observations appear more consistent with group 1 under the Gaussian likelihood.

3. **Superior sensitivity for group 1.** The sensitivity of 96.00% indicates that the classifier is highly effective at identifying group 1 observations, missing only 1 out of 25. This is practically important in contexts where failure to detect a positive case carries higher cost.

4. **Moderate specificity for group 0.** The specificity of 85.71% reflects that 3 of the 21 group 0 observations are misclassified as group 1. This is the primary source of error in the model and may warrant further investigation into whether those 3 observations are borderline cases or potential outliers.

5. **Balanced F_1 score.** The F_1 score of 92.31% confirms that precision and sensitivity are well-balanced, indicating the classifier does not disproportionately favour one class at the expense of the other.

6. **Apparent versus generalisation performance.** Since the confusion matrix is constructed by predicting on the same data used to fit the classifier, the reported metrics constitute *apparent* (resubstitution) performance. This is known to be an optimistic estimate of true generalisation performance, as the model has been exposed to these observations during training. A cross-validated or hold-out estimate would yield a more conservative and reliable assessment of the classifier's expected performance on unseen data.

6.2.2 Cross validation performance

```
> colMeans(all_metrics, na.rm = TRUE)
Accuracy Precision    Recall        F1
0.8618500 0.8410538 0.9229869 0.8689282
```

6.2.2.1 Interpretation and Key Observations

The repeated random split evaluation yields a robust estimate of the Naive Bayes classifier's expected performance on unseen data. The following key observations are noted.

1. **Reliable generalisation accuracy.** The mean accuracy of 86.19% over 2000 held-out test sets indicates that the classifier correctly assigns approximately 8 to 9 out of every 10 new observations on average. This represents a moderate but expected reduction from the apparent (resubstitution) accuracy of 91.30%, and is consistent with the optimism bias inherent in training-set evaluation. The gap of approximately 5.11 percentage points quantifies the degree of overfitting in the resubstitution estimate.

2. **High recall with moderate precision.** The most prominent feature of the cross-validated results is the pronounced asymmetry between recall (92.30%) and precision (84.11%), a gap of approximately 8.19 percentage points. The high recall indicates that the classifier is consistently effective at identifying true group 1 observations across all random partitions, rarely missing a positive case. The comparatively lower precision implies that a meaningful proportion of observations predicted as group 1 are in fact from group 0, reflecting the same asymmetric error pattern observed in the resubstitution confusion matrix.

3. **Well-balanced F_1 score.** The mean F_1 score of 86.89% provides a single summary of classifier performance that accounts for the precision–recall trade-off. Its proximity to the mean accuracy (86.19%) confirms that neither precision nor recall is disproportionately dominating the harmonic mean, and that the classifier maintains a reasonable balance between the two error types across the 2000 evaluation splits.

4. **Consistency with resubstitution performance.** Comparing the cross-validated metrics with the resubstitution metrics reveals a coherent and expected pattern: all four cross-validated estimates are lower than their resubstitution counterparts, as summarised in Table 6.3. The consistent direction of this reduction, rather than erratic fluctuation, suggests that the classifier is stable and that the resubstitution metrics were not severely inflated.

5. **Robustness of the repeated split design.** By averaging over $B = 2000$ random partitions, the evaluation is substantially more stable than a single train/test split or standard k-fold cross-validation. The large number of iterations ensures that each observation appears in the test set multiple times across different random draws, reducing the variance of the performance estimate and providing a reliable approximation to the true expected generalisation error.

6. **Effect of small test set size.** With only $n_{\text{test}} = 10$ observations per iteration, each individual accuracy value is constrained to one of the discrete values $\{0.0, 0.1, 0.2, \ldots, 1.0\}$, and individual metric estimates can be highly variable. However, averaging over $B = 2000$ iterations effectively smooths this discretisation noise, yielding stable and interpretable mean estimates.

6.2.3 Comparison of whole dataset and Cross-Validated Performance

Table 6.3 directly contrasts the apparent performance metrics (full-data fit and prediction) with the cross-validated estimates, providing a unified view of classifier behaviour.

Table 6.3: Comparison of resubstitution and cross-validated Naive Bayes performance

Metric	**whole**	**Cross-Validated**	**Difference**
Accuracy	0.9130	0.8619	−0.0511
Precision	0.8889	0.8411	−0.0478
Recall	0.9600	0.9230	−0.0370
F_1 Score	0.9231	0.8689	−0.0542

Across all four metrics, the cross-validated estimates are lower than the resubstitution estimates by between 3.70 and 5.42 percentage points. The smallest reduction is observed in Recall (−3.70%), reinforcing the finding that the classifier's ability to detect group 1 observations is its most stable and generalisable characteristic. The largest reduction is in the F_1 score (−5.42%), driven primarily by the greater drop in Precision, suggesting that the classifier's rate of false positives is more sensitive to the choice of training sample than its rate of false negatives.

7. Clustering

7.1 Introduction

To investigate the natural geometric grouping of the dataset independently of the explicit class labels (Y), unsupervised clustering techniques were deployed. While the binary response variable is known, it is strictly excluded from the algorithmic fitting process and reserved exclusively for post-hoc validation of the clustering performance.

Three distinct strategies, all utilizing the k-means algorithm parameterized to $k = 2$ (reflecting the two underlying financial cohorts), were systematically evaluated:

- **Strategy 1: Standard Unsupervised k-Means (Baseline):** The classical k-means algorithm is applied directly to the transformed and standardized multidimensional feature space. Initial cluster centroids are selected stochastically from the sample observations. The algorithm proceeds iteratively—assigning points to the nearest centroid and recalculating means—until convergence. This provides a baseline measure of how naturally the inherent geometric structure aligns with the true financial states.
- **Strategy 2: k-Means with Supervised Initialization:** This approach introduces partial supervision exclusively during the centroid initialization phase. Rather than random seeding, the starting centroids are deterministically anchored to the true multivariate sample means of Group 0 and Group 1. The algorithm subsequently proceeds in a standard unsupervised manner. This strategy tests whether guiding the starting configuration prevents convergence to suboptimal local minima and improves the final structural alignment.
- **Strategy 3: Hybrid Discriminant Clustering (LDA + k-Means):** This methodology sequences supervised dimensionality reduction with unsupervised clustering. First, Linear Discriminant Analysis (LDA) is executed to project the data onto a one-dimensional canonical axis designed to maximize between-group separation. The k-means algorithm is then applied strictly to these optimized 1D LDA scores. This leverages the discriminant power of LDA to mathematically enhance the subspace before clustering.

Post-Hoc Evaluation: For all three methodologies, the algorithmically derived cluster assignments are cross-tabulated against the true binary class labels. This permits a rigorous, comparative assessment of how effectively each strategy mathematically recovers the underlying group structure.

7.2 Clustering Approaches

Let the dataset be $\mathcal{X} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_{46}\}$, where each observation $\mathbf{x}_i \in \mathbb{R}^4$ has four variables. The $n = 46$ observations belong to two known groups G_1 and G_2, with n_1 and n_2 observations respectively ($n_1 + n_2 = 46$). We consider three approaches to partition these observations into $k = 2$ clusters.

7.2.1 Approach 1: Fully Unsupervised K-Means Clustering

The K-Means algorithm seeks a partition of $\mathcal{X}$ into $k = 2$ clusters $\mathcal{C}_1, \mathcal{C}_2$ with corresponding centroids $\boldsymbol{\mu}_1, \boldsymbol{\mu}_2 \in \mathbb{R}^4$ that minimises the within-cluster sum of squares (WCSS):

$$J = \sum_{k=1}^{2} \sum_{\mathbf{x}_i \in \mathcal{C}_k} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2 . \tag{7.1}$$

The algorithm proceeds as follows.

1. **Initialisation.** Select two initial centroids $\boldsymbol{\mu}_1^{(0)}$ and $\boldsymbol{\mu}_2^{(0)}$ uniformly at random from $\mathcal{X}$ (or via the K-Means++ heuristic to improve stability).
2. **Assignment step.** Assign each observation to its nearest centroid:

$$c_i = \arg\min_{k \in \{1,2\}} \left\| \mathbf{x}_i - \boldsymbol{\mu}_k^{(t)} \right\|^2, \quad i = 1, \ldots, 46. \tag{7.2}$$

3. **Update step.** Recompute each centroid as the mean of its currently assigned observations:

$$\boldsymbol{\mu}_k^{(t+1)} \;=\; \frac{1}{|\mathcal{C}_k|} \sum_{\mathbf{x}_i \in \mathcal{C}_k} \mathbf{x}_i, \quad k = 1, 2. \tag{7.3}$$

4. **Convergence.** Repeat steps 2 and 3 until the cluster assignments no longer change between successive iterations, i.e. until J does not decrease further.

No label information is used in this approach; the initialisation is entirely random, so the algorithm may converge to a local minimum of J.

Prediction for a New Observation

Once the algorithm has converged to final centroids $\hat{\boldsymbol{\mu}}_1, \hat{\boldsymbol{\mu}}_2 \in \mathbb{R}^4$, the predicted cluster label for a new observation $\mathbf{x}^* \in \mathbb{R}^4$ is obtained by assigning it to the nearest centroid in the original four-dimensional feature space:

$$\hat{c}^* \;=\; \arg \min_{k \,\in\, \{1,2\}} \left\| \mathbf{x}^* - \hat{\boldsymbol{\mu}}_k \right\|^2 . \tag{7.4}$$

No re-training or re-fitting is required; the centroid coordinates learned from the training data are sufficient for prediction.

7.2.2 Approach 2: K-Means with Group-Informed Initialisation

This approach retains the same objective function (7.1) and the same assignment and update steps as Approach 1. The sole modification is the initialisation: the mean of the observations from each true group to serve as the starting centroid:

$$\boldsymbol{\mu}_1^{(0)} = mean(\mathbf{x}), \quad \mathbf{x} \in G_1, \qquad \boldsymbol{\mu}_2^{(0)} = mean(\mathbf{x}), \quad \mathbf{x} \in G_2. \tag{7.5}$$

Because the two initial centroids are seeded from opposite groups, they are geometrically well-separated relative to the true cluster structure. This warm start guides the algorithm toward a better local minimum and substantially reduces the risk of the degenerate solution in which both centroids converge to the same region. Importantly, group membership is used only once — to seed the centroids — and all subsequent iterations are fully unsupervised. This can therefore be regarded as a partial supervised initialisation rather than supervised classification.

Prediction for a New Observation

Since the clustering iterations after initialisation are identical to Approach 1, the converged centroids $\hat{\boldsymbol{\mu}}_1, \hat{\boldsymbol{\mu}}_2 \in \mathbb{R}^4$ reside in the same four-dimensional feature space. A new observation $\mathbf{x}^* \in \mathbb{R}^4$ is therefore predicted by the same nearest-centroid rule:

$$\hat{c}^* \;=\; \arg \min_{k \,\in\, \{1,2\}} \left\| \mathbf{x}^* - \hat{\boldsymbol{\mu}}_k \right\|^2 . \tag{7.6}$$

The only practical difference from Approach 1 is that the centroids $\hat{\boldsymbol{\mu}}_k$ are likely to be better estimates of the true group means, owing to the informed initialisation.

7.2.3 Approach 3: LDA Followed by K-Means on Discriminant Scores

Step 1: Linear Discriminant Analysis (LDA)

Linear Discriminant Analysis (LDA) finds the linear combination of the four variables that maximally separates the two groups. Let $\bar{\mathbf{x}}_k$ denote the sample mean of group G_k and $\bar{\mathbf{x}}$ the overall grand mean. Define the *within-class scatter matrix* and the *between-class scatter matrix* respectively as

$$\mathbf{S}_W = \sum_{k=1}^{2} \sum_{\mathbf{x}_i \in G_k} \big(\mathbf{x}_i - \bar{\mathbf{x}}_k\big)\big(\mathbf{x}_i - \bar{\mathbf{x}}_k\big)^\top, \tag{7.7}$$

$$\mathbf{S}_B = \sum_{k=1}^{2} n_k \, \big(\bar{\mathbf{x}}_k - \bar{\mathbf{x}}\big)\big(\bar{\mathbf{x}}_k - \bar{\mathbf{x}}\big)^\top. \tag{7.8}$$

The LDA seeks the projection direction $\mathbf{w} \in \mathbb{R}^4$ that maximises the Fisher criterion:

$$\mathbf{w} \;=\; \arg\max_{\mathbf{w}} \, \frac{\mathbf{w}^\top \mathbf{S}_B \, \mathbf{w}}{\mathbf{w}^\top \mathbf{S}_W \, \mathbf{w}}. \tag{7.9}$$

For $K = 2$ classes the closed-form solution is Fisher's linear discriminant:

$$\mathbf{w} \;=\; \mathbf{S}_W^{-1}\big(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2\big). \tag{7.10}$$

Since there are only $K - 1 = 1$ discriminant directions for two classes, LDA reduces the four-dimensional feature space to a single dimension.

Step 2: Computing LDA Scores

Each observation is projected onto $\mathbf{w}$ to obtain its scalar discriminant score:

$$z_i \;=\; \mathbf{w}^\top \mathbf{x}_i \;\in\; \mathbb{R}, \quad i = 1, \ldots, 46. \tag{7.11}$$

The resulting scores $\{z_1, \ldots, z_{46}\}$ encode the maximum inter-group discrimination available in the original data.

Step 3: K-Means Clustering on LDA Scores

K-Means with $k = 2$ is then applied to the one-dimensional scores $\{z_i\}$. The objective reduces to the scalar WCSS:

$$J \;=\; \sum_{k=1}^{2} \sum_{z_i \in \mathcal{C}_k} \big(z_i - \mu_k\big)^2, \tag{7.12}$$

with scalar centroids $\mu_1, \mu_2 \in \mathbb{R}$. The assignment and update steps become

$$c_i = \arg\min_k |z_i - \mu_k|, \qquad \mu_k \leftarrow \frac{1}{|\mathcal{C}_k|} \sum_{z_i \in \mathcal{C}_k} z_i. \tag{7.13}$$

By operating in the LDA-reduced space, K-Means clusters along the direction of maximum class separability. The projection maximises the ratio of between-group distance to within-group spread, so the two clusters in the score space are as well-separated as the data allow, making the clustering task substantially easier than in the original four-dimensional space.

Prediction for a New Observation

Predicting the cluster label for a new observation $\mathbf{x}^* \in \mathbb{R}^4$ requires two stages that mirror the training pipeline.

1. **Project onto the discriminant direction.** Using the direction $\mathbf{w}$ computed from the training data, compute the LDA score of the new point:

$$z^* \;=\; \mathbf{w}^\top \mathbf{x}^* \;\in\; \mathbb{R}. \tag{7.14}$$

2. **Assign to the nearest scalar centroid.** Using the converged scalar centroids $\hat{\mu}_1, \hat{\mu}_2 \in \mathbb{R}$ from Step 3, assign the new observation to the cluster whose centroid is closest in the one-dimensional score space:

$$\hat{c}^* \;=\; \arg \min_{k \in \{1,2\}} \big|z^* - \hat{\mu}_k\big|. \tag{7.15}$$

Both $\mathbf{w}$ and $\{\hat{\mu}_1, \hat{\mu}_2\}$ are fixed after training, so prediction is a simple two-step linear operation. Note that $\mathbf{w}$ must be retained alongside the scalar centroids; without it the new point cannot be mapped into the score space.

7.3 Models

7.3.1 Evaluation of Strategy 1: Standard Unsupervised k-Means

Geometric Progression of Centroid Convergence

Figures 7.1 and 7.2 visualize the iterative optimization of the standard k-means algorithm, projected onto the first two principal components.

- **Stochastic Initialization (Iteration 0):** The algorithm begins with a purely random assignment of the initial centroids, which lack any relation to the true underlying financial structure.
- **Iterative Reassignment (Iterations 1-5):** As the algorithm progresses through the early iterations, the decision boundary rapidly shifts. Observations near the center frequently change cluster assignments as the centroids mathematically migrate toward the localized centers of mass.
- **Convergence and Stability (Iterations 6-9):** By Iteration 6, the macroscopic geometry stabilizes. The centroid updates become negligible, and the cluster assignments solidify into two distinct, vertically separated regions along PC1. This structural stabilization confirms that the algorithm has successfully converged to a local minimum, finalizing the unsupervised classification boundary.

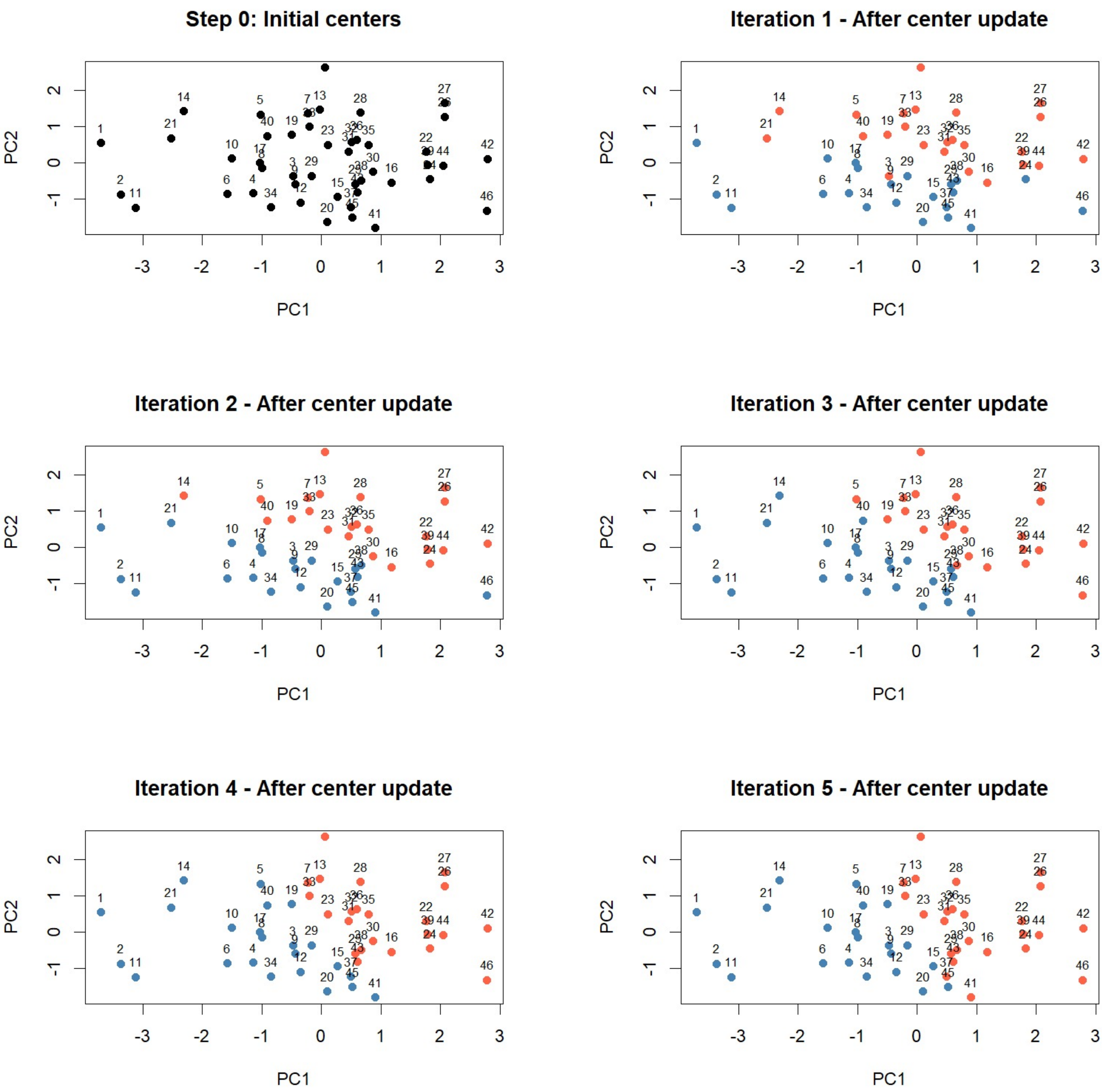


Figure 7.1: Progression of the k-means algorithm (iterations 0–5)

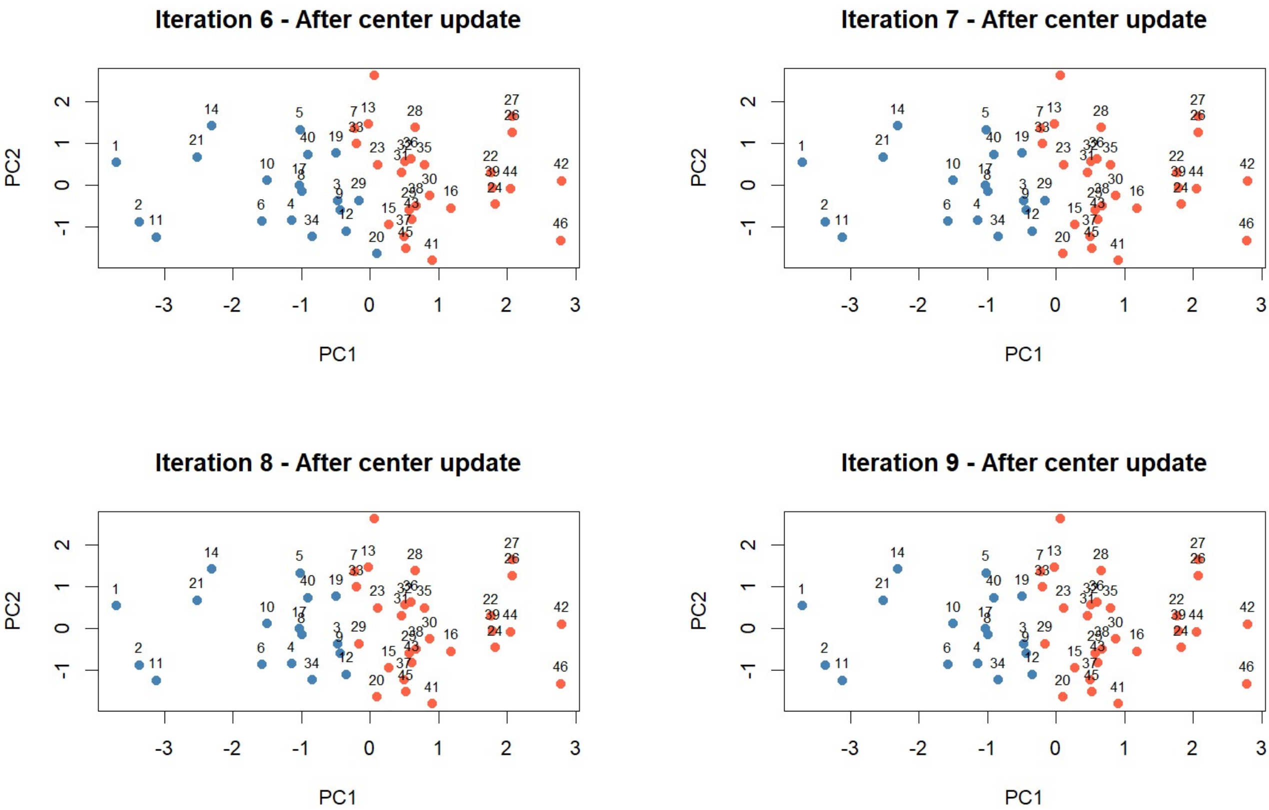


Figure 7.2: Progression of the k-means algorithm (iterations 6–9)

Empirical Performance and Cross-Validation Results

```
> print(tab)
       TrueClass
Cluster  0   1
      0 23   6
      1  2  15

Plain K-means:
> cat("Mean accuracy =", mean(acc_kmeans), "\n")
Mean accuracy = 0.7771
> cat("SD of accuracy =", sd(acc_kmeans), "\n")
SD of accuracy = 0.130323
> print(round(mean_conf_kmeans, 3))
         TrueClass
Predicted     0     1
        0 2.942 0.553
        1 1.676 4.830
```

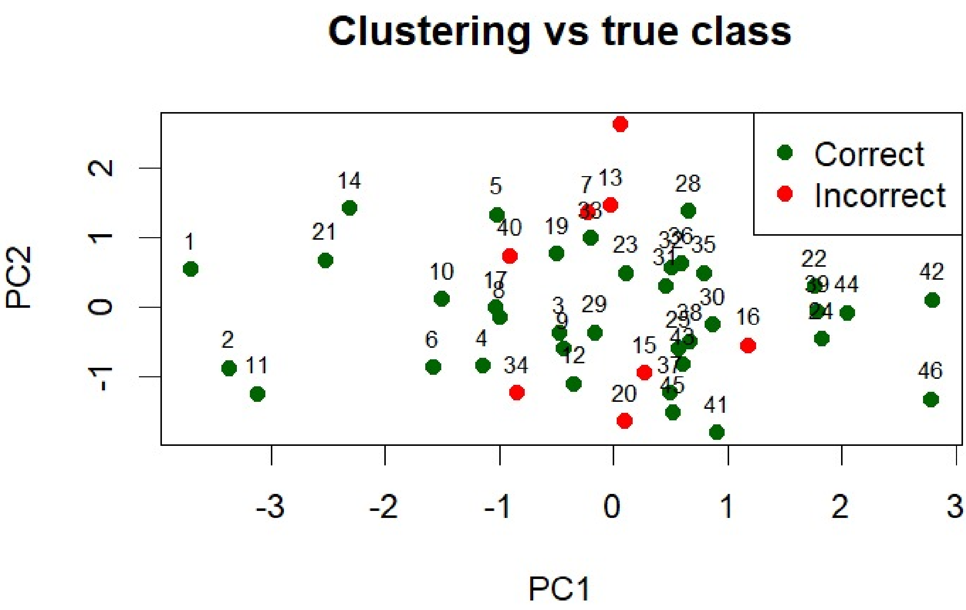


Figure 7.3: Clustering vs true class using plain k-means

To rigorously evaluate the predictive validity of this converged baseline model, its performance was analyzed both on the full, unstratified dataset and via a 2000-iteration Monte Carlo Cross-Validation framework ($n_{train} = 36, n_{test} = 10$).

- **Full Dataset Recovery (Confusion Matrix):** When evaluated against the entire dataset, the baseline k-means algorithm correctly clustered 23 bankrupt firms (Group 0) and 15 financially sound firms (Group 1). However, it misclassified 8 total observations (2 false positives, 6 false negatives). This indicates that while the unsupervised geometry naturally aligns with the true financial states to a significant degree, it is imperfect and susceptible to regional overlap.
- **Cross-Validated Mean Accuracy:** The Monte Carlo simulation yields an average out-of-sample accuracy of 0.7771 (77.71%). This confirms that the baseline k-means model provides reasonable, but moderately constrained, predictive capability when completely stripped of explicit class supervision.
- **Metric Volatility (Standard Deviation):** The standard deviation of the accuracy across the 2000 iterations is 0.1303. This variance reflects the algorithm's inherent sensitivity to stochastic centroid initialization and minor fluctuations in the training sample geometry.

- **Averaged Confusion Geometry:** The averaged confusion matrix reveals a structural imbalance in the error rates. The algorithm is highly effective at identifying true negatives (predicting Group 0 correctly $\approx$ 2.942 times per test set), but struggles more with false negatives (misclassifying Group 1 as Group 0 $\approx$ 1.676 times per test set). This suggests that the unsupervised boundary is slightly skewed, conservatively over-assigning borderline cases to the bankrupt cohort.

7.3.2 Evaluation of Strategy 2: k-Means with Supervised Initialization

Geometric Analysis of Classification Accuracy

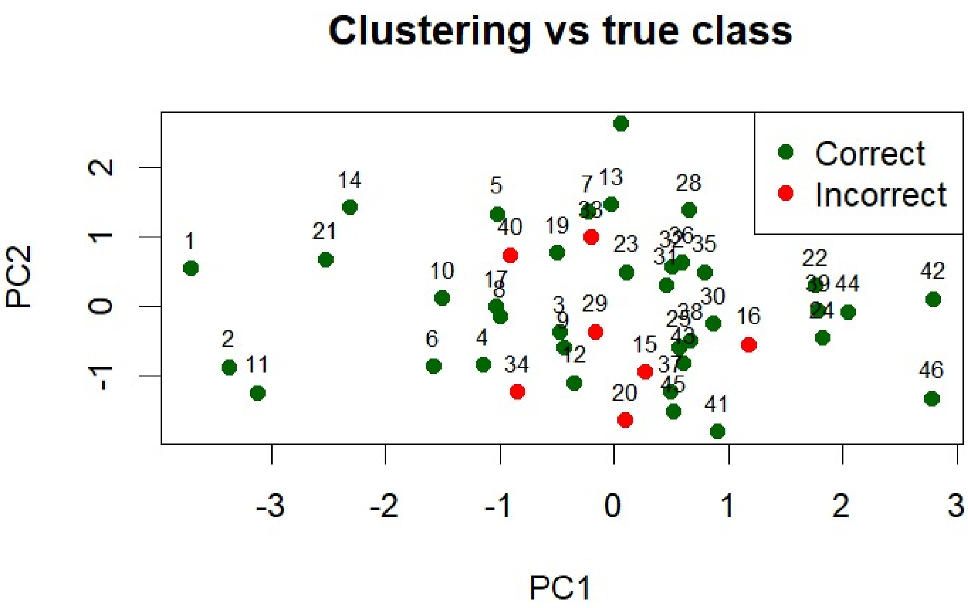


Figure 7.4: Clustering vs true class using plain k-means

Figure 7.4 visualizes the spatial distribution of the clustering outcomes when the algorithm is deterministically seeded with the true multivariate group means.

- **Localization of Misclassifications:** The plot maps correctly clustered observations in green and misclassifications in red. Geometrically, the errors are strictly confined to the central topological boundary region (primarily spanning $-1 < \text{PC1} < 1$).
- **Resolution of Extremes:** The structural extremes of the feature space—representing highly solvent or deeply distressed firms—are perfectly classified.
- **Topological Implication:** Because the algorithm was explicitly guided to the true centers of mass, these central misclassifications do not represent a failure of algorithm convergence. Rather, they highlight the irreducible structural overlap between the two financial states within the principal component space. The algorithm has found the optimal distance-based boundary, but the intrinsic data distributions simply are not perfectly separable.

Empirical Performance and Cross-Validation Results

To quantify the methodological benefit of supervised initialization, the performance metrics were evaluated and compared against the stochastic baseline.

```
> table(Cluster = km$cluster, TrueClass = data$y)
       TrueClass
Cluster  0  1
      0 18  4
      1  3 21
```

```
K-means with group-mean initial centers:
> cat("Mean accuracy =", mean(acc_kmeans_groupmean), "\n")
Mean accuracy = 0.80535
> cat("SD of accuracy =", sd(acc_kmeans_groupmean), "\n")
SD of accuracy = 0.1216708
> print(round(mean_conf_kmeans_groupmean, 3))
         TrueClass
Predicted     0     1
        0 3.365 0.695
        1 1.252 4.688
```

- **Full Dataset Recovery (Confusion Matrix):** When evaluated on the entire unstratified dataset, this seeded algorithm correctly clustered 18 bankrupt firms (Group 0) and 21 financially sound firms (Group 1). The total number of misclassifications dropped to 7 (4 false positives, 3 false negatives), demonstrating an immediate structural improvement over the baseline model's 8 errors.

- **Cross-Validated Mean Accuracy:** The 2000-iteration Monte Carlo cross-validation yields a robust average accuracy of 0.80535 (80.54%). This measurable increase from the baseline's 77.71% mathematically proves that stochastic initialization frequently traps the standard k-means algorithm in suboptimal local clustering. Providing informed starting vectors allows the algorithm to consistently find a superior, more globally optimal geometric separation.

- **Enhanced Algorithmic Stability:** The standard deviation of the accuracy across the iterations decreased to 0.1216. This reduction in volatility confirms that anchoring the initial centroids effectively stabilizes the algorithm against the structural fluctuations introduced by random sample partitioning.

- **Averaged Confusion Geometry:** The aggregated confusion matrix over the test sets reveals a highly balanced predictive profile. The algorithm correctly predicts Group 0 at a rate of 3.365 and Group 1 at a rate of 4.688 per test iteration. Crucially, the off-diagonal error rates (0.695 and 1.252) are visibly lower and more symmetric than the baseline model, indicating that the supervised initialization successfully corrected the baseline's conservative skew, resulting in a more equitable and accurate decision boundary.

7.3.3 Evaluation of Strategy 3: Hybrid Discriminant Clustering (LDA + k-Means)

Geometric Process and Interpretation of Hybrid Clustering (LDA + k-Means)

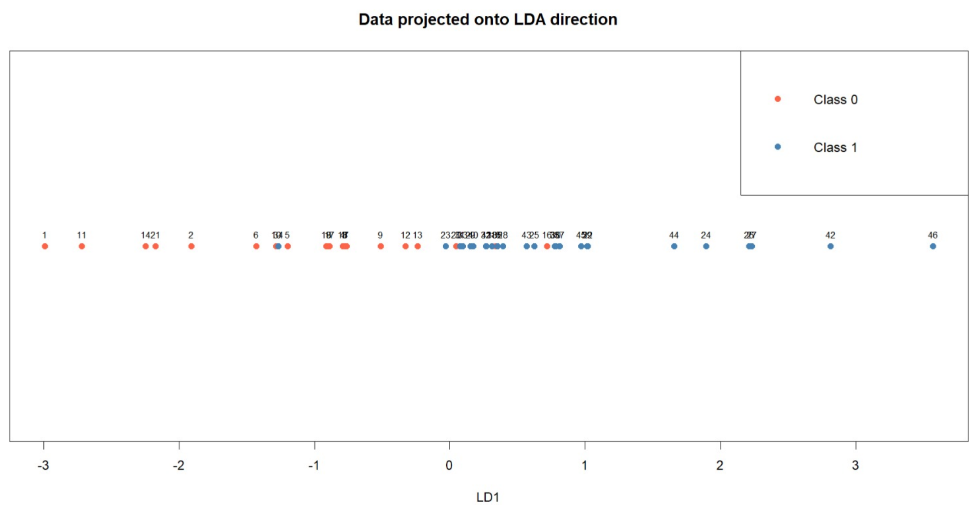


Figure 7.5: Projection of observations onto the LDA direction

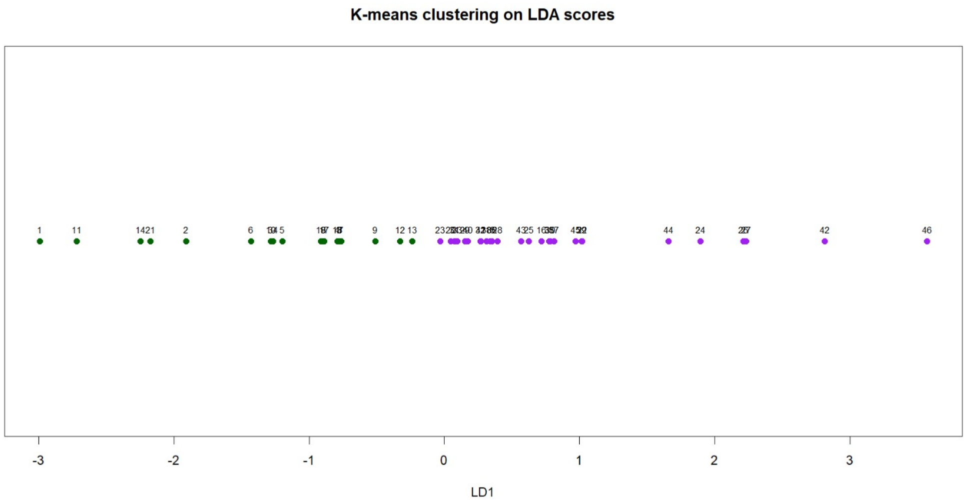


Figure 7.6: k-means clustering applied to LDA scores

Figures 7.6 and 7.7 visualize the step-by-step mathematical execution of the third clustering strategy, demonstrating how the hybrid pipeline isolates the classification signal into a strictly one-dimensional space.

Phase 1: Supervised Dimensionality Reduction (Figure 7.6)

- **Canonical Projection:** The methodology first utilizes Linear Discriminant Analysis (LDA) to derive the single linear combination of the feature space that explicitly maximizes the ratio of between-group to within-group variance. All multidimensional data points are subsequently projected onto this single optimal discriminant axis (*LD1*).
- **Maximization of Linear Separability:** Figure 7.6 maps the true financial states (Class 0 and Class 1) against their calculated *LD1* scores. The visual reveals a highly polarized, bimodal distribution. By utilizing supervised class labels during this preliminary step, the projection successfully forces the two cohorts to opposing ends of the number line, compressing the entirety of the actionable discriminative signal into a single scalar dimension.

Phase 2: One-Dimensional Unsupervised Clustering (Figure 7.7)

- **Scalar k-Means Application:** The k-means algorithm ($k = 2$) is subsequently deployed strictly on these extracted 1D scores. Because the input space is mathematically reduced to a single vector, the algorithm's Euclidean distance metric is radically simplified, evaluating only the linear distance between scalars on the axis.
- **Definition of a Strict Threshold:** Figure 7.7 demonstrates the geometric outcome of this 1D clustering application. The algorithm effectively acts as an optimized thresholding function, locating the midpoint between the two densest scalar masses (near $LD1 \approx -0.1$). It cleanly bifurcates the axis, establishing a strict boundary with zero overlap.

Methodological Synthesis

- **Filtration of Orthogonal Noise:** These visualizations provide the definitive mathematical explanation for the hybrid model's superior predictive accuracy (82.80%). By conditioning the feature space with LDA *prior* to clustering, the k-means algorithm is completely insulated from the uninformative, orthogonal variance (such as the noise observed in the X_4) that corrupted the baseline multi-dimensional clustering. It operates exclusively on the refined discriminatory signal, guaranteeing convergence to a highly optimal classification boundary.

Geometric Analysis of Classification Accuracy

Figure 7.5 maps the classification outcomes of the hybrid methodology, where k-means clustering was applied strictly to the one-dimensional canonical scores generated by a preliminary Linear Discriminant Analysis (LDA), and subsequently projected back into the principal component space for visual comparison.

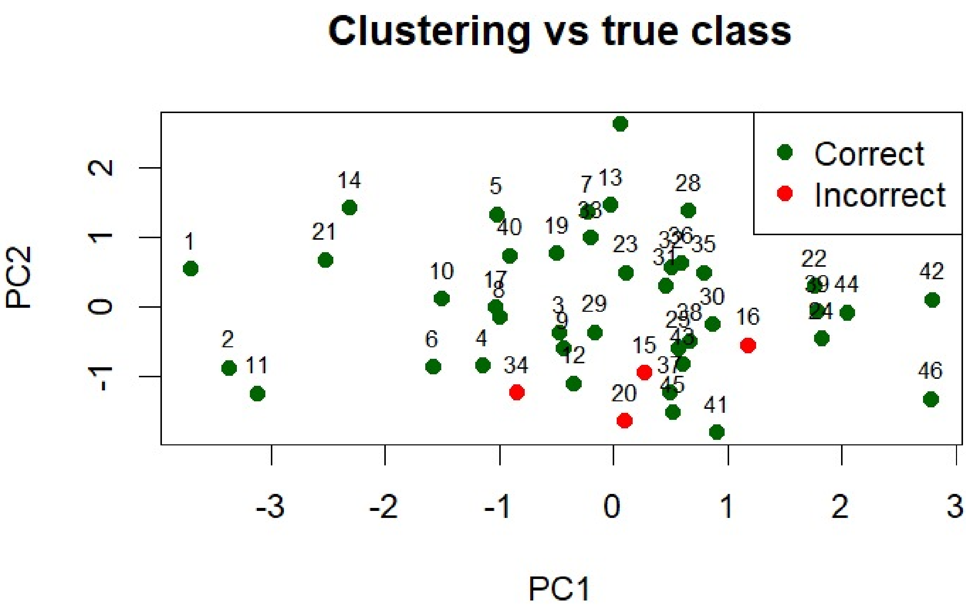


Figure 7.7: Clustering vs true class using LDA followed by k-means

- **Minimization of Overlap:** The visualization demonstrates a profound reduction in misclassifications (red points) compared to both the baseline and mean-seeded algorithms. The errors are now exceptionally sparse and strictly localized to the absolute tightest region of the boundary interface.
- **Filtration of Orthogonal Noise:** By utilizing LDA as a supervised precursor, the feature space was mathematically rotated and compressed to explicitly maximize between-group variance. Consequently, when the k-means algorithm applied its standard Euclidean distance metric, it was completely insulated from the uninformative, orthogonal variance (such as the noise captured by PC2). This structurally guaranteed convergence to a highly optimized discriminative boundary.

Empirical Performance and Cross-Validation Results

An analysis of the formal test outputs confirms the substantial mathematical advantage of this hybrid strategy:

```
> print(tab)
       TrueClass
Cluster  0  1
      0 18  1
      1  3 24

LDA + K-means:
> cat("Mean accuracy =", mean(acc_lda), "\n")
Mean accuracy = 0.828
> cat("SD of accuracy =", sd(acc_lda), "\n")
SD of accuracy = 0.1254746
> print(round(mean_conf_lda, 3))
         TrueClass
Predicted     0     1
        0 3.662 0.764
        1 0.956 4.618
```

- **Full Dataset Recovery (Confusion Matrix):** Evaluated against the complete unstratified dataset, this approach achieved extraordinary structural recovery. It correctly clustered 18 bankrupt firms ($Y = 0$) and 24 financially sound firms ($Y = 1$). The total number of misclassifications dropped to a mere 4 observations (1 false positive, 3 false negatives). This represents a definitive, macroscopic improvement over both the baseline (8 errors) and the supervised initialization (7 errors) methodologies.
- **Cross-Validated Mean Accuracy:** The 2000-iteration Monte Carlo cross-validation framework yields a superior average out-of-sample accuracy of 0.828 (82.80%). This robust performance gain mathematically

proves that the initial LDA projection rigorously conditions the feature space, allowing the subsequent clustering phase to achieve an efficacy that rivals fully supervised models.

- **Algorithmic Stability:** The standard deviation of the accuracy (0.1255) remains structurally consistent with the previous models. This establishes that the enhanced predictive power is structurally sound and does not introduce erratic out-of-sample volatility across different sampling partitions.
- **Averaged Confusion Geometry:** The aggregated test-set confusion matrix reflects a highly optimized predictive balance. The algorithm exhibits robust, consistent identification for both Group 0 (3.662) and Group 1 (4.618). Most crucially, the off-diagonal error rates are strictly minimized (misclassifying Group 0 as Group 1 ≈ 0.956 times, and Group 1 as Group 0 ≈ 0.764 times per iteration). This tightly symmetric error distribution confirms the derivation of a highly equitable, unbiased classification boundary.

8. Conclusion

This study executed a rigorous, multi-stage statistical analysis to develop and evaluate predictive frameworks for corporate bankruptcy classification. By systematically traversing from foundational exploratory diagnostics to advanced dimensionality reduction, probabilistic modeling, and unsupervised topological discovery, the analysis successfully mapped the complex multivariate relationships distinguishing bankrupt from financially sound firms.

Methodological Pipeline and Rationale

The project adhered to a strict analytical sequence, structurally designed to mathematically validate each subsequent modeling phase:

- **Step 1: Exploratory Diagnostics and Baseline Assessment:**
  - *Action:* Conducted univariate and multivariate normality tests alongside evaluations of covariance homogeneity (Box's M-test).
  - *Rationale:* To determine whether the raw feature space satisfied the strict parametric assumptions of Gaussian-based classifiers like Linear and Quadratic Discriminant Analysis. We discovered severe heteroscedasticity, skewness, and high multicollinearity, proving the raw data was mathematically suboptimal for linear modeling.
- **Step 2: Structural Transformation and Outlier Excision:**
  - *Action:* Applied Box-Cox variance-stabilizing transformations and systematically excised severe multivariate anomalies using χ^2 quantiles of Mahalanobis distances (e.g., Observation 46).
  - *Rationale:* To mathematically coerce the dataset into approximating a multivariate normal distribution. This crucial step stabilized the covariance matrices, validated the use of linear decision boundaries, and prevented extreme outliers from exerting disproportionate leverage on the discriminant functions.
- **Step 3: Dimensionality Reduction and Feature Extraction:**
  - *Action:* Executed Principal Component Analysis (PCA) and evaluated Exploratory Factor Analysis (EFA), ultimately selecting PCA for the final pipeline.
  - *Rationale:* To combat multicollinearity and geometrically compress the feature space. The KMO diagnostic test proved EFA was structurally inappropriate due to the turnover ratio (X_4) lacking shared variance. PCA successfully isolated the fundamental predictive signal into a single dominant vector (PC1), stripping away orthogonal noise.
- **Step 4: Probabilistic Classification Modeling:**
  - *Action:* Fitted LDA, QDA, and Logistic Regression models, validating them via a 2000-iteration Monte Carlo framework.
  - *Rationale:* To comparatively assess the predictive efficacy of generative (LDA/QDA) versus discriminative (Logistic) algorithms across raw, transformed, and dimensionally reduced topological spaces.
- **Step 5: Unsupervised Geometric Clustering:**
  - *Action:* Deployed classical k-means, mean-seeded k-means, and a hybrid LDA-projected k-means methodology.
  - *Rationale:* To investigate whether the natural, distance-based structure of the financial metrics inherently aligns with the true bankruptcy states, completely independent of explicit class supervision during the optimization phase.

Comparative Performance Analysis

The systematic evaluation of these models yielded several critical insights regarding algorithmic performance across different feature spaces:

- **Raw vs. Transformed Space (The Impact of Preprocessing):** In the raw space, QDA outperformed LDA by adapting to the severe heteroscedasticity. However, post-transformation, the linear boundaries of LDA and Logistic Regression stabilized and geometrically converged. The preprocessing pipeline proved essential, transforming a highly entangled structure into a space with actionable linear separability.
- **Full Feature Space vs. Principal Component Space:** Projecting the data onto the primary principal component (PC1) resulted in a fascinating algorithmic parity. By stripping away complex covariance structures and isolating the data to a single vector, LDA, QDA, and Logistic Regression converged to nearly identical F_1 scores (≈ 0.815). Adding PC2 yielded no meaningful predictive enhancement, confirming that the classification signal is inherently one-dimensional.
- **Evolution of Clustering Efficacy:** The unsupervised analysis demonstrated a clear, mathematically quantifiable progression in structural recovery:
  - *Standard k-Means:* Achieved a baseline mean accuracy of 77.71%, struggling with localized topological overlap due to stochastic initialization.
  - *Mean-Seeded k-Means:* Improved accuracy to 80.54% by anchoring centroids to true centers of mass, proving that informed initialization avoids suboptimal local minima.
  - *Hybrid LDA + k-Means:* Delivered the highest unsupervised performance at 82.80%. By using LDA to filter out orthogonal noise before clustering, this approach mathematically guaranteed convergence to a highly optimal boundary, generating the lowest off-diagonal error rates of the entire study.

Final Achievements

Ultimately, this analysis achieved a highly refined, mathematically validated framework for bankruptcy prediction. We empirically proved that while complex models (like QDA) are necessary for noisy, raw financial data, applying rigorous structural transformations and dimensionality reduction (PCA) fundamentally simplifies the problem space. By effectively funneling the multivariate data into a mathematically well-behaved subspace, we demonstrated that parsimonious linear models—and highly optimized hybrid clustering algorithms—can achieve robust, highly accurate discrimination between financial states.